CHAPTER C12
SEISMIC DESIGN REQUIREMENTS FOR BUILDING STRUCTURES

C12.1 STRUCTURAL DESIGN BASIS

The performance expectations for structures designed in accordance with this standard are described in Sections C11.1 and C11.5. Structures designed in accordance with the standard are likely to have a low probability of collapse but may suffer serious structural damage if subjected to the risk-targeted maximum considered earthquake (MCE_R) or stronger ground motion.

Although the seismic requirements of the standard are stated in terms of forces and loads, there are no external forces applied to the structure during an earthquake as, for example, is the case during a windstorm. The design forces are intended only as approximations to generate internal forces suitable for proportioning the strength and stiffness of structural elements and for estimating the deformations (when multiplied by the deflection amplification factor, C_d) that would occur in the same structure in the event of design earthquake (not MCE_R) ground motion.

C12.1.1 Basic Requirements. Chapter 12 of the standard sets forth a set of coordinated requirements that must be used together. The basic steps in structural design of a building structure for acceptable seismic performance are as follows:

1. Select gravity- and seismic force-resisting systems appropriate to the anticipated intensity of ground shaking. Section 12.2 sets forth limitations depending on the Seismic Design Category.
2. Configure these systems to produce a continuous, regular, and redundant load path so that the structure acts as an integral unit in responding to ground shaking. Section 12.3 addresses configuration and redundancy issues.
3. Analyze a mathematical model of the structure subjected to lateral seismic motions and gravity forces. Sections 12.6 and 12.7 set forth requirements for the method of analysis and for construction of the mathematical model. Sections 12.5, 12.8, and 12.9 set forth requirements for conducting a structural analysis to obtain internal forces and displacements.
4. Proportion members and connections to have adequate lateral and vertical strength and stiffness. Section 12.4 specifies how the effects of gravity and seismic loads are to be combined to establish required strengths, and Section 12.12 specifies deformation limits for the structure.

One- to three-story structures with shear wall or braced frame systems of simple configuration may be eligible for design under the simplified alternative procedure contained in Section 12.14. Any other deviations from the requirements of Chapter 12 are subject to approval by the authority having jurisdiction (AHJ) and must be rigorously justified, as specified in Section 11.1.4.

The baseline seismic forces used for proportioning structural elements (individual members, connections, and supports) are static horizontal forces derived from an elastic response spectrum procedure. A basic requirement is that horizontal motion can come from any direction relative to the structure, with detailed requirements for evaluating the response of the structure provided in Section 12.5. For most structures, the effect of vertical ground motions is not analyzed explicitly; it is implicitly included by adjusting the load factors (up and down) for permanent dead loads, as specified in Section 12.4. Certain conditions requiring more detailed analysis of vertical response are defined in Chapters 13 and 15 for nonstructural components and nonbuilding structures, respectively.

The basic seismic analysis procedure uses response spectra that are representative of, but substantially reduced from, the anticipated ground motions. As a result, at the MCE_R level of ground shaking, structural elements are expected to yield, buckle, or otherwise behave inelastically. This approach has substantial historical precedent. In past earthquakes, structures with appropriately ductile, regular, and continuous systems that were designed using *reduced* design forces have performed acceptably. In the standard, such design forces are computed by dividing the forces that would be generated in a structure behaving elastically when subjected to the design earthquake ground motion by the response modification coefficient, R, and this design ground motion is taken as two-thirds of the MCE_R ground motion.

The intent of R is to reduce the demand determined, assuming that the structure remains elastic at the design earthquake, to target the development of the first significant yield. This reduction accounts for the displacement ductility demand, R_d, required by the system and the inherent overstrength, Ω, of the seismic force-resisting system (SFRS) (Fig. C12.1-1). Significant yield is the point where complete plastification of a critical region of the SFRS first occurs (e.g., formation of the first plastic hinge in a moment frame), and the stiffness of the SFRS to further increases in lateral forces decreases as continued inelastic behavior spreads within the SFRS. This approach is consistent with member-level ultimate strength design practices. As such, first significant yield should not be misinterpreted as the point where first yield occurs in any member (e.g., 0.7 times the yield moment of a steel beam or either initial cracking or initiation of yielding in a reinforcing bar in a reinforced concrete beam or wall).

Fig. C12.1-1 shows the lateral force versus deformation relation for an archetypal moment frame used as an SFRS. First significant yield is shown as the lowest plastic hinge on the force–deformation diagram. Because of particular design rules and limits, including material strengths in excess of nominal or project-specific design requirements, structural elements are stronger by some degree than the strength required by analysis. The SFRS is therefore expected to reach first significant yield for forces in excess of design forces. With increased lateral loading,

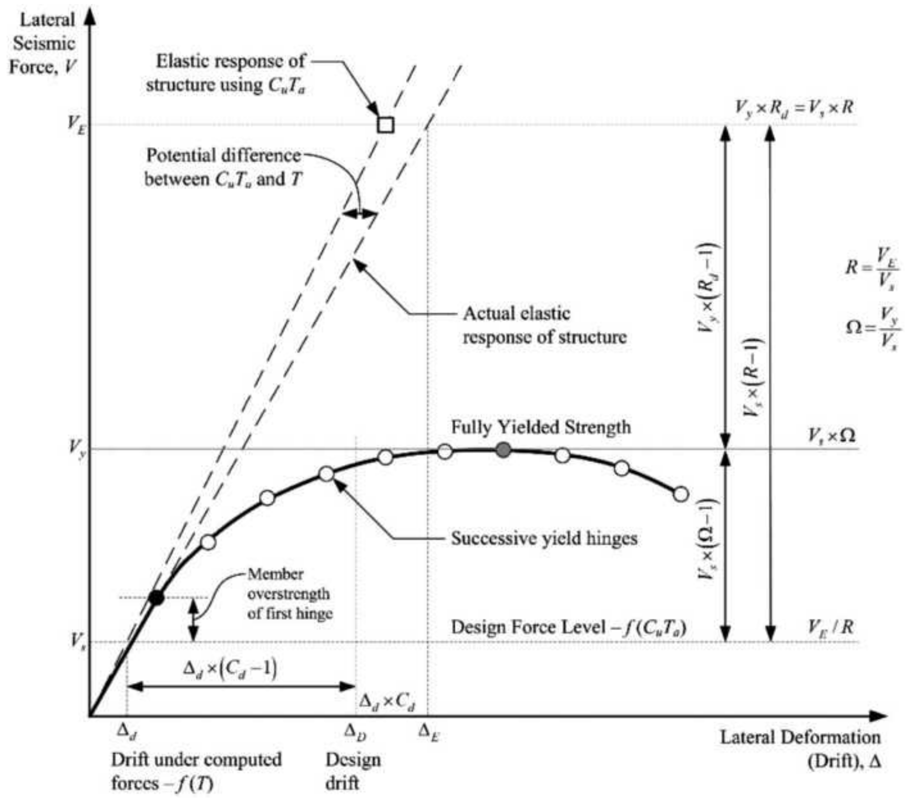


FIGURE C12.1-1 Inelastic Force–Deformation Curve

additional plastic hinges form and the resistance increases at a reduced rate (following the solid curve) until the maximum strength is reached, representing a fully yielded system. The maximum strength developed along the curve is substantially higher than that at first significant yield, and this margin is referred to as the system overstrength capacity. The ratio of these strengths is denoted as Ω. Furthermore, the figure illustrates the potential variation that can exist between the actual elastic response of a system and that considered using the limits on the fundamental period (assuming 100% mass participation in the fundamental mode—see Section C12.8.6). Although not a concern for strength design, this variation can have an effect on the expected drifts.

The system overstrength described above is the direct result of overstrength of the elements that form the SFRS and, to a lesser extent, the lateral force distribution used to evaluate the inelastic force–deformation curve. These two effects interact with applied gravity loads to produce sequential plastic hinges, as illustrated in the figure. This member overstrength is the consequence of several sources. First, material overstrength (i.e., actual material strengths higher than the nominal material strengths specified in the design) may increase the member overstrength significantly. For example, a recent survey shows that the mean yield strength of ASTM A36 steel is about 30% to 40% higher than the specified yield strength used in design calculations. Second, member design strengths usually incorporate a strength reduction or resistance factor, ϕ, to produce a low probability of failure under design loading. It is common to not include this factor in the member load-deformation relation when evaluating the seismic response of a structure in a nonlinear structural analysis. Third, designers can introduce additional strength by selecting sections or specifying reinforcing patterns that exceed those required by the computations. Similar situations occur where prescriptive minimums of the standard, or of the referenced design standards, control the design. Finally, the design of many flexible structural systems (e.g., moment-resisting frames) can be controlled by the drift rather than strength, with sections selected to control lateral deformations rather than to provide the specified strength.

The result is that structures typically have a much higher lateral strength than that specified as the minimum by the standard, and the first significant yielding of structures may occur at lateral load levels that are 30% to 100% higher than the prescribed design seismic forces. If provided with adequate ductile detailing, redundancy, and regularity, full yielding of structures may occur at load levels that are two to four times the prescribed design force levels.

Most structural systems have some elements whose action cannot provide reliable inelastic response or energy dissipation. Similarly, some elements are required to remain essentially elastic to maintain the structural integrity of the structure (e.g., columns supporting a discontinuous SFRS). Such elements and actions must be protected from undesirable behavior by considering that the actual forces within the structure can be significantly larger than those at first significant yield. The standard specifies an overstrength factor, Ω_0, to amplify the prescribed seismic forces for use in design of such elements and for such actions. This approach is a simplification to determining the maximum forces that could be developed in a system and the distribution of these forces within the structure. Thus, this specified overstrength factor is neither an upper nor a lower bound; it is simply an approximation specified to provide a nominal degree of protection against undesirable behavior.

The elastic deformations calculated under these reduced forces (*see* Section C12.8.6) are multiplied by the deflection amplification factor, C_d, to estimate the deformations likely to result from the design earthquake ground motion. This factor was first introduced in ATC 3-06 (ATC 1978). For a vast majority of systems, C_d is less than R, with a few notable exceptions, where inelastic drift is strongly coupled with an increased risk of collapse (e.g., reinforced concrete bearing walls). Research over the past 30 years has illustrated that inelastic displacements may be significantly greater than Δ_E for many structures and less than Δ_E for others. Where C_d is substantially less than R, the system is considered to have damping greater than the nominal 5% of critical damping. As set forth in Section 12.12 and Chapter 13, the amplified deformations are used to assess story drifts and to determine seismic demands on elements of the structure that are not part of the seismic force-resisting system and on nonstructural components within structures.

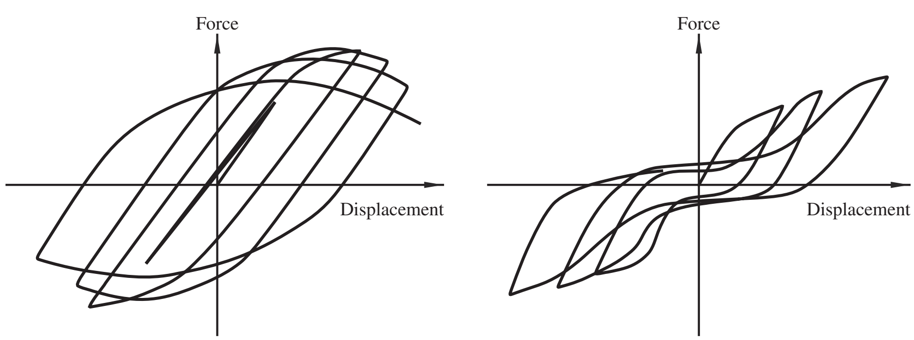
Fig. C12.1-1 illustrates the significance of seismic design parameters contained in the standard, including the response modification coefficient, R; the deflection amplification factor, C_d; and the overstrength factor, Ω_0. The values of these parameters, provided in Table 12.2-1, as well as the criteria for story drift and P-delta effects, have been established considering the characteristics of typical properly designed structures. The provisions of the standard anticipate an SFRS with redundant characteristics wherein significant system strength above the level of first significant yield can be obtained by plastification at other critical locations in the structure before the formation of a collapse mechanism. If excessive "optimization" of a structural design is performed with lateral resistance provided by only a few elements, the successive yield hinge behavior depicted in Fig. C12.1-1 is not able to form, the actual overstrength (Ω) is small, and use of the seismic design parameters in the standard may not provide the intended seismic performance.

The response modification coefficient, R, represents the ratio of the forces that would develop under the specified ground motion if the structure had an entirely linear-elastic response to the prescribed design forces (Fig. C12.1-1). The structure must be designed so that the level of significant yield exceeds the prescribed design force. The ratio R_d, expressed as $R_d = V_E/V_S$, where V_E is the elastic seismic force demand and V_S is the prescribed seismic force demand, is always larger than 1.0; thus, all structures are designed for forces smaller than those the design ground motion would produce in a structure with a completely linear-elastic response. This reduction is possible for a number of reasons. As the structure begins to yield and deform inelastically, the effective period of response of the structure lengthens, which results in a reduction in strength demand for most structures. Furthermore, the inelastic action results in a significant amount of energy dissipation (hysteretic damping) in addition to other sources of damping present below significant yield. The combined effect, which is known as the ductility reduction, explains why a properly designed structure with a fully yielded strength (V_y in Fig. C12.1-1) that is significantly lower than V_E can be capable of providing satisfactory performance under the design ground motion excitations.

The energy dissipation resulting from hysteretic behavior can be measured as the area enclosed by the force–deformation curve of the structure as it experiences several cycles of excitation. Some structures have far more energy dissipation capacity than others. The extent of energy dissipation capacity available depends largely on the amount of stiffness and strength degradation the structure undergoes as it experiences repeated cycles of inelastic deformation. Fig. C12.1-2 shows representative load deformation curves for two simple substructures, such as a beam–column assembly in a frame. Hysteretic curve (a) in the figure represents the behavior of substructures that have been detailed for ductile behavior. The substructure can maintain almost all of its strength and stiffness over several large cycles of inelastic deformation. The resulting force–deformation "loops" are quite wide and open, resulting in a large amount of energy dissipation. Hysteretic curve (b) represents the behavior of a substructure that has much less energy dissipation than that for the substructure (a) but has a greater change in response period. The structural response is determined by a combination of energy dissipation and period modification.

The principles of this section outline the conceptual intent behind the seismic design parameters used by the standard. However, these parameters are based largely on engineering judgment of the various materials and performance of structural systems in past earthquakes and cannot be directly computed using the relationships presented in Fig. C12.1-1. The seismic design parameters chosen for a specific project or system should be chosen with care. For example, lower values should be used for structures possessing a low degree of redundancy wherein all the plastic hinges required for the formation of a mechanism may be formed essentially simultaneously and at a force level close to the specified design strength. This situation can result in considerably more detrimental P-delta effects. Because it is difficult for individual designers to judge the extent to which the value of R should be adjusted based on the inherent redundancy of their designs, Section 12.3.4 provides the redundancy factor, ρ, that is typically determined by being based on the removal of individual seismic force-resisting elements.

Higher order seismic analyses are permitted for any structure and are required for some structures (*see* Section 12.6); lower limits based on the equivalent lateral force procedure may, however, still apply.



(a) Ductile hysteresis loops (b) Pinched hysteresis loops

FIGURE C12.1-2 Typical Hysteretic Curves

C12.1.2 Member Design, Connection Design, and Deformation Limit. Given that key elements of the seismic force-resisting system are likely to yield in response to ground motions, as discussed in Section C12.1.1, it might be expected that structural connections would be required to develop the strength of connected members. Although that is a logical procedure, it is not a general requirement. The actual requirement varies by system and generally is specified in the standards for design of the various structural materials cited by reference in Chapter 14. Good seismic design requires careful consideration of this issue.

C12.1.3 Continuous Load Path and Interconnection. In effect, Section 12.1.3 calls for the seismic design to be complete and in accordance with the principles of structural mechanics. The loads must be transferred rationally from their point of origin to the final point of resistance. This requirement should be obvious, but it often is overlooked by those inexperienced in earthquake engineering. Design consideration should be given to potentially adverse effects where there is a lack of redundancy. Given the many unknowns and uncertainties in the magnitude and characteristics of earthquake loading, in the materials and systems of construction for resisting earthquake loadings, and in the methods of analysis, good earthquake engineering practice has been to provide as much redundancy as possible in the seismic force-resisting system of buildings. Redundancy plays an important role in determining the ability of the building to resist earthquake forces. In a structural system without redundant elements, every element must remain operative to preserve the integrity of the building structure. However, in a highly redundant system, one or more redundant elements may fail and still leave a structural system that retains its integrity and can continue to resist lateral forces, albeit with diminished effectiveness.

Although a redundancy requirement is included in Section 12.3.4, overall system redundancy can be improved by making all joints of the vertical load-carrying frame moment resisting and incorporating them into the seismic force-resisting system. These multiple points of resistance can prevent a catastrophic collapse caused by distress or failure of a member or joint. (The overstrength characteristics of this type of frame are discussed in Section C12.1.1.)

The minimum connection forces are not intended to be applied simultaneously to the entire seismic force-resisting system.

C12.1.4 Connection to Supports. The requirement is similar to that given in Section 1.4 on connections to supports for general structural integrity. See Section C1.4.

C12.1.5 Foundation Design. Most foundation design criteria are still stated in terms of allowable stresses, and the forces computed in the standard are all based on the strength level of response. When developing strength-based criteria for foundations, all the factors cited in Section 12.1.5 require careful consideration. Section C12.13 provides specific guidance.

C12.1.6 Material Design and Detailing Requirements. The design limit state for resistance to an earthquake is unlike that for any other load within the scope of the standard. The earthquake limit state is based on overall system performance, not member performance, where repeated cycles of inelastic straining are accepted as an energy-dissipating mechanism. Provisions that modify customary requirements for proportioning and detailing structural members and systems are provided to produce the desired performance.

C12.2 STRUCTURAL SYSTEM SELECTION

C12.2.1 Selection and Limitations. For the purpose of seismic analysis and design requirements, seismic force-resisting systems are grouped into categories as shown in Table 12.2-1. These categories are subdivided further for various types of vertical elements used to resist seismic forces. In addition, the sections for detailing requirements are specified.

Specification of response modification coefficients, R, requires considerable judgment based on knowledge of actual earthquake performance and research studies. The coefficients and factors in Table 12.2-1 continue to be reviewed in light of recent research results. The values of R for the various systems were selected considering observed performance during past earthquakes, the toughness (ability to dissipate energy without serious degradation) of the system, and the amount of damping typically present in the system when it undergoes inelastic response. FEMA P-695 (2009b) has been developed with the purpose of establishing and documenting a methodology for quantifying seismic force-resisting system performance and response parameters for use in seismic design. Whereas R is a key parameter being addressed, related design parameters such as the overstrength factor, Ω_0, and the deflection amplification factor, C_d, also are addressed. Collectively, these terms are referred to as "seismic design coefficients (or factors)." Future systems are likely to derive their seismic design coefficients (or factors) using this methodology, and existing system coefficients (or factors) also may be reviewed in light of this new procedure.

Height limits have been specified in codes and standards for more than 50 years. The structural system limitations and limits on structural height, h_n, specified in Table 12.2-1, evolved from these initial limitations and were further modified by the collective expert judgment of the NEHRP Provisions Update Committee (PUC) and the ATC-3 project team (the forerunners of the PUC). They have continued to evolve over the past 30 years based on observations and testing, but the specific values are based on subjective judgment.

In a bearing wall system, major load-carrying columns are omitted and the walls carry a major portion of the gravity (dead and live) loads. The walls supply in-plane lateral stiffness and strength to resist wind and earthquake loads and other lateral loads. In some cases, vertical trusses are used to augment lateral stiffness. In general, lack of redundancy for support of vertical and horizontal loads causes values of R to be lower for this system compared with R values of other systems.

In a building frame system, gravity loads are carried primarily by a frame supported on columns rather than by bearing walls. Some portions of the gravity load may be carried on bearing walls, but the amount carried should represent a relatively small percentage of the floor or roof area. Lateral resistance is provided by shear walls or braced frames. Light-framed walls with shear panels are intended for use only with wood and steel building frames. Although gravity load-resisting systems are not required to provide lateral resistance, most of them do. To the extent that the gravity load-resisting system provides additional lateral resistance, it enhances the building's seismic performance capability, so long as it is capable of resisting the resulting stresses and undergoing the associated deformations.

In a moment-resisting frame system, moment-resisting connections between the columns and beams provide lateral resistance. In Table 12.2-1, such frames are classified as ordinary, intermediate, or special. In high seismic design categories, the anticipated ground motions are expected to produce large inelastic demands, so special moment frames designed and detailed for ductile response in accordance with Chapter 14 are required. In low Seismic Design Categories, the inherent overstrength in typical structural designs is such that the anticipated inelastic demands are somewhat reduced, and less ductile systems may be used safely. Because these less ductile ordinary framing systems do not possess as much toughness, lower values of R are specified.

The values for R, Ω_0, and C_d at the composite systems in Table 12.2-1 are similar to those for comparable systems of structural steel and reinforced concrete. Use of the tabulated values is allowed only when the design and detailing requirements in Section 14.3 are followed.

In a dual system, a three-dimensional space frame made up of columns and beams provides primary support for gravity loads. Primary lateral resistance is supplied by shear walls or braced frames, and secondary lateral resistance is provided by a moment frame complying with the requirements of Chapter 14.

Where a beam–column frame or slab–column frame lacks special detailing, it cannot act as an effective backup to a shear wall subsystem, so there are no dual systems with ordinary moment frames. Instead, Table 12.2-1 permits the use of a shear wall–frame interactive system with ordinary reinforced concrete moment frames and ordinary reinforced concrete shear walls. Use of this defined system, which requires compliance with Section 12.2.5.8, offers a significant advantage over a simple combination of the two constituent ordinary reinforced concrete systems. Where those systems are simply combined, Section 12.2.3.3 would require use of seismic design parameters for an ordinary reinforced concrete moment frame.

In a cantilevered column system, stability of mass at the top is provided by one or more columns with base fixity acting as a single-degree-of-freedom system.

Cantilever column systems are essentially a special class of moment-resisting frame, except that they do not possess the redundancy and overstrength that most moment-resisting frames derive from sequential formation of yield or plastic hinges. Where a typical moment-resisting frame must form multiple plastic hinges in members to develop a yield mechanism, a cantilever column system develops hinges only at the base of the columns to form a mechanism. As a result, their overstrength is limited to that provided by material overstrength and by design conservatism.

It is permitted to construct cantilever column structures using any of the systems that can be used to develop moment frames, including ordinary and special steel; ordinary, intermediate, and special concrete; and timber frames. The system limitations for cantilever column systems reflect the type of moment frame detailing provided but with a limit on structural height, h_n, of 35 ft (10.7 m).

The value of R for cantilever column systems is derived from moment-resisting frame values where R is divided by Ω_0 but is not taken as less than 1 or greater than 2 1/2. This range accounts for the lack of sequential yielding in such systems. C_d is taken as equal to R, recognizing that damping is quite low in these systems and inelastic displacement of these systems is not less than the elastic displacement.

C12.2.1.1 Alternative Structural Systems. Historically, this standard has permitted the use of alternative seismic force-resisting systems subject to satisfactory demonstration that the proposed systems' lateral force resistance and energy dissipation capacity is equivalent to structural systems listed in Table 12.2-1, for equivalent values of the response modification coefficient, R, overstrength factor, Ω_0, and deflection amplification coefficient, C_d. These design factors were established based on limited analytical and laboratory data and the engineering judgment of the developers of the standard.

Under funding from the Federal Emergency Management Agency, the Applied Technology Council developed a rational methodology for validation of design criteria for seismic force-resisting systems under its ATC-63 project. Published as *FEMA P-695* (2009b), this methodology recognizes that the fundamental goal of seismic design rules contained in the standard is to limit collapse probability to acceptable levels. The *FEMA P-695* methodology uses nonlinear response history analysis to predict an adjusted collapse margin ratio (ACMR) for a suite of archetypical structures designed in accordance with a proposed set of system-specific design criteria and subjected to a standard series of ground motion accelerograms. The suite of archetypical structures is intended to represent the typical types and sizes of structures that are likely to incorporate the system. The ACMR relates to the conditional probability of collapse given MCE_R shaking and considers uncertainties associated with the record-to-record variability of ground motions, the quality of the design procedure, the comprehensiveness and quality of the laboratory data upon which the analytical modeling is based; and uncertainties associated with the analytical modeling. Subsequent studies have been used to benchmark this methodology against selected systems contained in Table 12.2-1 and have demonstrated that the methodology provides rational results consistent with past engineering judgment for many systems. The *FEMA P-695* methodology is therefore deemed to constitute the preferred procedure for demonstrating adequate collapse resistance for new structural systems not currently contained in Table 12.2-1.

Under the *FEMA P-695* methodology, the archetypes used to evaluate seismic force-resisting systems are designed using the criteria for Risk Category II structures and are evaluated to demonstrate that the conditional probability of collapse of such structures conforms to the 10% probability of collapse goal stated in this section and also described in Section C1.3.1 of the commentary to this standard. It is assumed that application of the seismic importance factors and more restrictive drift limits associated with the design requirements for structures assigned to Risk Categories III and IV will provide such structures with the improved resistance to collapse described in Section C1.3.1 for those Risk Categories.

In addition to providing a basis for establishment of design criteria for structural systems that can be used for design of a wide range of structures, the *FEMA P-695* methodology also contains a building-specific methodology intended for application to individual structures. The rigor associated with application of the *FEMA P-695* methodology may not be appropriate to the design of individual structures that conform with limited and clearly defined exceptions to the criteria contained in the standard for a defined structural system. Nothing contained in this section is intended to require the use of *FEMA P-695* or similar methodologies for such cases.

C12.2.1.2 Elements of Seismic Force-Resisting Systems. This standard and its referenced standards specify design and detailing criteria for members and their connections (elements) of seismic force-resisting systems defined in Table 12.2-1. Substitute elements replace portions of the defined seismic force-resisting systems. Examples include proprietary products made up of special steel moment-resisting connections or proprietary shear walls for use in light-frame construction. Requirements for qualification of substitute elements of seismic force-resisting systems are intended to ensure equivalent seismic performance of the element and the system as a whole. The evaluation of suitability for substitution is based on comparison of key performance parameters of the code-defined (conforming) element and the substitute element.

FEMA P-795, *Quantification of Building Seismic Performance Factors: Component Equivalency Methodology* (2011) is an acceptable methodology to demonstrate equivalence of substitute elements and their connections and provides methods for component testing, calculation of parameter statistics from test data, and acceptance criteria for evaluating equivalency. Key performance parameters include strength ratio, stiffness ratio, deformation capacity, and cyclic strength and stiffness characteristics.

Section 12.2.1.2, item f, requires independent design review as a condition of approval of the use of substitute elements. It is not the intent that design review be provided for every project that incorporates a substitute component, but rather that such review would be performed one time, as part of the general qualification of such substitute components. When used on individual projects, evidence of such review could include an evaluation service report or review letter indicating the conditions under which use of the substitute component is acceptable.

C12.2.2 Combinations of Framing Systems in Different Directions. Different seismic force-resisting systems can be used along each of the two orthogonal axes of the structure, as long as the respective values of R, Ω_0, and C_d are used. Depending on the combination selected, it is possible that one of the two systems may limit the extent of the overall system with regard to structural system limitations or structural height, h_n; the more restrictive of these would govern.

C12.2.3 Combinations of Framing Systems in the Same Direction. The intent of the provision requiring use of the most stringent seismic design parameters (R, Ω_0, and C_d) is to prevent mixed seismic force-resisting systems that could concentrate inelastic behavior in the lower stories.

C12.2.3.1 R, C_d, and Ω_0 Values for Vertical Combinations. This section expands upon Section 12.2.3 by specifying the requirements specific to the cases where (a) the value of R for the lower seismic force-resisting system is lower than that for the upper system, and (b) the value of R for the upper seismic force-resisting system is lower than that for the lower system.

The two cases are intended to account for all possibilities of vertical combinations of seismic force-resisting systems in the same direction. For a structure with a vertical combination of three or more seismic force-resisting systems in the same direction, Section 12.2.3.1 must be applied to the adjoining pairs of systems until the vertical combinations meet the requirements therein.

There are also exceptions to these requirements for conditions that do not affect the dynamic characteristics of the structure or that do not result in concentration of inelastic demand in critical areas.

C12.2.3.2 Two-Stage Analysis Procedure. A two-stage equivalent lateral force procedure is permitted where the lower portion of the structure has a minimum of 10 times the stiffness of the upper portion of the structure. In addition, the period of the entire structure is not permitted to be greater than 1.1 times the period of the upper portion considered as a separate structure supported at the transition from the upper to the lower portion. An example would be a concrete podium under a wood- or steel-framed upper portion of a structure. The upper portion may be analyzed for seismic forces and drifts using the values of R, Ω_0, and C_d for the upper portion as a separate structure. The seismic forces (e.g., shear and overturning) at the base of the upper portion are applied to the top of the lower portion and scaled up by the ratio of $(R/\rho)_{\text{upper}}$ to $(R/\rho)_{\text{lower}}$. The lower portion, which now includes the seismic forces from the upper portion, may then be analyzed using the values of R, Ω_0, and C_d for the lower portion of the structure.

C12.2.3.3 R, C_d, and Ω_0 Values for Horizontal Combinations. For almost all conditions, the least value of R of different seismic force-resisting systems in the same direction must be used in design. This requirement reflects the expectation that the entire system will undergo the same deformation with its behavior controlled by the least ductile system. However, for light-frame construction or flexible diaphragms meeting the listed conditions, the value of R for each independent line of resistance can be used. This exceptional condition is consistent with light-frame construction that uses the ground for parking with residential use above.

C12.2.4 Combination Framing Detailing Requirements. This requirement is provided so that the seismic force-resisting system with the highest value of R has the necessary ductile detailing throughout. The intent is that details common to both systems be designed to remain functional throughout the response to earthquake load effects to preserve the integrity of the seismic force-resisting system.

C12.2.5 System-Specific Requirements

C12.2.5.1 Dual System. The moment frame of a dual system must be capable of resisting at least 25% of the design seismic forces; this percentage is based on judgment. The purpose of the 25% frame is to provide a secondary seismic force-resisting system with higher degrees of redundancy and ductility to improve the ability of the building to support the service loads (or at least the effect of gravity loads) after strong earthquake shaking. The primary system (walls or bracing) acting together with the moment frame must be capable of resisting all of the design seismic forces. The following analyses are required for dual systems:

1. The moment frame and shear walls or braced frames must resist the design seismic forces, considering fully the force and deformation interaction of the walls or braced frames and the moment frames as a single system. This analysis must be made in accordance with the principles of structural mechanics that consider the relative rigidities of the elements and torsion in the system. Deformations imposed upon members of the moment frame by their interaction with the shear walls or braced frames must be considered in this analysis.
2. The moment frame must be designed with sufficient strength to resist at least 25% of the design seismic forces.

C12.2.5.2 Cantilever Column Systems. Cantilever column systems are singled out for special consideration because of their unique characteristics. These structures often have limited redundancy and overstrength and concentrate inelastic behavior at their bases. As a result, they have substantially less energy dissipation capacity than other systems. A number of apartment buildings incorporating this system experienced severe damage and, in some cases, collapsed in the 1994 Northridge (California) earthquake. Because the ductility of columns that have large axial stress is limited, cantilever column systems may not be used where individual column axial demands from seismic load effects exceed 15% of their available axial strength, including slenderness effects.

Elements providing restraint at the base of cantilever columns must be designed for seismic load effects, including overstrength, so that the strength of the cantilever columns is developed.

C12.2.5.3 Inverted Pendulum-Type Structures. Inverted pendulum-type structures do not have a unique entry in Table 12.2-1 because they can be formed from many structural systems. Inverted pendulum-type structures have more than half of their mass concentrated near the top (producing one degree of freedom in horizontal translation) and rotational compatibility of the mass with the column (producing vertical accelerations acting in opposite directions). Dynamic response amplifies this rotation; hence, the bending moment induced at the top of the column can exceed that computed using the procedures of Section 12.8. The requirement to design for a top moment that is one-half of the base moment calculated in accordance with Section 12.8 is based on analyses of inverted pendulums covering a wide range of practical conditions.

C12.2.5.4 Increased Structural Height Limit for Steel Eccentrically Braced Frames, Steel Special Concentrically Braced Frames, Steel Buckling-Restrained Braced Frames, Steel Special Plate Shear Walls, and Special Reinforced Concrete Shear Walls. The first criterion for an increased limit on structural height, h_n, precludes extreme torsional irregularity because premature failure of one of the shear walls or braced frames could lead to excessive inelastic torsional response. The second criterion, which is similar to the redundancy requirements, is to limit the structural height of systems that are too strongly dependent on any single line of shear walls or braced frames. The inherent torsion resulting from the distance between the center of mass and the center of rigidity must be included, but accidental torsional effects are neglected for ease of implementation.

C12.2.5.5 Special Moment Frames in Structures Assigned to Seismic Design Categories D through F. Special moment frames, either alone or as part of a dual system, are required to be used in Seismic Design Categories D through F where the structural height, h_n, exceeds 160 ft (48.8 m) (or 240 ft [73.2 m]) for buildings that meet the provisions of Section 12.2.5.4) as indicated in Table 12.2-1. In shorter buildings where special moment frames are not required to be used, the special moment frames may be discontinued and supported on less ductile systems as long as the requirements of Section 12.2.3 for framing system combinations are followed.

For the situation where special moment frames are required, they should be continuous to the foundation. In cases where the foundation is located below the building's base, provisions for discontinuing the moment frames can be made as long as the seismic forces are properly accounted for and transferred to the supporting structure.

C12.2.5.6 Steel Ordinary Moment Frames. Steel ordinary moment frames (OMFs) are less ductile than steel special moment frames; consequently, their use is prohibited in structures assigned to Seismic Design Categories D, E, and F (Table 12.2-1). Structures with steel OMFs, however, have exhibited acceptable behavior in past earthquakes where the structures were sufficiently limited in their structural height, number of stories, and seismic mass. The provisions in the standard reflect these observations. The exception is discussed separately below. Table C12.2-1 summarizes the provisions.

C12.2.5.6.1 Seismic Design Category D or E. Single-story steel OMFs are permitted, provided that (a) the structural height, h_n, is a maximum of 65 ft (20 m), (b) the dead load supported by and tributary to the roof is a maximum of 20 lb/ft^2 (0.96 kN/m^2), and (c) the dead load of the exterior walls more than 35 ft (10.6 m) above the seismic base tributary to the moment frames is a maximum of 20 lb/ft^2 (0.96 kN/m^2).

In structures of light-frame construction, multistory steel OMFs are permitted, provided that (a) the structural height, h_n, is a maximum of 35 ft (10.6 m), (b) the dead load of the roof and each floor above the seismic base supported by and tributary to the moment frames are each a maximum of 35 lb/ft^2 (1.68 kN/m^2), and (c) the dead load of the exterior walls tributary to the moment frames is a maximum of 20 lb/ft^2 (0.96 kN/m^2).

EXCEPTION: Industrial structures, such as aircraft maintenance hangars and assembly buildings, with steel OMFs have performed well in past earthquakes with strong ground motions (EQE Inc. 1983, 1985, 1986a, 1986b, 1986c, and 1987); the exception permits single-story steel OMFs to be unlimited in height provided that (a) the structure is limited to the enclosure of equipment or machinery; (b) its occupants are limited to maintaining and monitoring the equipment, machinery, and their associated processes; (c) the sum of the dead load and equipment loads supported by and tributary to the roof is a maximum of 20 lb/ft^2 (0.96 kN/m^2); and (d) the dead load of the exterior wall system, including exterior columns more than 35 ft (10.6 m) above the seismic base is a maximum of 20 lb/ft^2 (0.96 kN/m^2). Though the latter two load limits (Items C and D) are similar to those described in this section, there are meaningful differences.

The exception further recognizes that these facilities often require large equipment or machinery, and associated systems, not supported by or considered tributary to the roof, that support the intended operational functions of the structure, such as top running bridge cranes, jib cranes, and liquid storage containment and distribution systems. To limit the seismic interaction between the seismic force-resisting systems and these components, the exception requires the weight of equipment or machinery that is not self-supporting (i.e., not freestanding) for all loads (e.g., dead, live, or seismic) to be included when determining compliance with the roof or exterior wall load limits. This *equivalent* equipment load shall be in addition to the loads listed above.

To determine the equivalent equipment load, the exception requires the weight to be considered fully (100%) tributary to an area not exceeding 600 ft^2 (55.8 m^2). This limiting area can be taken either to an adjacent exterior wall for cases where the weight is supported by an exterior column (which may also span to the first interior column) or to the adjacent roof for cases where the weight is supported entirely by an interior column or columns, but not both; nor can a fraction of the weight be allocated to each zone. Equipment loads within overlapping tributary areas should be combined in the same limiting area. Other provisions in the standard, as well as in past editions, require satisfying wall load limits tributary to the moment frame, but this requirement is not included in the exception in that it is based on a component-level approach that does not consider the interaction between systems in the structure. As such, the limiting area is considered to be a reasonable approximation of the tributary area of a moment frame segment for the purpose of this conversion. Although this weight allocation procedure may not represent an accurate physical distribution, it is considered to be a reasonable method for verifying compliance with the specified load limits to limit seismic interactions. The engineer must still be attentive to actual mass distributions when computing seismic loads. Further information is discussed in Section C11.1.3.

C12.2.5.6.2 Seismic Design Category F. Single-story steel OMFs are permitted, provided that they meet conditions (a) and (b) described in Section C12.2.5.6.1 for single-story frames and (c) the dead load of the exterior walls tributary to the moment frames is a maximum of 20 lb/ft^2 (0.96 kN/m^2).

C12.2.5.7 Steel Intermediate Moment Frames. Steel intermediate moment frames (IMFs) are more ductile than steel ordinary moment frames (OMFs) but less ductile than steel special moment frames; consequently, restrictions are placed

Table C12.2-1 Summary of Conditions for OMFs and IMFs in Structures Assigned to Seismic Design Category D, E, or F (Refer to the Standard for Additional Requirements)

Section	Frame	SDC	Max. Number Stories	Light-Frame Construction	Max. h_n ft	Max. roof/floor D_L (lb/ft^2)	Exterior Wall DL Max. (lb/ft^2)	Exterior Wall DL Wall[a] Height (ft)
12.2.5.6.1(a)	OMF	D, E	1	NA	65	20	20	35
12.2.5.6.1(a)-Exc	OMF	D, E	1	NA	NL	20	20	35
12.2.5.6.1(b)	OMF	D, E	NL	Required	35	35	20	0
12.2.5.6.2	OMF	F	1	NA	65	20	20	0
12.2.5.7.1(a)	IMF	D	1	NA	65	20	20	35
12.2.5.7.1(a)-Exc	IMF	D	1	NA	NL	20	20	35
12.2.5.7.1(b)	IMF	D	NL	NA	35	NL	NL	NA
12.2.5.7.2(a)	IMF	E	1	NA	65	20	20	35
12.2.5.7.2(a)-Exc	IMF	E	1	NA	NL	20	20	35
12.2.5.7.2(b)	IMF	E	NL	NA	35	35	20	0
12.2.5.7.3(a)	IMF	F	1	NA	65	20	20	0
12.2.5.7.3(b)	IMF	F	NL	Required	35	35	20	0

Note: NL means No Limit; NA means Not Applicable. For metric units, use 20 m for 65 ft and use 10.6 m for 35 ft. For 20 lb/ft^2, use 0.96 kN/m^2 and for 30 lb/ft^2, use 1.68 kN/m^2.
[a]Applies to portion of wall above listed wall height.

on their use in structures assigned to Seismic Design Category D and their use is prohibited in structures assigned to Seismic Design Categories E and F (Table 12.2-1). As with steel OMFs, steel IMFs have also exhibited acceptable behavior in past earthquakes where the structures were sufficiently limited in their structural height, number of stories, and seismic mass. The provisions in the standard reflect these observations. The exceptions are discussed separately (following). Table C12.2-1 summarizes the provisions.

C12.2.5.7.1 Seismic Design Category D. Single-story steel IMFs are permitted without limitations on dead load of the roof and exterior walls, provided that the structural height, h_n, is a maximum of 35 ft (10.6 m). An increase to 65 ft (20 m) is permitted for h_n, provided that (a) the dead load supported by and tributary to the roof is a maximum of 20 lb/ft^2 (0.96 kN/m^2), and (b) the dead load of the exterior walls more than 35 ft (10.6 m) above the seismic base tributary to the moment frames is a maximum of 20 lb/ft^2 (0.96 kN/m^2).

The exception permits single-story steel IMFs to be unlimited in height, provided that they meet all of the conditions described in the exception to Section C12.2.5.6.1 for the same structures.

C12.2.5.7.2 Seismic Design Category E. Single-story steel IMFs are permitted, provided that they meet all of the conditions described in Section C12.2.5.6.1 for single-story OMFs.

The exception permits single-story steel IMFs to be unlimited in height, provided that they meet all of the conditions described in Section C12.2.5.6.1 for the same structures.

Multistory steel IMFs are permitted, provided that they meet all of the conditions described in Section C12.2.5.6.1 for multistory OMFs, except that the structure is not required to be of light-frame construction.

C12.2.5.7.3 Seismic Design Category F. Single-story steel IMFs are permitted, provided that (a) the structural height, h_n, is a maximum of 65 ft (20 m), (b) the dead load supported by and tributary to the roof is a maximum of 20 lb/ft^2 (0.96 kN/m^2), and (c) the dead load of the exterior walls tributary to the moment frames is a maximum of 20 lb/ft^2 (0.96 kN/m^2).

Multistory steel IMFs are permitted, provided that they meet all of the conditions described in the exception to Section C12.2.5.6.1 for multistory OMFs in structures of light-frame construction.

C12.2.5.8 Shear Wall–Frame Interactive Systems. For structures assigned to Seismic Design Category A or B (where seismic hazard is low), it is usual practice to design shear walls and frames of a shear wall–frame structure to resist lateral forces in proportion to their relative rigidities, considering interaction between the two subsystems at all levels. As discussed in Section C12.2.1, this typical approach would require use of a lower response modification coefficient, R, than that defined for shear wall–frame interactive systems. Where the special requirements of this section are satisfied, more reliable performance is expected, justifying a higher value of R.

C12.3 DIAPHRAGM FLEXIBILITY, CONFIGURATION IRREGULARITIES, AND REDUNDANCY

C12.3.1 Diaphragm Flexibility. Most seismic force-resisting systems have two distinct parts: the horizontal system, which distributes lateral forces to the vertical elements and the vertical system, which transmits lateral forces between the floor levels and the base of the structure.

The horizontal system may consist of diaphragms or a horizontal bracing system. For the majority of buildings, diaphragms offer the most economical and positive method of resisting and distributing seismic forces in the horizontal plane. Typically, diaphragms consist of a metal deck (with or without concrete), concrete slabs, and wood sheathing and/or decking. Although most diaphragms are flat, consisting of the floors of buildings, they also may be inclined, curved, warped, or folded configurations, and most diaphragms have openings.

The diaphragm stiffness relative to the stiffness of the supporting vertical seismic force-resisting system ranges from flexible to rigid and is important to define. Provisions defining diaphragm flexibility are given in Sections 12.3.1.1 through 12.3.1.3. If a diaphragm cannot be idealized as either flexible or rigid, explicit consideration of its stiffness must be included in the analysis.

The diaphragms in most buildings braced by wood light-frame shear walls are semirigid. Because semirigid diaphragm modeling is beyond the capability of available software for wood light-frame buildings, it is anticipated that this requirement will be met by evaluating force distribution using both rigid and flexible diaphragm models and taking the worse case of the two. Although this procedure is in conflict with common design practice, which typically includes only flexible diaphragm force distribution for wood light-frame buildings, it is one method of capturing the effect of the diaphragm stiffness.

C12.3.1.1 Flexible Diaphragm Condition. Section 12.3.1.1 defines broad categories of diaphragms that may be idealized as flexible, regardless of whether the diaphragm meets the calculated conditions of Section 12.3.1.3. These categories include the following:

a. Construction with relatively stiff vertical framing elements, such as steel-braced frames and concrete or masonry shear walls;
b. One- and two-family dwellings; and
c. Light-frame construction (e.g., construction consisting of light-frame walls and diaphragms) with or without nonstructural toppings of limited stiffness.

For item c above, compliance with story drift limits along each line of shear walls is intended as an indicator that the shear walls are substantial enough to share load on a tributary area basis and not require torsional force redistribution.

C12.3.1.2 Rigid Diaphragm Condition. Span-to-depth ratio limits are included in the deemed-to-comply condition as an indirect measure of the flexural contribution to diaphragm stiffness.

C12.3.1.3 Calculated Flexible Diaphragm Condition. A diaphragm is permitted to be idealized as flexible if the calculated diaphragm deflection (typically at midspan) between supports (lines of vertical elements) is greater than two times the average story drift of the vertical lateral force-resisting elements located at the supports of the diaphragm span.

Fig. 12.3-1 depicts a distributed load, conveying the intent that the tributary lateral load be used to compute δ_{MDD}, consistent with the Section 11.3 symbols. A diaphragm opening is illustrated, and the shorter arrows in the portion of the diaphragm with the opening indicate lower load intensity because of lower tributary seismic mass.

C12.3.2 Irregular and Regular Classification. The configuration of a structure can significantly affect its performance during a strong earthquake that produces the ground motion contemplated in the standard. Structural configuration can be divided into two aspects: horizontal and vertical. Most seismic design provisions were derived for buildings that have regular configurations, but earthquakes have shown repeatedly that buildings that have irregular configurations suffer greater damage. This situation prevails even with good design and construction.

There are several reasons for the poor behavior of irregular structures. In a regular structure, the inelastic response, including energy dissipation and damage, produced by strong ground shaking tends to be well distributed throughout the structure. However, in irregular structures, inelastic behavior can be concentrated by irregularities and can result in rapid failure of structural elements in these areas. In addition, some irregularities introduce unanticipated demands into the structure, which designers frequently overlook when detailing the structural system. Finally, the elastic analysis methods typically used in the design of structures often cannot predict the distribution of earthquake demands in an irregular structure very well, leading to inadequate design in the areas associated with the irregularity. For these reasons, the standard encourages regular structural configurations and prohibits gross irregularity in buildings located on sites close to major active faults, where very strong ground motion and extreme inelastic demands are anticipated. The termination of seismic force-resisting elements at the foundation, however, is not considered to be a discontinuity.

C12.3.2.1 Horizontal Irregularity. A building may have a symmetric geometric shape without reentrant corners or wings but still be classified as irregular in plan because of its distribution of mass or vertical seismic force-resisting elements. Torsional effects in earthquakes can occur even where the centers of mass and rigidity coincide. For example, ground motion waves acting on a skew with respect to the building axis can cause torsion. Cracking or yielding in an asymmetric fashion also can cause torsion. These effects also can magnify the torsion caused by eccentricity between the centers of mass and rigidity. Torsional structural irregularities (Types 1a and 1b) are defined to address this concern.

A square or rectangular building with minor reentrant corners would still be considered regular, but large reentrant corners creating a crucifix form would produce an irregular structural configuration (Type 2). The response of the wings of this type of building generally differs from the response of the building as a whole, and this difference produces higher local forces than would be determined by application of the standard without modification. Other winged plan configurations (e.g., H-shapes) are classified as irregular even if they are symmetric because of the response of the wings.

Significant differences in stiffness between portions of a diaphragm at a level are classified as Type 3 structural irregularities because they may cause a change in the distribution of seismic forces to the vertical components and may create torsional forces not accounted for in the distribution normally considered for a regular building.

Where there are discontinuities in the path of lateral force resistance, the structure cannot be considered regular. The most critical discontinuity defined is the out-of-plane offset of vertical elements of the seismic force-resisting system (Type 4). Such offsets impose vertical and lateral load effects on horizontal elements that are difficult to provide for adequately.

Where vertical lateral force-resisting elements are not parallel to the major orthogonal axes of the seismic force-resisting system, the equivalent lateral force procedure of the standard cannot be applied appropriately, so the structure is considered to have an irregular structural configuration (Type 5).

Fig. C12.3-1 illustrates horizontal structural irregularities.

C12.3.2.2 Vertical Irregularity. Vertical irregularities in structural configuration affect the responses at various levels and induce loads at these levels that differ significantly from the distribution assumed in the equivalent lateral force procedure given in Section 12.8.

A moment-resisting frame building might be classified as having a soft story irregularity (Type 1) if one story is much taller than the adjoining stories and the design did not compensate for the resulting decrease in stiffness that normally would occur.

A building is classified as having a weight (mass) irregularity (Type 2) where the ratio of mass to stiffness in adjacent stories differs significantly. This difference typically occurs where a heavy mass (e.g., an interstitial mechanical floor) is placed at one level.

A vertical geometric irregularity (Type 3) applies regardless of whether the larger dimension is above or below the smaller one.

Vertical lateral force-resisting elements at adjoining stories that are offset from each other in the vertical plane of the elements and impose overturning demands on supporting structural elements, such as beams, columns, trusses, walls, or slabs, are classified as in-plane discontinuity irregularities (Type 4).

Buildings with a weak-story irregularity (Type 5) tend to develop all of their inelastic behavior and consequent damage at the weak story, possibly leading to collapse.

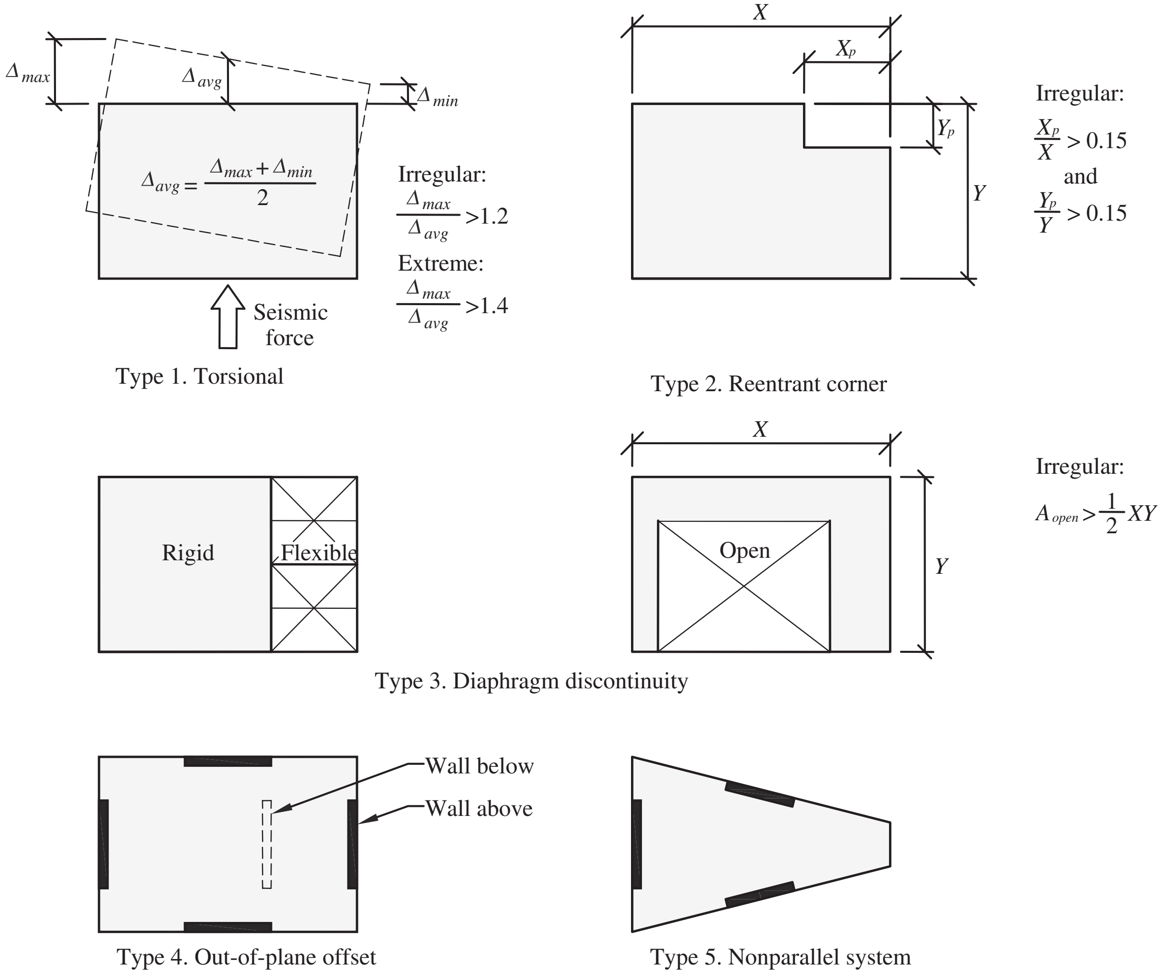


FIGURE C12.3-1 Horizontal Structural Irregularity Examples

Fig. C12.3-2 illustrates examples of vertical structural irregularities.

C12.3.3 Limitations and Additional Requirements for Systems with Structural Irregularities

C12.3.3.1 Prohibited Horizontal and Vertical Irregularities for Seismic Design Categories D through F. The prohibitions and limits caused by structural irregularities in this section stem from poor performance in past earthquakes and the potential to concentrate large inelastic demands in certain portions of the structure. Even where such irregularities are permitted, they should be avoided whenever possible in all structures.

C12.3.3.2 Extreme Weak Stories. Because extreme weak story irregularities are prohibited in Section 12.3.3.1 for buildings located in Seismic Design Categories D, E, and F, the limitations and exceptions in this section apply only to buildings assigned to Seismic Design Category B or C. Weak stories of structures assigned to Seismic Design Category B or C that are designed for seismic load effects including overstrength are exempted because reliable inelastic response is expected.

C12.3.3.3 Elements Supporting Discontinuous Walls or Frames. The purpose of requiring elements (e.g., beams, columns, trusses, slabs, and walls) that support discontinuous walls or frames to be designed to resist seismic load effects including overstrength is to protect the gravity load-carrying system against possible overloads caused by overstrength of the seismic force-resisting system. Either columns or beams may be subject to such failure; therefore, both should include this design requirement. Beams may be subject to failure caused by overloads in either the downward or upward directions of force. Examples include reinforced concrete beams, the weaker top laminations of glued laminated beams, or unbraced flanges of steel beams or trusses. Hence, the provision has not been limited simply to downward force, but instead to the larger context of "vertical load." Additionally, walls that support isolated point loads from frame columns or discontinuous perpendicular walls or walls with significant vertical offsets, as shown in Figs. C12.3-3 and C12.3-4, can be subject to the same type of failure caused by overload.

The connection between the discontinuous element and the supporting member must be adequate to transmit the forces required for the design of the discontinuous element. For example, where the discontinuous element is required to comply with the seismic load effects, including overstrength in Section 12.4.3, as is the case for a steel column in a braced frame or a moment frame, its connection to the supporting member is required to be designed to transmit the same forces. These same seismic load effects are not required for shear walls, and thus, the connection between the shear wall and the supporting member would only need to be designed to transmit the loads associated with the shear wall.

For wood light-frame shear wall construction, the final sentence of Section 12.3.3.3 results in the shear and overturning connections at the base of a discontinued shear wall (i.e., shear

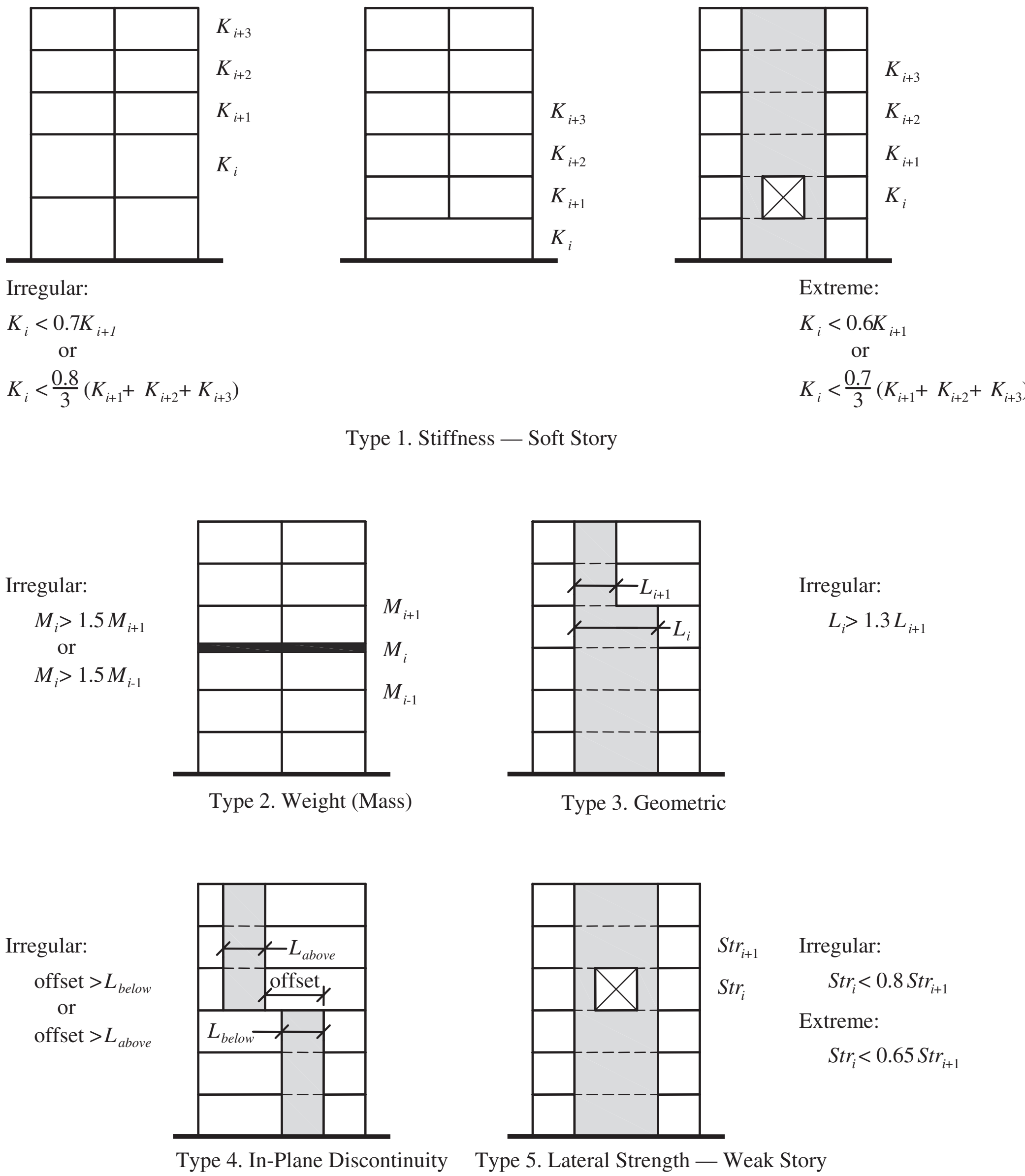


FIGURE C12.3-2 Vertical Structural Irregularities

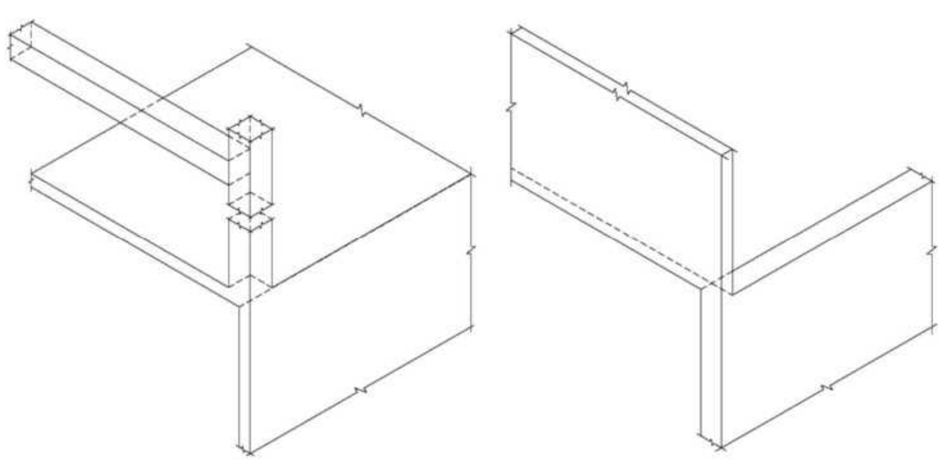


FIGURE C12.3-3 Vertical In-Plane-Discontinuity Irregularity from Columns or Perpendicular Walls (Type 4)

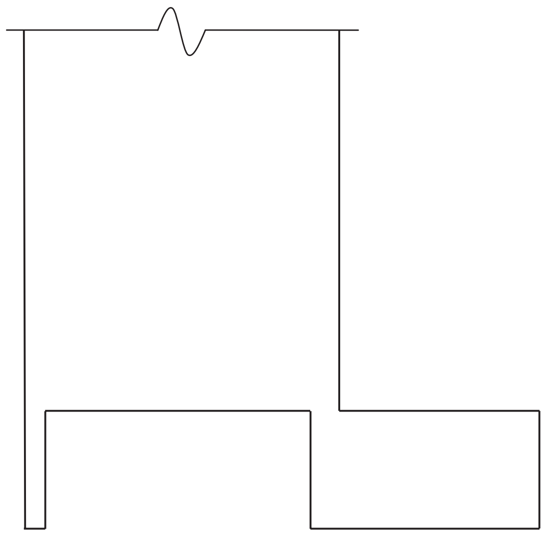


FIGURE C12.3-4 Vertical In-Plane-Discontinuity Irregularity from Walls with Significant Offsets (Type 4)

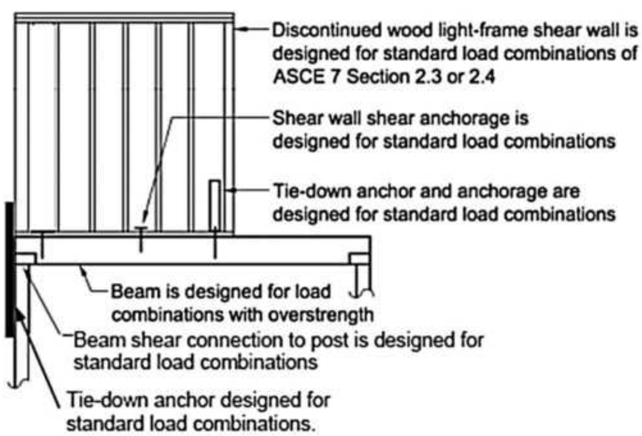


FIGURE C12.3-5 Discontinued Wood Light-Frame Shear Wall

fasteners and tie-downs) being designed using the load combinations of Section 2.3 or 2.4 rather than the load combinations with overstrength of Section 12.4.3 (Fig. C12.3-5). The intent of the first sentence of Section 12.3.3.3 is to protect the system providing resistance to forces transferred from the shear wall by designing the system for seismic load effects including overstrength; strengthening of the shear wall anchorage to this system is not required to meet this intent.

C12.3.3.4 Increase in Forces Caused by Irregularities for Seismic Design Categories D through F. The listed irregularities may result in loads that are distributed differently than those assumed in the equivalent lateral force procedure of Section 12.8, especially as related to the interconnection of the diaphragm with vertical elements of the seismic force-resisting system. The 25% increase in force is intended to account for this difference. Where the force is calculated using the seismic load effects including overstrength, no further increase is warranted.

C12.3.4 Redundancy. The standard introduces a revised redundancy factor, ρ, for structures assigned to Seismic Design Category D, E, or F to quantify redundancy. The value of this factor is either 1.0 or 1.3. This factor has the effect of reducing the response modification coefficient, R, for less redundant structures, thereby increasing the seismic demand. The factor is specified in recognition of the need to address the issue of redundancy in the design.

The desirability of redundancy, or multiple lateral force-resisting load paths, has long been recognized. The redundancy provisions of this section reflect the belief that an excessive loss of story shear strength or development of an extreme torsional irregularity (Type 1b) may lead to structural failure. The value of ρ determined for each direction may differ.

C12.3.4.1 Conditions Where Value of ρ is 1.0. This section provides a convenient list of conditions where ρ is 1.0.

C12.3.4.2 Redundancy Factor, ρ, for Seismic Design Categories D through F. There are two approaches to establishing a redundancy factor, ρ, of 1.0. Where neither condition is satisfied, ρ is taken as equal to 1.3. It is permitted to take ρ equal to 1.3 without checking either condition. A reduction in the value of ρ from 1.3 is not permitted for structures assigned to Seismic Design Category D that have an extreme torsional irregularity (Type 1b). Seismic Design Categories E and F are not also specified because extreme torsional irregularities are prohibited (see Section 12.3.3.1).

The first approach is a check of the elements outlined in Table 12.3-3 for cases where the seismic design story shear exceeds 35% of the base shear. Parametric studies (conducted by Building Seismic Safety Council Technical Subcommittee 2 but unpublished) were used to select the 35% value. Those studies indicated that stories with story shears of at least 35% of the base shear include all stories of low-rise buildings (buildings up to five to six stories) and about 87% of the stories of tall buildings. The intent of this limit is to exclude penthouses of most buildings and the uppermost stories of tall buildings from the redundancy requirements.

This approach requires the removal (or loss of moment resistance) of an individual lateral force-resisting element to determine its effect on the remaining structure. If the removal of elements, one by one, does not result in more than a 33% reduction in story strength or an extreme torsional irregularity, ρ may be taken as 1.0. For this evaluation, the determination of story strength requires an in-depth calculation. The intent of the check is to use a simple measure (elastic or plastic) to determine whether an individual member has a significant effect on the overall system. If the original structure has an extreme torsional irregularity to begin with, the resulting ρ is 1.3. Fig. C12.3-6 presents a flowchart for implementing the redundancy requirements.

As indicated in Table 12.3-3, braced frame, moment frame, shear wall, and cantilever column systems must conform to redundancy requirements. Dual systems also are included but, in most cases, are inherently redundant. Shear walls or wall piers with a height-to-length aspect ratio greater than 1.0 within any story have been included; however, the required design of collector elements and their connections for Ω_0 times the design force may address the key issues. To satisfy the collector force requirements, a reasonable number of shear walls usually is required. Regardless, shear wall systems are addressed in this section so that either an adequate number of wall elements is included or the proper redundancy factor is applied. For wall piers, the height is taken as the height of the adjacent opening and generally is less than the story height.

The second approach is a deemed-to-comply condition wherein the structure is regular and has a specified arrangement of seismic force-resisting elements to qualify for a ρ of 1.0. As part of the parametric study, simplified braced frame and moment frame systems were investigated to determine their sensitivity to the analytical redundancy criteria. This simple deemed-to-comply condition is consistent with the results of the study.

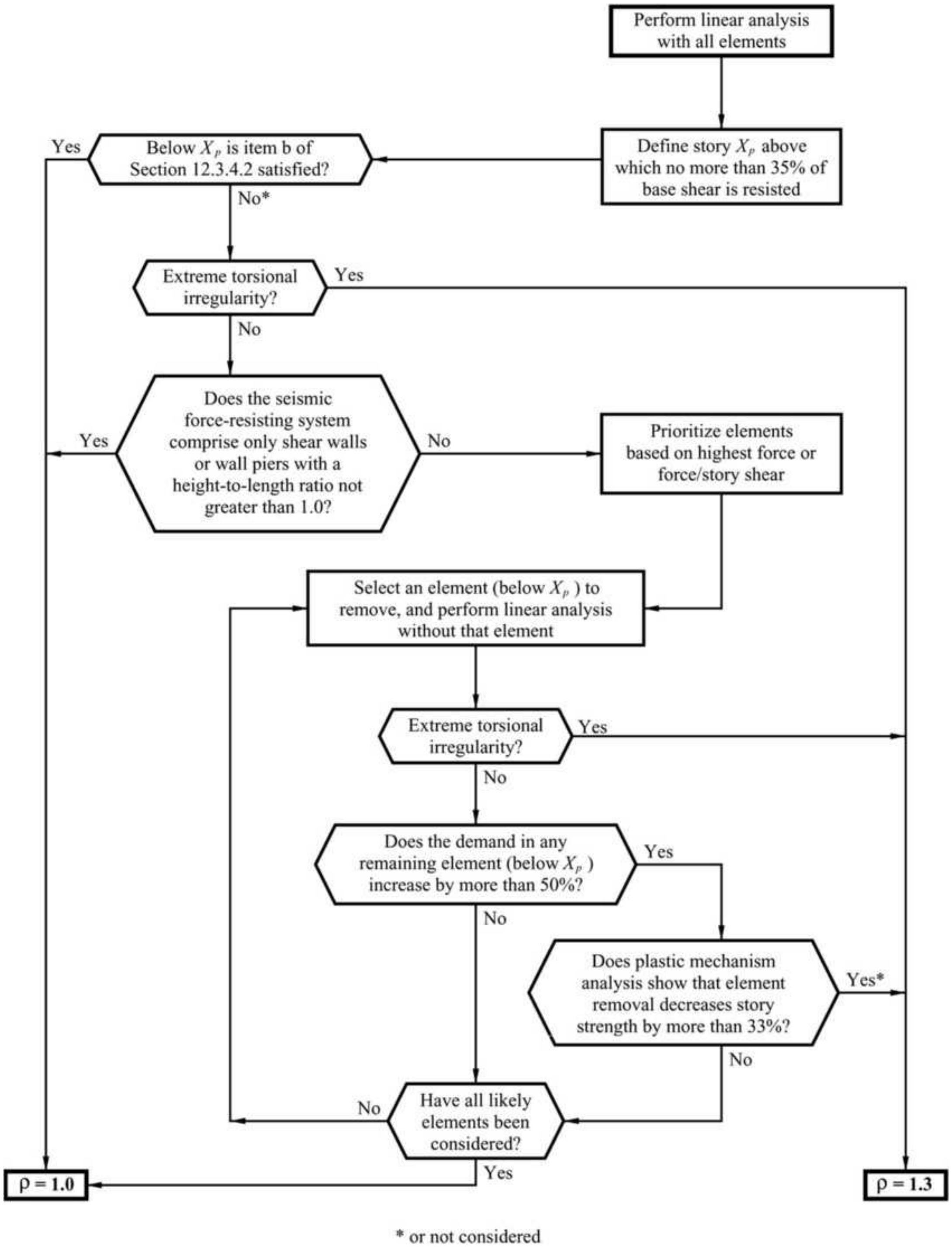


FIGURE C12.3-6 Calculation of the Redundancy Factor, ρ

C12.4 SEISMIC LOAD EFFECTS AND COMBINATIONS

C12.4.1 Applicability. Structural elements designated by the engineer as part of the seismic force-resisting system typically are designed directly for seismic load effects. None of the seismic forces associated with the design base shear are formally assigned to structural elements that are not designated as part of the seismic force-resisting system, but such elements must be designed using the load conditions of Section 12.4 and must accommodate the deformations resulting from application of seismic loads.

C12.4.2 Seismic Load Effect. The seismic load effect includes horizontal and vertical components. The horizontal seismic load effects, E_h, are caused by the response of the structure to horizontal seismic ground motions, whereas the vertical seismic load effects are caused by the response of the structure to vertical seismic ground motions. The basic load combinations in Chapter 2 were duplicated and reformulated in Section 12.4 to clarify the intent of the provisions for the vertical seismic load effect term, E_v.

The concept of using an equivalent static load coefficient applied to the dead load to represent vertical seismic load effects was first introduced in ATC 3-06 (1978), where it was defined as simply $\pm 0.2D$. The load combinations where the vertical seismic load coefficient was to be applied assumed strength design load combinations. Neither ATC 3-06 (1978) nor the early versions of the NEHRP provisions (FEMA 2009a) clearly explained how the values of 0.2 were determined, but it is reasonable to assume that it was based on the judgment of the writers of those documents. It is accepted by the writers of this standard that vertical ground motions do occur and that the value of $\pm 0.2S_{DS}$ was determined based on consensus judgment. Many issues enter into the development of the vertical coefficient, including phasing of vertical ground motion and appropriate R factors, which make determination of a more precise value difficult. Although no specific rationale or logic is provided in editions of the NEHRP provisions (FEMA 2009a) on how the value of $0.2S_{DS}$ was determined, one possible way to rationalize the selection of the $0.2S_{DS}$ value is to recognize that it is equivalent to $(2/3)(0.3)S_{DS}$, where the 2/3 factor represents the often-assumed ratio between the vertical and horizontal components of motion, and the 0.3 factor represents the 30% in the 100% to 30% orthogonal load combination rule used for horizontal motions.

For situations where the vertical component of ground motion is explicitly included in design analysis, the vertical ground motion spectra definition that is provided in Section 11.9 should be used. Following the rationale described above, the alternate vertical ground motion component determined in Section 11.9, S_{av}, is combined with the horizontal component of ground motion by using the 100%–30% orthogonal load combination rule used for horizontal motions resulting in the vertical seismic load effect determined with Eq. (12.4-4b), $E_v = 0.3S_{av}D$.

C12.4.2.1 Horizontal Seismic Load Effect. Horizontal seismic load effects, E_h, are determined in accordance with Eq. (12.4-3) as $E_h = \rho Q_E$. Q_E is the seismic load effect of horizontal seismic forces from V or F_p. The purpose of E_h is to approximate the horizontal seismic load effect from the design basis earthquake to be used in load combinations including E for the design of lateral force-resisting elements including diaphragms, vertical elements of seismic force-resisting systems as defined in Table 12.2-1, the design and anchorage of elements such as structural walls, and the design of nonstructural components.

C12.4.2.2 Vertical Seismic Load Effect. The vertical seismic load effect, E_v, is determined with Eq. (12.4-4a) as $E_v = 0.2S_{DS}D$ or with Eq. (12.4-4b) as $E_v = 0.3S_{av}D$. E_v is permitted to be taken as zero in Eqs. (12.4-1), (12.4-2), (12.4-5), and (12.4-6) for structures assigned to Seismic Design Category B and in Eq. (12.4-2) for determining demands on the soil–structure interface of foundations. E_v increases the load on beams and columns supporting seismic elements and increases the axial load in the P–M interaction of walls resisting seismic load effects.

C12.4.3 Seismic Load Effects Including Overstrength. Some elements of properly detailed structures are not capable of safely resisting ground-shaking demands through inelastic behavior. To ensure safety, these elements must be designed with sufficient strength to remain elastic.

The horizontal load effect including overstrength may be calculated in either of two ways. The load effect may be approximated by use of an overstrength factor, Ω_0, which approximates the inherent overstrength in typical structures based on the structure's seismic force-resisting systems. This approach is addressed in Section 12.4.3.1. Alternatively, the expected system strength may be directly calculated based on actual member sizes and expected material properties, as addressed in Section 12.4.3.2.

C12.4.3.1 Horizontal Seismic Load Effect Including Overstrength. Horizontal seismic load effects including overstrength, E_{mh}, are determined in accordance with Eq. (12.4-7) as $E_{mh} = \Omega_0 Q_E$. Q_E is the effect of horizontal seismic forces from V, F_{px}, or F_p. The purpose for E_{mh} is to approximate the maximum seismic load for the design of critical elements, including discontinuous systems, transfer beams and columns supporting discontinuous systems, and collectors. Forces calculated using this approximate method need not be used if a more rigorous evaluation as permitted in Section 12.4.3.2 is used.

C12.4.3.2 Capacity-Limited Horizontal Seismic Load Effect. The standard permits the horizontal seismic load effect including overstrength to be calculated directly using actual member sizes and expected material properties where it can be determined that yielding of other elements in the structure limits the force that can be delivered to the element in question. When calculated this way, the horizontal seismic load effect including overstrength is termed the capacity-limited seismic load effect, E_{cl}.

As an example, the axial force in a column of a moment-resisting frame results from the shear forces in the beams that connect to this column. The axial forces caused by seismic loads need never be taken as greater than the sum of the shear forces in these beams at the development of a full structural mechanism, considering the probable strength of the materials and strain-hardening effects. For frames controlled by beam hinge-type mechanisms, these shear forces would typically be calculated as $2M_{pr}/L_h$, where M_{pr} is the probable flexural strength of the beam considering expected material properties and strain hardening, and L_h is the distance between plastic hinge locations. Both ACI 318 and AISC 341 require that beams in special moment frames be designed for shear calculated in this manner, and both standards include many other requirements that represent the capacity-limited seismic load effect instead of the use of a factor approximating overstrength. This design approach is sometimes termed "capacity design." In this design method, the capacity (expected strength) of one or more elements is used to generate the demand (required strength) for other elements, because the yielding of the former limits the forces delivered to the latter. In this context, the capacity of the yielding element is its expected or mean anticipated strength, considering potential variation in material yield strength and strain-hardening effects. When calculating the capacity of elements for this purpose, expected member strengths should not be reduced by strength reduction or resistance factors, ϕ.

The capacity-limited design is not restricted to yielding limit states (axial, flexural, or shear); other examples include flexural buckling (axial compression) used in steel special concentrically braced frames, or lateral-torsional buckling in steel ordinary moment frame beams, as confirmed by testing.

C12.4.4 Minimum Upward Force for Horizontal Cantilevers for Seismic Design Categories D through F. In Seismic Design Categories D, E, and F, horizontal cantilevers are designed for an upward force that results from an effective vertical acceleration of 1.2 times gravity. This design requirement is meant to provide some minimum strength in the upward direction and to account for possible dynamic amplification of vertical ground motions resulting from the vertical flexibility of the cantilever. The requirement is not applied to downward forces on cantilevers, for which the typical load combinations are used.

C12.5 DIRECTION OF LOADING

Seismic forces are delivered to a building through ground accelerations that may approach from any direction relative to the orthogonal directions of the building; therefore, seismic effects are expected to develop in both directions simultaneously. The standard requires structures to be designed for the most critical loading effects from seismic forces applied in any direction. The procedures outlined in this section are deemed to satisfy this requirement.

For horizontal structural elements such as beams and slabs, orthogonal effects may be minimal; however, design of vertical elements of the seismic force-resisting system that participate in both orthogonal directions is likely to be governed by these effects.

C12.5.1 Direction of Loading Criteria. For structures with orthogonal seismic force-resisting systems, the most critical load effects can typically be computed using a pair of orthogonal directions that coincide with the principal axes of the structure. Structures with nonparallel or nonorthogonal systems may require a set of orthogonal direction pairs to determine the most critical load effects. If a three-dimensional mathematical model is used, the analyst must be attentive to the orientation of the global axes in the model in relation to the principal axes of the structure.

C12.5.2 Seismic Design Category B. Recognizing that design of structures assigned to Seismic Design Category (SDC) B is often controlled by nonseismic load effects and, therefore, is not sensitive to orthogonal loadings regardless of any horizontal structural irregularities, it is permitted to determine the most critical load effects by considering that the maximum response can occur in any single direction; simultaneous application of response in the orthogonal direction is not required. Typically, the two directions used for analysis coincide with the principal axes of the structure.

C12.5.3 Seismic Design Category C. Design of structures assigned to SDC C often parallels the design of structures assigned to SDC B and, therefore, as a minimum conforms to Section 12.5.2. Although it is not likely that design of the seismic force-resisting systems in regular structures assigned to SDC C would be sensitive to orthogonal loadings, special consideration must be given to structures with nonparallel or nonorthogonal systems (Type 5 horizontal structural irregularity) to avoid overstressing by different directional loadings. In this case, the standard provides two methods to approximate simultaneous orthogonal loadings and requires a three-dimensional mathematical model of the structure for analysis in accordance with Section 12.7.3.

The orthogonal combination procedure in item (a) of Section 12.5.3.1 combines the effects from 100% of the seismic load applied in one direction with 30% of the seismic load applied in the perpendicular direction. This general approximation—the "30% rule"—was introduced by Rosenblueth and Contreras (1977) based on earlier work by A. S. Veletsos and also N. M. Newmark (cited in Rosenblueth and Contreras 1977) as an alternative to performing the more rational, yet computationally demanding, response history analysis, and is applicable to any elastic structure. Combining effects for seismic loads in each direction, and accidental torsion in accordance with Sections 12.8.4.2 and 12.8.4.3, results in the following 16 load combinations:

- $Q_E = \pm Q_{E.X+AT} \pm 0.3 Q_{E.Y}$ where $Q_{E.Y}$ = effect of Y-direction load at the center of mass (Section 12.8.4.2);
- $Q_E = \pm Q_{E.X-AT} \pm 0.3 Q_{E.Y}$ where $Q_{E.X}$ = effect of X-direction load at the center of mass (Section 12.8.4.2);
- $Q_E = \pm Q_{E.Y+AT} \pm 0.3 Q_{E.X}$ where AT = accidental torsion computed in accordance with Sections 12.8.4.2 and 12.8.4.3; and
- $Q_E = \pm Q_{E.Y-AT} \pm 0.3 Q_{E.X}$.

Though the standard permits combining effects from forces applied independently in any pair of orthogonal directions (to approximate the effects of concurrent loading), accidental torsion need not be considered in the direction that produces the lesser effect, per Section 12.8.4.2. This provision is sometimes disregarded when using a mathematical model for three-dimensional analysis that can automatically include accidental torsion, which then results in 32 load combinations.

The maximum effect of seismic forces, Q_E, from orthogonal load combinations is modified by the redundancy factor, ρ, or the overstrength factor, Ω_0, where required, and the effects of vertical seismic forces, E_v, are considered in accordance with Section 12.4 to obtain the seismic load effect, E.

These orthogonal combinations should not be confused with uniaxial modal combination rules, such as the square root of the sum of the squares (SRSS) or the complete quadratic combination (CQC) method. In past standards, an acceptable alternative to the above was to use the SRSS method to combine effects of the two orthogonal directions, where each term computed is assigned the sign that resulted in the most conservative result. This method is no longer in common use. Although both approaches described for considering orthogonal effects are approximations, it is important to note that they were developed with consideration of results for a square building.

Orthogonal effects can alternatively be considered by performing three-dimensional response history analyses (see Chapter 16) with application of orthogonal ground motion pairs applied simultaneously in any two orthogonal directions. If the structure is located within 3 mi (5 km) of an active fault, the ground motion pair should be rotated to the fault-normal and fault-parallel directions of the causative fault.

C12.5.4 Seismic Design Categories D through F. The direction of loading for structures assigned to SDCs D, E, or F conforms to Section 12.5.3 for structures assigned to SDC C. If a Type 5 horizontal structural irregularity exists, then orthogonal effects are similarly included in design. Recognizing the higher seismic risk associated with structures assigned to SDCs D, E, or F, the standard provides additional requirements for vertical members coupled between intersecting seismic force-resisting systems.

C12.6 ANALYSIS PROCEDURE SELECTION

Table 12.6-1 provides the permitted analysis procedures for all Seismic Design Categories. The table is applicable only to buildings without seismic isolation (Chapter 17) or passive energy devices (Chapter 18) for which there are additional requirements in Sections 17.4 and 18.2.4, respectively.

The four basic procedures provided in Table 12.6-1 are the equivalent lateral force (ELF, Section 12.8), the modal response spectrum (MRS analysis, Section 12.9.1), the linear response history (LRH, Section 12.9.2), and the nonlinear response history (NRH, Section 16.1) analysis procedures. Nonlinear static pushover analysis is not provided as an "approved" analysis procedure in the standard.

The ELF procedure is allowed for all buildings assigned to Seismic Design Category B or C and for all buildings assigned

Table C12.6-1 Values of 3.5T_s for Various Cities and Site Classes

Location	S_s (g)	S_1 (g)	3.5T_s (s) for Site Class A & B	C	D	E
Denver	0.219	0.057	0.91	1.29	1.37	1.07
Boston	0.275	0.067	0.85	1.21	1.30	1.03
New York City	0.359	0.070	0.68	0.97	1.08	0.93
Las Vegas	0.582	0.179	1.08	1.50	1.68	1.89
St. Louis	0.590	0.169	1.00	1.40	1.60	1.81
San Diego	1.128	0.479	1.31	1.73	1.99	2.91
Memphis	1.341	0.368	0.96	1.38	1.59	2.25
Charleston	1.414	0.348	0.86	1.25	1.47	2.08
Seattle	1.448	0.489	1.18	1.55	1.78	2.63
San Jose	1.500	0.600	1.40	1.82	2.10	2.12
Salt Lake City	1.672	0.665	1.39	1.81	2.09	3.10

to Seismic Design Category D, E, or F, except for the following:

- Structures with structural height, $h_n > 160$ ft ($h_n > 48.8$ m) and $T > 3.5T_s$;
- Structures with structural height, $h_n > 160$ ft ($h_n > 48.8$ m) and $T \leq 3.5T_s$ but with one or more of the structural irregularities in Table 12.3-1 or 12.3-2; and
- Structures with structural height, $h_n < 160$ ft ($h_n < 48.8$ m) and with one or more of the following structural irregularities: torsion or extreme torsion (Table 12.3-1); or soft story, extreme soft story, weight (mass), or vertical geometric (Table 12.3-2).

$T_s = S_{D1}/S_{DS}$ is the period at which the horizontal and descending parts of the design response spectrum intersect (Fig. 11.4-1). The value of T_s depends on the site class because S_{DS} and S_{D1} include such effects. Where the ELF procedure is not allowed, the analysis must be performed using modal response spectrum or response history analysis.

The use of the ELF procedure is limited to buildings with the listed structural irregularities because the procedure is based on an assumption of a gradually varying distribution of mass and stiffness along the height and negligible torsional response. The basis for the $3.5T_s$ limitation is that the higher modes become more dominant in taller buildings (Lopez and Cruz 1996, Chopra 2007a,b), and as a result, the ELF procedure may underestimate the seismic base shear and may not correctly predict the vertical distribution of seismic forces in taller buildings.

As Table C12.6-1 demonstrates, the value of $3.5T_s$ generally increases as ground motion intensity increases and as soils become softer. Assuming a fundamental period of approximately 0.1 times the number of stories, the maximum structural height, h_n, for which the ELF procedure applies ranges from about 10 stories for low seismic hazard sites with firm soil to 30 stories for high seismic hazard sites with soft soil. Because this trend was not intended, the 160-ft (48.8-m) height limit is introduced.

C12.7 MODELING CRITERIA

C12.7.1 Foundation Modeling. Structural systems consist of three interacting subsystems: the structural framing (girders, columns, walls, and diaphragms), the foundation (footings, piles, and caissons), and the supporting soil. The ground motion that a structure experiences, as well as the response to that ground motion, depends on the complex interaction among these subsystems.

Those aspects of ground motion that are affected by site characteristics are assumed to be independent of the structure–foundation system because these effects would occur in the free field in the absence of the structure. Hence, site effects are considered separately (Sections 11.4.3 through 11.4.5 and Chapters 20 and 21).

Given a site-specific ground motion or response spectrum, the dynamic response of the structure depends on the foundation system and on the characteristics of the soil that support the system. The dependence of the response on the structure–foundation–soil system is referred to as soil–structure interaction (SSI). Such interactions usually, but not always, result in a reduction of seismic base shear. This reduction is caused by the flexibility of the foundation–soil system and an associated lengthening of the fundamental period of vibration of the structure. In addition, the soil system may provide an additional source of damping. However, that total displacement typically increases with soil–structure interaction.

If the foundation is considered to be rigid, the computed base shears are usually conservative, and it is for this reason that rigid foundation analysis is permitted. The designer may neglect soil–structure interaction or may consider it explicitly in accordance with Section 12.13.3 or implicitly in accordance with Chapter 19.

As an example, consider a moment-frame building without a basement and with moment-frame columns supported on footings designed to support shear and axial loads (i.e., pinned column bases). If foundation flexibility is not considered, the columns should be restrained horizontally and vertically, but not rotationally. Consider a moment-frame building with a basement. For this building, horizontal restraint may be provided at the level closest to grade, as long as the diaphragm is designed to transfer the shear out of the moment frame. Because the columns extend through the basement, they may also be restrained rotationally and vertically at this level. However, it is often preferable to extend the model through the basement and provide the vertical and rotational restraints at the foundation elements, which is more consistent with the actual building geometry.

C12.7.2 Effective Seismic Weight. During an earthquake, the structure accelerates laterally, and these accelerations of the structural mass produce inertial forces. These inertial forces, accumulated over the height of the structure, produce the seismic base shear.

When a building vibrates during an earthquake, only that portion of the mass or weight that is physically tied to the structure needs to be considered as effective. Hence, live loads (e.g., loose furniture, loose equipment, and human occupants) need not be included. However, certain types of live loads, such as storage loads, may develop inertial forces, particularly where they are densely packed.

Also considered as contributing to effective seismic weight are the following:

1. All permanent equipment (e.g., air conditioners, elevator equipment, and mechanical systems);
2. Partitions to be erected or rearranged as specified in Section 4.3.2 (greater of actual partition weight and 10 lb/ft^2 (0.5 kN/m^2) of floor area);
3. 20% of significant snow load, $p_f > 30$ lb/ft^2 ($p_f > 1.4$ kN/m^2) and
4. The weight of landscaping and similar materials.

The full snow load need not be considered because maximum snow load and maximum earthquake load are unlikely to occur simultaneously and loose snow does not move with the roof.

C12.7.3 Structural Modeling. The development of a mathematical model of a structure is always required because the story drifts and the design forces in the structural members cannot be determined without such a model. In some cases, the mathematical model can be as simple as a free-body diagram as long as the model can appropriately capture the strength and stiffness of the structure.

The most realistic analytical model is three-dimensional, includes all sources of stiffness in the structure and the soil–foundation system as well as P-delta effects, and allows for nonlinear inelastic behavior in all parts of the structure–foundation–soil system. Development of such an analytical model is time-consuming, and such analysis is rarely warranted for typical building designs performed in accordance with the standard. Instead of performing a nonlinear analysis, inelastic effects are accounted for indirectly in the linear analysis methods by means of the response modification coefficient, R, and the deflection amplification factor, C_d.

Using modern software, it often is more difficult to decompose a structure into planar models than it is to develop a full three-dimensional model, so three-dimensional models are now commonplace. Increased computational efficiency also allows efficient modeling of diaphragm flexibility. Three-dimensional models are required where the structure has horizontal torsional (Type 1), out-of-plane offset (Type 4), or nonparallel system (Type 5) irregularities.

Analysis using a three-dimensional model is not required for structures with flexible diaphragms that have horizontal out-of-plane offset irregularities. It is not required because the irregularity imposes seismic load effects in a direction other than the direction under consideration (orthogonal effects) because of eccentricity in the vertical load path caused by horizontal offsets of the vertical lateral force-resisting elements from story to story. This situation is not likely to occur, however, with flexible diaphragms to an extent that warrants such modeling. The eccentricity in the vertical load path causes a redistribution of seismic design forces from the vertical elements in the story above to the vertical elements in the story below in essentially the same direction. The effect on the vertical elements in the orthogonal direction in the story below is minimal. Three-dimensional modeling may still be required for structures with flexible diaphragms caused by other types of horizontal irregularities (e.g., nonparallel system).

In general, the same three-dimensional model may be used for the equivalent lateral force, the modal response spectrum, and the linear response history analysis procedures. Modal response spectrum and linear response history analyses require a realistic modeling of structural mass; the response history method also requires an explicit representation of inherent damping. Five percent of critical damping is automatically included in the modal response spectrum approach. Chapter 16 and the related commentary have additional information on linear and nonlinear response history analysis procedures.

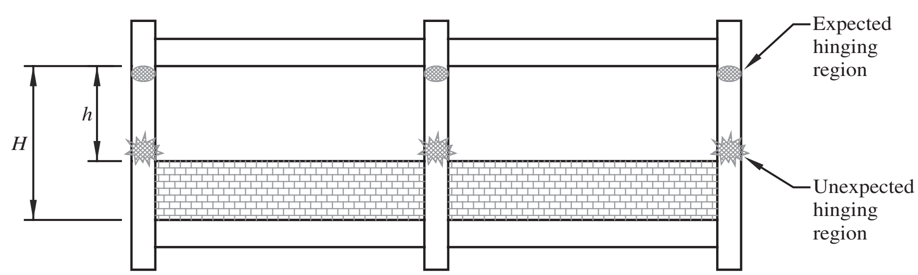
It is well known that deformations in the panel zones of the beam–column joints of steel moment frames are a significant source of flexibility. Two different mechanical models for including such deformations are summarized in Charney and Marshall (2006). These methods apply to both elastic and inelastic systems. For elastic structures, centerline analysis provides reasonable, but not always conservative, estimates of frame flexibility. Fully rigid end zones should not be used because this method always results in an overestimation of lateral stiffness in steel moment-resisting frames. Partially rigid end zones may be justified in certain cases, such as where doubler plates are used to reinforce the panel zone.

Including the effect of composite slabs in the stiffness of beams and girders may be warranted in some circumstances. Where composite behavior is included, due consideration should be paid to the reduction in effective composite stiffness for portions of the slab in tension (Schaffhausen and Wegmuller 1977, Liew et al. 2001).

For reinforced concrete buildings, it is important to address the effects of axial, flexural, and shear cracking in modeling the effective stiffness of the structural elements. Determining appropriate effective stiffness of the structural elements should take into consideration the anticipated demands on the elements, their geometry, and the complexity of the model. Recommendations for computing cracked section properties may be found in Paulay and Priestley (1992) and similar texts.

When dynamic analysis is performed, at least three dynamic degrees of freedom must be present at each level consistent with language in Section 16.2.2. Depending on the analysis software and modal extraction technique used, dynamic degrees of freedom and static degrees of freedom are not identical. It is possible to develop an analytical model that has many static degrees of freedom but only one or two dynamic degrees of freedom. Such a model does not capture response properly.

C12.7.4 Interaction Effects. The interaction requirements are intended to prevent unexpected failures in members of moment-resisting frames. Fig. C12.7-1 illustrates a typical situation where masonry infill is used and this masonry is fitted tightly against reinforced concrete columns. Because the masonry is much stiffer than the columns, hinges in a column form at the top of the column and at the top of the masonry rather than at the top



Expected plastic hinge capacity = M_p

Expected column shear = $2M_p/H$

Actual column shear = $2M_p/h$

FIGURE C12.7-1 Undesired Interaction Effects

and bottom of the column. If the column flexural capacity is M_p, the shear in the columns increases by the factor H/h, and this increase may cause an unexpected nonductile shear failure in the columns. Many building collapses have been attributed to this effect.

C12.8 EQUIVALENT LATERAL FORCE PROCEDURE

The equivalent lateral force (ELF) procedure provides a simple way to incorporate the effects of inelastic dynamic response into a linear static analysis. This procedure is useful in preliminary design of all structures and is allowed for final design of the vast majority of structures. The procedure is valid only for structures without significant discontinuities in mass and stiffness along the height, where the dominant response to ground motions is in the horizontal direction without significant torsion.

The ELF procedure has three basic steps:

1. Determine the seismic base shear, V;
2. Distribute V vertically along the height of the structure; and
3. Distribute V horizontally across the width and breadth of the structure.

Each of these steps is based on a number of simplifying assumptions. A broader understanding of these assumptions may be obtained from any structural dynamics textbook that emphasizes seismic applications.

C12.8.1 Seismic Base Shear. Treating the structure as a single-degree-of-freedom system with 100% mass participation in the fundamental mode, Eq. (12.8-1) simply expresses V as the product of the effective seismic weight, W, and the seismic response coefficient, C_s, which is a period-dependent, spectral pseudoacceleration, in g units. C_s is modified by the response modification coefficient, R, and the Importance Factor, I_e, as appropriate, to account for inelastic behavior and to provide for improved performance for high-occupancy or essential structures.

C12.8.1.1 Calculation of Seismic Response Coefficient. The standard prescribes five equations for determining C_s. Eqs. (12.8-2), (12.8-3), and (12.8-4) are illustrated in Fig. C12.8-1.

Eq. (12.8-2) controls where $0.0 < T < T_s$ and represents the constant acceleration part of the design response spectrum (Section 11.4.5). In this region, C_s is independent of period. Although the theoretical design response spectrum shown in Fig. 11.4-1 illustrates a transition in pseudoacceleration to the peak ground acceleration as the fundamental period, T, approaches zero from T_0, this transition is not used in the ELF procedure. One reason is that simple reduction of the response spectrum by $(1/R)$ in the short-period region would exaggerate inelastic effects.

Eq. (12.8-3), representing the constant velocity part of the spectrum, controls where $T_s < T < T_L$. In this region, the seismic response coefficient is inversely proportional to period, and the pseudovelocity (pseudoacceleration divided by circular frequency, ω, assuming steady-state response) is constant. T_L, the long-period transition period, represents the transition to constant displacement and is provided in Figs. 22-12 through 22-16. T_L ranges from 4 s in the north-central conterminous states and western Hawaii to 16 s in the Pacific Northwest and in western Alaska.

Eq. (12.8-4), representing the constant displacement part of the spectrum, controls where $T > T_L$. Given the current mapped values of T_L, this equation only affects long-period structures. The transition period has recently received increased attention because displacement response spectra from the 2010 magnitude 8.8 Chilean earthquake indicate that a considerably lower transition period is possible in locations controlled by subduction zone earthquakes.

The final two equations represent minimum base shear levels for design. Eq. (12.8-5) is the minimum base shear and primarily affects sites in the far field. This equation provides an allowable strength of approximately 3% of the weight of the structure. This minimum base shear was originally enacted in 1933 by the state of California (Riley Act). Based on research conducted in the ATC-63 project (FEMA 2009b), it was determined that this equation provides an adequate level of collapse resistance for long-period structures when used in conjunction with other provisions of the standard.

Eq. (12.8-6) applies to sites near major active faults (as reflected by values of S_1) where pulse-type effects can increase long-period demands.

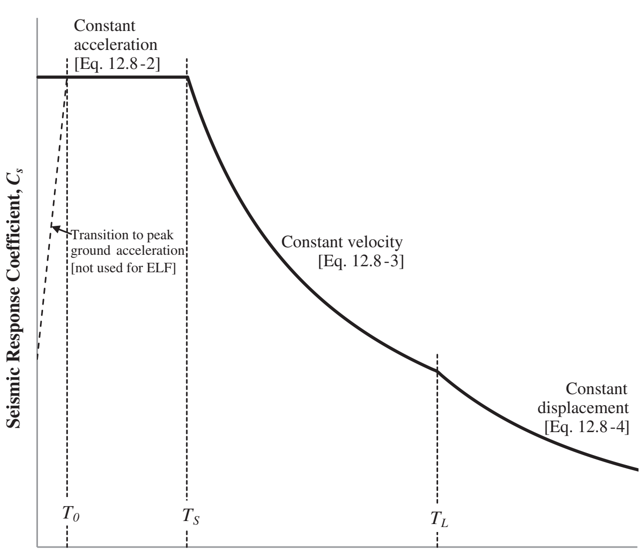


FIGURE C12.8-1 Seismic Response Coefficient Versus Period

C12.8.1.2 Soil–Structure Interaction Reduction. Soil–structure interaction, which can significantly influence the dynamic response of a structure during an earthquake, is addressed in Chapter 19.

C12.8.1.3 Maximum S_{DS} Value in Determination of C_s and E_v. This cap on the maximum value of S_{DS} reflects engineering judgment about performance of code-complying, regular, low-rise buildings in past earthquakes. It was created during the update from the 1994 UBC to the 1997 UBC and has been carried through to this standard. At that time, near-source factors were introduced, which increased the design force for buildings in Zone 4, which is similar to Seismic Design Categories D through F in this standard. The near-source factor was based on observations of instrument recording during the 1994 Northridge earthquake and new developments in seismic hazard and ground motion science. The cap placed on S_{DS} for design reflected engineering judgment by the SEAOC Seismology Committee about performance of code-complying low-rise structures based on anecdotal evidence from past California earthquakes, specifically the 1971 San Fernando, 1989 Loma Prieta, and 1994 Northridge earthquakes.

In the 1997 UBC, the maximum reduction of the cap provided was 30%. Since the change from seismic zones in the 1997 UBC to probabilistic and deterministic seismic hazard in ASCE 7-02 (2003) and subsequent editions, S_{DS} values in some parts of the country can exceed $S_{DS} = 2.0$, creating reductions well beyond the original permitted reduction. That is the rationale for this provision providing a maximum reduction in design force of 30%.

The structural height, period, redundancy, and regularity conditions required for use of the limit are important qualifiers. Additionally, the observations of acceptable performance have been with respect to collapse and life safety, not damage control or preservation of function, so this cap on the design force is limited to Risk Category I and II structures, not Risk Category III and IV structures, where higher performance is expected. Also, because past earthquake experience has indicated that buildings on very soft soils, Site Classes E and F, have performed noticeably more poorly than buildings on more competent ground, this cap cannot be used on those sites.

C12.8.2 Period Determination. The fundamental period, T, for an elastic structure is used to determine the design base shear, V, as well as the exponent, k, that establishes the distribution of V along the height of the structure (see Section 12.8.3). T may be computed using a mathematical model of the structure that incorporates the requirements of Section 12.7 in a properly substantiated analysis. Generally, this type of analysis is performed using a computer program that incorporates all deformational effects (e.g., flexural, shear, and axial) and accounts for the effect of gravity load on the stiffness of the structure. For many structures, however, the sizes of the primary structural members are not known at the outset of design. For preliminary design, as well as instances where a substantiated analysis is not used, the standard provides formulas to compute an approximate fundamental period, T_a (see Section 12.8.2.1). These periods represent lower-bound estimates of T for different structure types. Period determination is typically computed for a mathematical model that is fixed at the base. That is, the base where seismic effects are imparted into the structure is globally restrained (e.g., horizontally, vertically, and rotationally). Column base modeling (i.e., pinned or fixed) for frame-type seismic force-resisting systems is a function of frame mechanics, detailing, and foundation (soil) rigidity; attention should be given to the adopted assumption. However, this conceptual restraint is not the same for the structure as is stated above. Soil flexibility may be considered for computing T (typically assuming a rigid foundation element). The engineer should be attentive to the equivalent linear soil-spring stiffness used to represent the deformational characteristics of the soil at the base (see Section 12.13.3). Similarly, pinned column bases in frame-type structures are sometimes used to conservatively account for soil flexibility under an assumed rigid foundation element. Period shifting of a fixed-base model of a structure caused by soil–structure interaction is permitted in accordance with Chapter 19.

The fundamental mode of a structure with a geometrically complex arrangement of seismic force-resisting systems determined with a three-dimensional model may be associated with the torsional mode of response of the system, with mass participating in both horizontal directions (orthogonal) concurrently. The analyst must be attentive to this mass participation and recognize that the period used to compute the design base shear should be associated to the mode with the largest mass participation in the direction being considered. Often in this situation, these periods are close to each other. Significant separation between the torsional mode period (when fundamental) and the shortest translational mode period may be an indicator of an ill-conceived structural system or potential modeling error. The standard requires that the fundamental period, T, used to determine the design base shear, V, does not exceed the approximate fundamental period, T_a, times the upper limit coefficient, C_u, provided in Table 12.8-1. This period limit prevents the use of an unusually low base shear for design of a structure that is, analytically, overly flexible because of mass and stiffness inaccuracies in the analytical model. C_u has two effects on T_a. First, recognizing that project-specific design requirements and design assumptions can influence T, C_u lessens the conservatism inherent in the empirical formulas for T_a to more closely follow the mean curve (Fig. C12.8-2). Second, the values for C_u recognize that the formulas for T_a are targeted to structures in high seismic hazard locations. The stiffness of a structure is most likely to decrease in areas of lower seismicity, and this decrease is accounted for in the values of C_u. The response modification coefficient, R, typically decreases to account for reduced ductility demands, and the relative wind effects increase in lower seismic hazard locations. The design engineer must therefore be attentive to the value used for design of seismic force-resisting systems in structures that are controlled by wind effects. Although the value for C_u is most likely to be independent of the governing design forces in high wind areas, project-specific serviceability requirements may add considerable stiffness to a structure and decrease the value of C_u from considering seismic effects alone. This effect should be assessed where design forces for seismic and wind effects are almost equal. Lastly, if T from a properly substantiated analysis (Section 12.8.2) is less than $C_u T_a$, then the lower value of T and $C_u T_a$ should be used for the design of the structure.

C12.8.2.1 Approximate Fundamental Period. Eq. (12.8-7) is an empirical relationship determined through statistical analysis of the measured response of building structures in small- to moderate-sized earthquakes, including response to wind effects (Goel and Chopra 1997, 1998). Fig. C12.8-2 illustrates such data

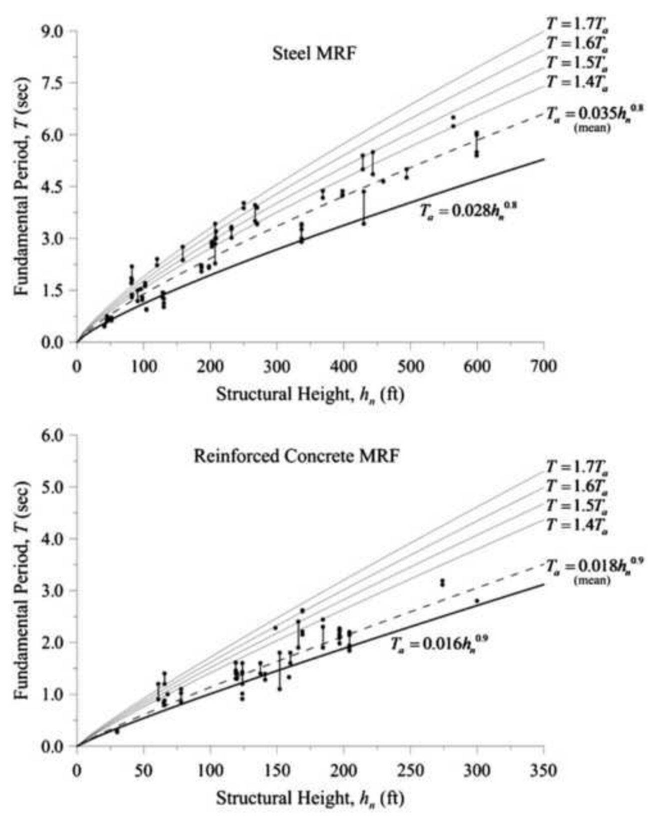
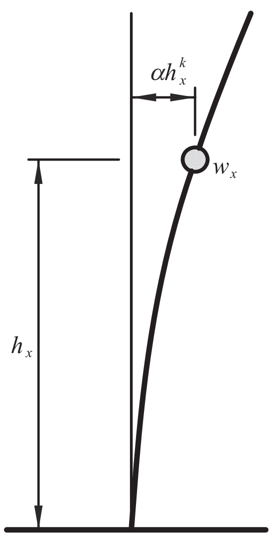


FIGURE C12.8-2 Variation of Fundamental Period with Structural Height

for various building structures with steel and reinforced concrete moment-resisting frames. Historically, the exponent, x, in Eq. (12.8-7) has been taken as 0.75 and was based on the assumption of a linearly varying mode shape while using Rayleigh's method. The exponents provided in the standard, however, are based on actual response data from building structures, thus more accurately reflecting the influence of mode shape on the exponent. Because the empirical expression is based on the lower bound of the data, it produces a lower bound estimate of the period for a building structure of a given height. This lower bound period, when used in Eqs. (12.8-3) and (12.8-4) to compute the seismic response coefficient, C_s, provides a conservative estimate of the seismic base shear, V.

C12.8.3 Vertical Distribution of Seismic Forces. Eq. (12.8-12) is based on the simplified first mode shape shown in Fig. C12.8-3. In the figure, F_x is the inertial force at level x, which is simply the absolute acceleration at level x times the mass at level x. The base shear is the sum of these inertial forces, and Eq. (12.8-11) simply gives the ratio of the lateral seismic force at level x, F_x, to the total design lateral force or shear at the base, V.

The deformed shape of the structure in Fig. C12.8-3 is a function of the exponent k, which is related to the fundamental period of the structure, T. The variation of k with T is illustrated in Fig. C12.8-4. The exponent k is intended to approximate the effect of higher modes, which are generally more dominant in structures with a longer fundamental period of vibration. Lopez and Cruz (1996) discuss the factors that influence higher modes



$$F_x = \omega^2 \alpha h_x^k \frac{w_x}{g}$$

$$V_b = \sum_{i=1}^{n} \omega^2 \alpha h_i^k \frac{w_i}{g}$$

$$C_{vx} = \frac{F_x}{V_b} = \frac{w_x h_x^k}{\sum_{i=1}^{n} w_i h_i^k}$$

FIGURE C12.8-3 Basis of Eq. (12.8-12)

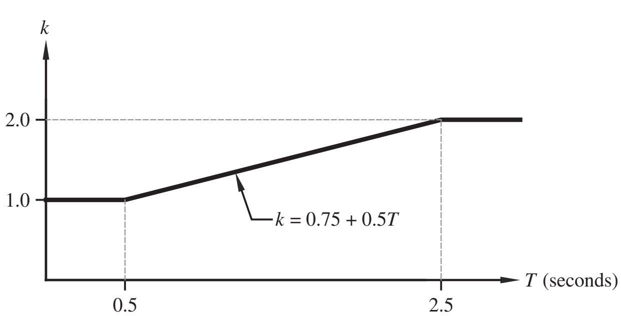


FIGURE C12.8-4 Variation of Exponent *k* with Period *T*

of response. Although the actual first mode shape for a structure is also a function of the type of seismic force-resisting system, that effect is not reflected in these equations. Also, because T is limited to $C_u T_a$ for design, this mode shape may differ from that corresponding to the statistically based empirical formula for the approximate fundamental period, T_a. A drift analysis in accordance with Section 12.8.6 can be conducted using the actual period (see Section C12.8.6). As such, k changes to account for the variation between T and the actual period.

The horizontal forces computed using Eq. (12.8-11) do not reflect the actual inertial forces imparted on a structure at any particular point in time. Instead, they are intended to provide lateral seismic forces at individual levels that are consistent with enveloped results from more accurate analyses (Chopra and Newmark 1980).

C12.8.4 Horizontal Distribution of Forces. Within the context of an ELF analysis, the horizontal distribution of lateral forces in a given story to various seismic force-resisting elements in that story depends on the type, geometric arrangement, and vertical extents of the structural elements and on the shape and flexibility of the floor or roof diaphragm. Because some elements of the seismic force-resisting system are expected to respond inelastically to the design ground motion, the distribution of forces to the various structural elements and other systems also depends on the strength of the yielding elements and their sequence of yielding (see Section C12.1.1). Such effects cannot be captured accurately by a linear elastic static analysis (Paulay 1997), and a nonlinear dynamic analysis is too computationally cumbersome to be applied to the design of most buildings. As such, approximate methods are used to account for uncertainties in horizontal distribution in an elastic static analysis, and to a lesser extent in elastic dynamic analysis.

Of particular concern in regard to the horizontal distribution of lateral forces is the torsional response of the structure during the earthquake. The standard requires that the inherent torsional moment be evaluated for every structure with diaphragms that are not flexible (see Section C12.8.4.1). Although primarily a factor for torsionally irregular structures, this mode of response has also been observed in structures that are designed to be symmetric in plan and layout of seismic force-resisting systems (De La Llera and Chopra 1994). This torsional response in the case of a torsionally regular structure is caused by a variety of "accidental" torsional moments caused by increased eccentricities between the centers of rigidity and mass that exist because of uncertainties in quantifying the mass and stiffness distribution of the structure, as well as torsional components of earthquake ground motion that are not included explicitly in code-based designs (Newmark and Rosenblueth 1971). Consequently, the accidental torsional moment can affect any structure, and potentially more so for a torsionally irregular structure. The standard requires that the accidental torsional moment be considered for every structure (see Section C12.8.4.2) as well as the amplification of this torsion for structures with torsional irregularity (see Section C12.8.4.3).

C12.8.4.1 Inherent Torsion. Where a rigid diaphragm is in the analytical model, the mass tributary to that floor or roof can be idealized as a lumped mass located at the resultant location on the floor or roof—termed the center of mass (CoM). This point represents the resultant of the inertial forces on the floor or roof. This diaphragm model simplifies structural analysis by reducing what would be many degrees of freedom in the two principal directions of a structure to three degrees of freedom (two horizontal and one rotational about the vertical axis). Similarly, the resultant stiffness of the structural members providing lateral stiffness to the structure tributary to a given floor or roof can be idealized as the center of rigidity (CoR).

It is difficult to accurately determine the center of rigidity for a multistory building because the center of rigidity for a particular story depends on the configuration of the seismic force-resisting elements above and below that story and may be load dependent (Chopra and Goel 1991). Furthermore, the location of the CoR is more sensitive to inelastic behavior than the CoM. If the CoM of a given floor or roof does not coincide with the CoR of that floor or roof, an inherent torsional moment, M_t, is created by the eccentricity between the resultant seismic force and the CoR. In addition to this *idealized* inherent torsional moment, the standard requires that an accidental torsional moment, M_{ta}, be considered (see Section C12.8.4.2).

Similar principles can be applied to models of semirigid diaphragms that explicitly model the in-plane stiffness of the diaphragm, except that the deformation of the diaphragm needs to be included in computing the distribution of the resultant seismic force and inherent torsional moment to the seismic force-resisting system.

This inherent torsion is included automatically when performing a three-dimensional analysis using either a rigid or semirigid diaphragm. If a two-dimensional planar analysis is used, where permitted, the CoR and CoM for each story must be determined explicitly and the applied seismic forces must be adjusted accordingly.

For structures with flexible diaphragms (as defined in Section 12.3), vertical elements of the seismic force-resisting system are assumed to resist inertial forces from the mass that is tributary to the elements with no explicitly computed torsion. No diaphragm is perfectly flexible; therefore some torsional forces develop even when they are neglected.

C12.8.4.2 Accidental Torsion. The locations of the centers of mass and rigidity for a given floor or roof typically cannot be established with a high degree of accuracy because of mass and stiffness uncertainty and deviations in design, construction, and loading from the idealized case. To account for this inaccuracy, the standard requires the consideration of a minimum eccentricity of 5% of the width of a structure perpendicular to the direction being considered to any static eccentricity computed using idealized locations of the centers of mass and rigidity. Where a structure has a geometrically complex or nonrectangular floor plan, the eccentricity is computed using the diaphragm extents perpendicular to the direction of loading (see Section C12.5).

One approach to account for this variation in eccentricity is to shift the CoM each way from its calculated location and apply the seismic lateral force at each shifted location as separate seismic load cases. It is typically conservative to assume that the CoM offsets at all floors and roof occur simultaneously and in the same direction. This offset produces an "accidental" static torsional moment, M_{ta}, at each story. Most computer programs can automate this offset for three-dimensional analysis by automatically applying these static moments in the autogenerated seismic load case (along the global coordinate axes used in the computer model—see Section C12.5). Alternatively, user-defined torsional moments can be applied as separate load cases and then added to the seismic lateral force load case. For two-dimensional analysis, the accidental torsional moment is distributed to each seismic force-resisting system as an applied static lateral force in proportion to its relative elastic lateral stiffness and distance from the CoR.

Shifting the CoM is a static approximation and thus does not affect the dynamic characteristics of the structure, as would be the case were the CoM to be physically moved by, for example, altering the horizontal mass distribution and mass moment of inertia. Although this "dynamic" approach can be used to adjust the eccentricity, it can be too computationally cumbersome for static analysis and therefore is reserved for dynamic analysis (see Section C12.9.1.5).

The previous discussion is applicable only to a rigid diaphragm model. A similar approach can be used for a semirigid diaphragm model except that the accidental torsional moment is decoupled into nodal moments or forces that are placed throughout the diaphragm. The amount of nodal action depends on how sensitive the diaphragm is to in-plane deformation. As the in-plane stiffness of the diaphragm decreases, tending toward a flexible diaphragm, the nodal inputs decrease proportionally.

The physical significance of this mass eccentricity should not be confused with the physical meaning of the eccentricity required for representing nonuniform wind pressures acting on a structure. However, this accidental torsion also incorporates to a lesser extent the potential torsional motion input into structures with large footprints from differences in ground motion within the footprint of the structure.

Torsionally irregular structures whose fundamental mode is potentially dominated by the torsional mode of response can be more sensitive to dynamic amplification of this accidental torsional moment. Consequently, the 5% minimum can underestimate the accidental torsional moment. In these cases, the standard requires the amplification of this moment for design when using an elastic static analysis procedure, including satisfying the drift limitations (see Section C12.8.4.3).

Accidental torsion results in forces that are combined with those obtained from the application of the seismic design story shears, V_x, including inherent torsional moments. All elements are designed for the maximum effects determined, considering positive accidental torsion, negative accidental torsion, and no accidental torsion (see Section C12.5). Where consideration of earthquake forces applied concurrently in any two orthogonal directions is required by the standard, it is permitted to apply the 5% eccentricity of the center of mass along the single orthogonal direction that produces the greater effect, but it need not be applied simultaneously in the orthogonal direction.

The exception in this section provides relief from accidental torsion requirements for buildings that are deemed to be relatively insensitive to torsion. It is supported by research (Debock et al. 2014) that compared the collapse probability (using nonlinear dynamic response history analysis) of buildings designed with and without accidental torsion requirements. The research indicated that, while accidental torsion requirements are important for most torsionally sensitive buildings (i.e., those with plan torsional irregularities arising from torsional flexibility or irregular plan layout), and especially for buildings in Seismic Design Category D, E or F, the implementation of accidental torsion provisions has little effect on collapse probability for Seismic Design Category B buildings without Type 1b horizontal structural irregularity and for Seismic Design Category D buildings without Type1a or 1b irregularity.

C12.8.4.3 Amplification of Accidental Torsional Moment. For structures with torsional or extreme torsional irregularity (Type 1a or 1b horizontal structural irregularity) analyzed using the equivalent lateral force procedure, the standard requires amplification of the accidental torsional moment to account for increases in the torsional moment caused by potential yielding of the perimeter seismic force-resisting systems (i.e., shifting of the center of rigidity), as well as other factors potentially leading to dynamic torsional instability. For verifying torsional irregularity requirements in Table 12.3-1, story drifts

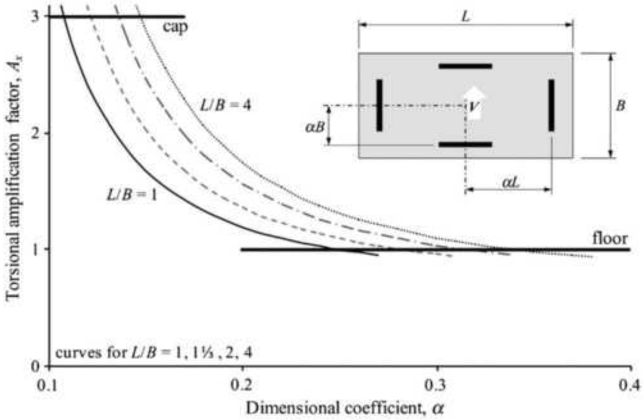


FIGURE C12.8-5 Torsional Amplification Factor for Symmetric Rectangular Buildings

resulting from the applied loads, which include both the inherent and accidental torsional moments, are used with no amplification of the accidental torsional moment ($A_x = 1$). The same process is used when computing the amplification factor, A_x, except that displacements (relative to the base) at the level being evaluated are used in lieu of story drifts. Displacements are used here to indicate that amplification of the accidental torsional moment is primarily a system-level phenomenon, proportional to the increase in acceleration at the extreme edge of the structure, and not explicitly related to an individual story and the components of the seismic force-resisting system contained therein.

Eq. (12.8-14) was developed by the SEAOC Seismology Committee to encourage engineers to design buildings with good torsional stiffness; it was first introduced in the UBC (1988). Fig. C12.8-5 illustrates the effect of Eq. (12.8-14) for a symmetric rectangular building with various aspect ratios (L/B) where the seismic force-resisting elements are positioned at a variable distance (defined by α) from the center of mass in each direction. Each element is assumed to have the same stiffness. The structure is loaded parallel to the short direction with an eccentricity of $0.05L$.

For α equal to 0.5, these elements are at the perimeter of the building, and for α equal to 0.0, they are at the center (providing no torsional resistance). For a square building ($L/B = 1.00$), A_x is greater than 1.0 where α is less than 0.25 and increases to its maximum value of 3.0 where α is equal to 0.11. For a rectangular building with L/B equal to 4.00, A_x is greater than 1.0 where α is less than 0.34 and increases to its maximum value of 3.0 where α is equal to 0.15.

C12.8.5 Overturning. The overturning effect on a vertical lateral force-resisting element is computed based on the calculation of lateral seismic force, F_x, times the height from the base to the level of the horizontal lateral force-resisting element that transfers F_x to the vertical element, summed over each story. Each vertical lateral force-resisting element resists its portion of overturning based on its relative stiffness with respect to all vertical lateral force-resisting elements in a building or structure. The seismic forces used are those from the equivalent lateral force procedure determined in Section 12.8.3 or based on a dynamic analysis of the building or structure. The overturning forces may be resisted by dead loads and can be combined with dead and live loads or other loads, in accordance with the load combinations of Section 2.3.7.

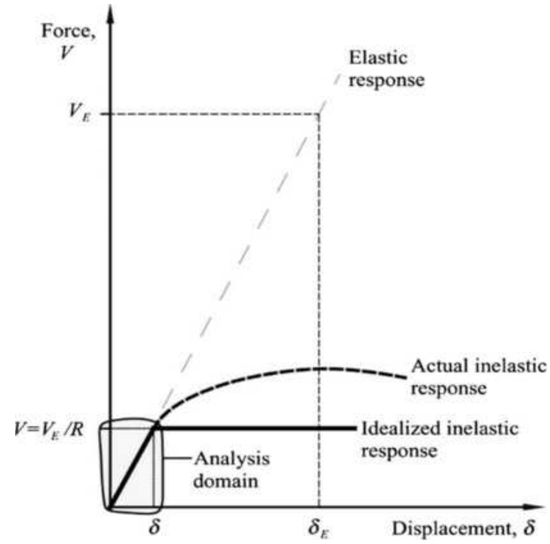


FIGURE C12.8-6 Displacements Used to Compute Drift

C12.8.6 Story Drift Determination. Eq. (12.8-15) is used to estimate inelastic deflections (δ_x), which are then used to calculate design story drifts, Δ. These story drifts must be less than the allowable story drifts, Δ_a, of Table 12.12-1. For structures without torsional irregularity, computations are performed using deflections of the centers of mass of the floors bounding the story. If the eccentricity between the centers of mass of two adjacent floors, or a floor and a roof, is more than 5% of the width of the diaphragm extents, it is permitted to compute the deflection for the bottom of the story at the point on the floor that is vertically aligned with the location of the center of mass of the top floor or roof. This situation can arise where a building has story offsets and the diaphragm extents of the top of the story are smaller than the extents of the bottom of the story. For structures assigned to Seismic Design Category C, D, E, or F that are torsionally irregular, the standard requires that deflections be computed along the edges of the diaphragm extents using two vertically aligned points.

Fig. C12.8-6 illustrates the force-displacement relationships between elastic response, response to reduced design-level forces, and the expected inelastic response. If the structure remained elastic during an earthquake, the force developed would be V_E, and the corresponding displacement would be δ_E. V_E does not include R, which accounts primarily for ductility and system overstrength. According to the equal displacement approximation rule of seismic response, the maximum displacement of an inelastic system is approximately equal to that of an elastic system with the same initial stiffness. This condition has been observed for structures idealized with bilinear inelastic response and a fundamental period, T, greater than T_s (see Section 11.4.6). For shorter period structures, peak displacement of an inelastic system tends to exceed that of the corresponding elastic system. Because the forces are reduced by R, the resulting displacements are representative of an elastic system and need to be amplified to account for inelastic response.

The deflection amplification factor, C_d, in Eq. (12.8-15) amplifies the displacements computed from an elastic analysis using prescribed forces to represent the expected inelastic displacement for the design-level earthquake and is typically less than R (Section C12.1.1). It is important to note that C_d is a story-level amplification factor and does not represent displacement amplification of the elastic response of a structure, either modeled as an *effective* single-degree-of-freedom structure (fundamental mode) or a constant amplification to represent the deflected shape of a multiple-degree-of-freedom structure, in effect, implying that the mode shapes do not change during inelastic response. Furthermore, drift-level forces are different than design-level forces used for strength compliance of the structural elements. Drift forces are typically lower because the computed fundamental period can be used to compute the base shear (see Section C12.8.6.2).

When conducting a drift analysis, the analyst should be attentive to the applied gravity loads used in combination with the strength-level earthquake forces so that consistency between the forces used in the drift analysis and those used for stability verification (P-Δ) in Section 12.8.7 is maintained, including consistency in computing the fundamental period if a second-order analysis is used. Further discussion is provided in Section C12.8.7.

The design forces used to compute the elastic deflection (δ_{xe}) include the Importance Factor, I_e, so Eq. (12.8-15) includes I_e in the denominator. This inclusion is appropriate because the allowable story drifts (except for masonry shear wall structures) in Table 12.12-1 are more stringent for higher Risk Categories.

C12.8.6.1 Minimum Base Shear for Computing Drift. Except for period limits (as described in Section C12.8.6.2), all of the requirements of Section 12.8 must be satisfied when computing drift for an ELF analysis, except that the minimum base shear determined from applying Eq. (12.8-5) does not need to be considered. This equation represents a minimum strength that needs to be provided to a system (see Section C12.8.1.1). Eq. (12.8-6) needs to be considered, when triggered, because it represents the increase in the response spectrum in the long-period range from near-fault effects.

C12.8.6.2 Period for Computing Drift. Where the design response spectrum of Section 11.4.6 or the corresponding equations of Section 12.8.1 are used and the fundamental period of the structure, T, is less than the long-period transition period, T_L, displacements increase with increasing period (even though forces may decrease). Section 12.8.2 applies an upper limit on T so that design forces are not underestimated, but if the lateral forces used to compute drifts are inconsistent with the forces corresponding to T, then displacements can be overestimated. To account for this variation in dynamic response, the standard allows the determination of displacements using forces that are consistent with the computed fundamental period of the structure without the upper limit of Section 12.8.2.

The analyst must still be attentive to the period used to compute drift forces. The same analytical representation (see Section C12.7.3) of the structure used for strength design must also be used for computing displacements. Similarly, the same analysis method (Table 12.6-1) used to compute design forces must also be used to compute drift forces. It is generally appropriate to use 85% of the computed fundamental period to account for mass and stiffness inaccuracies as a precaution against overly flexible structures, but it need not be taken as less than that used for strength design. The more flexible the structure, the more likely it is that P-delta effects ultimately control the design (see Section C12.8.7). Computed values of T that are significantly greater than (perhaps more than 1.5 times in high seismic areas) $C_u T_a$ may indicate a modeling error. Similar to the discussion in Section C12.8.2, the analyst should assess the value of C_u used where serviceability constraints from wind effects add significant stiffness to the structure.

C12.8.7 P-Delta Effects. Fig. C12.8-7 shows an idealized static force-displacement response for a simple one-story structure

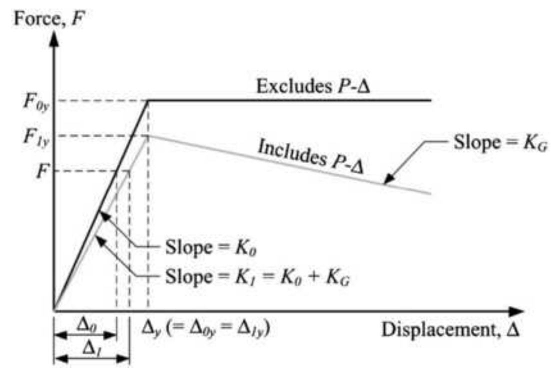


FIGURE C12.8-7 Idealized Response of a One-Story Structure with and without P-Δ

(e.g., idealized as an inverted pendulum-type structure). As the top of the structure displaces laterally, the gravity load, P, supported by the structure acts through that displacement and produces an increase in overturning moment by P times the story drift, Δ, that must be resisted by the structure—the so-called "P-delta (P-Δ) effect." This effect also influences the lateral displacement response of the structure from an applied lateral force, F.

The response of the structure not considering the P-Δ effect is depicted by Condition 0 in the figure with a slope of K_0 and lateral first-order yield force F_{0y}. This condition characterizes the first-order response of the structure (the response of the structure from an analysis not including P-delta effects). Where the P-Δ effect is included (depicted by Condition 1 in the figure), the related quantities are K_1 and F_{1y}. This condition characterizes the second-order response of the structure (the response of the structure from an analysis including P-delta effects).

The geometric stiffness of the structure, K_G, in this example is equal to the gravity load, P, divided by the story height, h_{sx}. K_G is used to represent the change in lateral response by analytically reducing the elastic stiffness, K_0. K_G is negative where gravity loads cause compression in the structure. Because the two response conditions in the figure are for the same structure, the inherent yield displacement of the structure is the same ($\Delta_{0y} = \Delta_{1y} = \Delta_y$).

Two consequential points taken from the figure are (1) the increase in required strength and stiffness of the seismic force-resisting system where the P-Δ effect influences the lateral response of the structure must be accounted for in design, and (2) the P-Δ effect can create a negative stiffness condition during postyield response, which could initiate instability of the structure. Where the postyield stiffness of the structure may become negative, dynamic displacement demands can increase significantly (Gupta and Krawinkler 2000).

One approach that can be used to assess the influence of the P-Δ effect on the lateral response of a structure is to compare the first-order response to the second-order response, which can be done using an elastic stability coefficient, θ, defined as the absolute value of K_G divided by K_0.

$$\theta = \frac{|K_G|}{K_0} = \left| \frac{P\Delta_{0y}}{F_{0y} h_{sx}} \right| \quad \text{(C12.8-1)}$$

Given the above, and the geometric relationships shown in Fig. C12.8-7, it can be shown that the force producing yield in condition 1 (with P-Δ effects) is

$$F_{1y} = F_{0y}(1-\theta) \quad \text{(C12.8-2)}$$

and that for a force, F, less than or equal to F_{1y}

$$\Delta_1 = \frac{\Delta_0}{1-\theta} \quad \text{(C12.8-3)}$$

Therefore, the stiffness ratio, K_0/K_1, is

$$\frac{K_0}{K_1} = \frac{1}{1-\theta} \quad \text{(C12.8-4)}$$

In the previous equations,

F_{0y} = the lateral first-order yield force;
F_{1y} = the lateral second-order yield force;
h_{sx} = the story height (or structure height in this example);
K_G = the geometric stiffness;
K_0 = the elastic first-order stiffness;
K_1 = the elastic second-order stiffness;
P = the total gravity load supported by the structure;
Δ_0 = the lateral first-order drift;
Δ_{0y} = the lateral first-order yield drift;
Δ_1 = the lateral second-order drift;
Δ_{1y} = the lateral second-order yield drift; and
θ = the elastic stability coefficient.

A physical interpretation of this effect is that to achieve the second-order response depicted in the figure, the seismic force-resisting system must be designed to have the increased stiffness and strength depicted by the first-order response. As θ approaches unity, Δ_1 approaches infinity and F_1 approaches zero, defining a state of static instability.

The intent of Section 12.8.7 is to determine whether P-Δ effects are significant when considering the first-order response of a structure and, if so, to increase the strength and stiffness of the structure to account for P-Δ effects. Some material-specific design standards require P-Δ effects to always be included in the elastic analysis of a structure and strength design of its members. The amplification of first-order member forces in accordance with Section 12.8.7 should not be misinterpreted to mean that these other requirements can be disregarded; nor should they be applied concurrently. Therefore, Section 12.8.7 is primarily used to verify compliance with the allowable drifts and check against potential postearthquake instability of the structure, while provisions in material-specific design standards are used to increase member forces for design, if provided. In so doing, the analyst should be attentive to the stiffness of each member used in the mathematical model so that synergy between standards is maintained.

Eq. (12.8-16) is used to determine the elastic stability coefficient, θ, of each story of a structure.

$$\theta = \left|\frac{P\Delta_0}{F_0 h_{sx}}\right| = \frac{P\Delta I_e}{V_x h_{sx} C_d} \quad \text{(C12.8-5)}$$

where

h_{sx}, I_e, and V_x are the same as defined in the standard and
F_0 = the force in a story causing $\Delta_0 = \sum F_x = V_x$;
Δ_0 = the elastic lateral story drift = $\Delta I_e/C_d$;
Δ = the inelastic story drift determined in accordance with Section 12.8.6; and
P = the total *point-in-time* gravity load supported by the structure.

Structures with θ less than 0.10 generally are expected to have a positive monotonic postyield stiffness. Where θ for any story exceeds 0.10, P-Δ effects must be considered for the entire structure using one of the two approaches in the standard. Either first-order displacements and member forces are multiplied by $1/(1-\theta)$ or the P-Δ effect is explicitly included in the structural analysis and the resulting θ is multiplied by $1/(1+\theta)$ to verify compliance with the first-order stability limit. Most commercial computer programs can perform second-order analysis. The analyst must therefore be attentive to the algorithm incorporated in the software and cognizant of any limitations, including suitability of iterative and noniterative methods, inclusion of second-order effects (P-Δ and P-δ) in automated modal analyses, and appropriateness of superposition of design forces.

Gravity load drives the increase in lateral displacements from the equivalent lateral forces. The standard requires the total vertical design load, and the largest vertical design load for combination with earthquake loads is given by combination 6 from Section 2.3.6, which is transformed to

$$(1.2 + 0.2S_{DS})D + 1.0L + 0.2S + 1.0E$$

where the 1.0 factor on L is actually 0.5 for many common occupancies. The provision of Section 12.8.7 allows the factor on dead load D to be reduced to 1.0 for the purpose of P-delta analysis under seismic loads. The vertical seismic component need not be considered for checking θ_{max}.

As explained in the commentary for Chapter 2, the 0.5 and 0.2 factors on L and S, respectively, are intended to capture the arbitrary point-in-time values of those loads. The factor 1.0 results in the dead load effect being fairly close to best estimates of the arbitrary point-in-time value for dead load. L is defined in Chapter 4 of the standard to include the reduction in live load based on floor area. Many commercially available computer programs do not include live load reduction in the basic structural analysis. In such programs, live reduction is applied only in the checking of design criteria; this difference results in a conservative calculation with regard to the requirement of the standard.

The seismic story shear, V_x (in accordance with Section 12.8.4), used to compute θ includes the Importance Factor, I_e. Furthermore, the design story drift, Δ (in accordance with Section 12.8.6), does not include this factor. Therefore, I_e has been added to Eq. (12.8-16) to correct an apparent omission in previous editions of the standard. Nevertheless, the standard has always required V_x and Δ used in this equation to be those occurring simultaneously.

Eq. (12.8-17) establishes the maximum stability coefficient, θ_{max}, permitted. The intent of this requirement is to protect structures from the possibility of instability triggered by post-earthquake residual deformation. The danger of such failures is real and may not be eliminated by apparently available overstrength. This problem is particularly true of structures designed in regions of lower seismicity.

For the idealized system shown in Fig. C12.8-7, assume that the maximum displacement is $C_d\Delta_0$. Assuming that the unloading stiffness, K_u, is equal to the elastic stiffness, K_0, the residual displacement is

$$\left(C_d - \frac{1}{\beta}\right)\Delta_0 \quad \text{(C12.8-6)}$$

Additionally, assume that there is a factor of safety, FS, of 2 against instability at the maximum residual drift, $\Delta_{r,\max}$. Evaluating the overturning and resisting moments ($F_0 = V_0$ in this example),

$$P\Delta_{r,\max} \le \frac{V_0}{\beta FS}h \quad \text{where} \quad \beta = \frac{V_0}{V_{0y}} \le 1.0 \qquad \text{(C12.8-7)}$$

Therefore,

$$\frac{P[\Delta_0(\beta C_d - 1)]}{V_0 h} \le 0.5 \rightarrow \theta_{\max}(\beta C_d - 1) = 0.5 \rightarrow \theta_{\max} = \frac{0.5}{\beta C_d - 1} \qquad \text{(C12.8-8)}$$

Conservatively assume that $\beta C_d - 1 \approx \beta C_d$

$$\theta_{\max} = \frac{0.5}{\beta C_d} \le 0.25 \qquad \text{(C12.8-9)}$$

In the previous equations,

C_d = the displacement amplification factor;
FS = the factor of safety;
h_{sx} = the story height (or height of the structure in this example);
P = the total point-in-time gravity load supported by the structure;
V_0 = the first-order story shear demand;
V_{0y} = the first-order yield strength of the story;
β = the ratio of shear demand to shear capacity;
Δ_0 = the elastic lateral story drift;
$\Delta_{r,\max}$ = the maximum residual drift at $V_0 = 0$; and
$\theta_{\max}$ = the maximum elastic stability coefficient.

The standard requires that the computed stability coefficient, θ, not exceed 0.25 or $0.5/\beta C_d$, where βC_d is an adjusted ductility demand that takes into account the variation between the story strength demand and the story strength supplied. The story strength demand is simply V_x. The story strength supplied may be computed as the shear in the story that occurs simultaneously with the attainment of the development of first significant yield of the overall structure. To compute first significant yield, the structure should be loaded with a seismic force pattern similar to that used to compute story strength demand and iteratively increased until first yield. Alternatively, a simple and conservative procedure is to compute the ratio of demand to strength for each member of the seismic force-resisting system in a particular story and then use the largest such ratio as β.

The principal reason for inclusion of β is to allow for a more equitable analysis of those structures in which substantial extra strength is provided, whether as a result of added stiffness for drift control, code-required wind resistance, or simply a feature of other aspects of the design. Some structures inherently possess more strength than required, but instability is not typically a concern. For many flexible structures, the proportions of the structural members are controlled by drift requirements rather than strength requirements; consequently, β is less than 1.0 because the members provided are larger and stronger than required. This method has the effect of reducing the inelastic component of total seismic drift, and thus, β is placed as a factor on C_d.

Accurate evaluation of β would require consideration of all pertinent load combinations to find the maximum ratio of demand to capacity caused by seismic load effects in each member. A conservative simplification is to divide the total demand with seismic load effects included by the total capacity; this simplification covers all load combinations in which dead and live load effects add to seismic load effects. If a member is controlled by a load combination where dead load counteracts seismic load effects, to be correctly computed, β must be based only on the seismic component, not the total. The gravity load, P, in the P-Δ computation would be less in such a circumstance and, therefore, θ would be less. The importance of the counteracting load combination does have to be considered, but it rarely controls instability.

Although the P-Δ procedure in the standard reflects a simple static idealization as shown in Fig. C12.8-7, the real issue is one of dynamic stability. To adequately evaluate second-order effects during an earthquake, a nonlinear response history analysis should be performed that reflects variability of ground motions and system properties, including initial stiffness, strain hardening stiffness, initial strength, hysteretic behavior, and magnitude of point-in-time gravity load, P. Unfortunately, the dynamic response of structures is highly sensitive to such parameters, causing considerable dispersion to appear in the results (Vamvatsikos 2002). This dispersion, which increases dramatically with stability coefficient θ, is caused primarily by the incrementally increasing residual deformations (ratcheting) that occur during the response. Residual deformations may be controlled by increasing either the initial strength or the secondary stiffness. Gupta and Krawinkler (2000) give additional information.

C12.9 LINEAR DYNAMIC ANALYSIS

C12.9.1 Modal Response Spectrum Analysis. In the modal response spectrum analysis method, the structure is decomposed into a number of single-degree-of-freedom systems, each having its own mode shape and natural period of vibration. The number of modes available is equal to the number of mass degrees of freedom of the structure, so the number of modes can be reduced by eliminating mass degrees of freedom. For example, rigid diaphragm constraints may be used to reduce the number of mass degrees of freedom to one per story for planar models and to three per story (two translations and rotation about the vertical axis) for three-dimensional structures. However, where the vertical elements of the seismic force-resisting system have significant differences in lateral stiffness, rigid diaphragm models should be used with caution because relatively small in-plane diaphragm deformations can have a significant effect on the distribution of forces.

For a given direction of loading, the displacement in each mode is determined from the corresponding spectral acceleration, modal participation, and mode shape. Because the sign (positive or negative) and the time of occurrence of the maximum acceleration are lost in creating a response spectrum, there is no way to recombine modal responses exactly. However, statistical combination of modal responses produces reasonably accurate estimates of displacements and component forces. The loss of signs for computed quantities leads to problems in interpreting force results where seismic effects are combined with gravity effects, produce forces that are not in equilibrium, and make it impossible to plot deflected shapes of the structure.

C12.9.1.1 Number of Modes. The key motivation to perform modal response spectrum analysis is to determine how the actual distribution of mass and stiffness of a structure affects the elastic displacements and member forces. Where at least 90% of the modal mass participates in the response, the distribution of forces and displacements is sufficient for design. The scaling required by Section 12.9.1.4 controls the overall magnitude of design values so that incomplete mass participation does not produce nonconservative results.

The number of modes required to achieve 90% modal mass participation is usually a small fraction of the total number of modes. Lopez and Cruz (1996) contribute further discussion of the number of modes to use for modal response spectrum analysis.

In general, the provisions require modal analysis to determine all individual modes of vibration, but permit modes with periods less than or equal to 0.05 s to be collectively treated as a single, rigid mode of response with an assumed period of 0.05 s. In general, structural modes of interest to building design have periods greater than 0.05 s (frequencies greater than 20 Hz), and earthquake records tend to have little, if any, energy, at frequencies greater than 20 Hz. Thus, only "rigid" response is expected for modes with frequencies above 20 Hz. Although not responding dynamically, the "residual mass" of modes with frequencies greater than 20 Hz should be included in the analysis to avoid underestimation of earthquake design forces.

Section 4.3 of ASCE 4 (ASCE 2000) provides formulas that may be used to calculate the modal properties of the residual-mass mode. When using the formulas of ASCE 4 to calculate residual-mass mode properties, the "cut-off" frequency should be taken as 20 Hz and the response spectral acceleration at 20 Hz (0.05 s) should be assumed to govern response of the residual-mass mode. It may be noted that the properties of residual-mass mode are derived from the properties of modes with frequencies less than or equal to 20 Hz, such that modal analysis need only determine properties of modes of vibration with periods greater than 0.05 s (when the residual-mass mode is included in the modal analysis). The design response spectral acceleration at 0.05 s (20 Hz) should be determined using Eq. (11.4-5) of this standard where the design response spectrum shown in Fig. 11.4-1 is being used for the design analysis. Substituting 0.05 s for T and $0.2T_s$ for T_0 in Eq. (11.4-5), one obtains the residual-mode response spectral acceleration as $S_a = S_{DS}\ [0.4 + 0.15/T_s]$. Most general-purpose linear structural analysis software has the capacity to consider residual mass modes in order to meet the existing requirements ASCE 4 (ASCE 2000).

The exception permits excluding modes of vibration when such would result in a modal mass in each orthogonal direction of at least 90% of the actual mass. This approach has been included in ASCE 7 (2003, 2010) for many years and is still considered adequate for most building structures that typically do not have significant modal mass in the very short period range.

C12.9.1.2 Modal Response Parameters. The design response spectrum (whether the general spectrum from Section 11.4.6 or a site-specific spectrum determined in accordance with Section 21.2) is representative of linear elastic structures. Division of the spectral ordinates by the response modification coefficient, R, accounts for inelastic behavior, and multiplication of spectral ordinates by the Importance Factor, I_e, provides the additional strength needed to improve the performance of important structures. The displacements that are computed using the response spectrum that has been modified by R and I_e (for strength) must be amplified by C_d and reduced by I_e to produce the expected inelastic displacements (see Section C12.8.6.)

C12.9.1.3 Combined Response Parameters. Most computer programs provide for either the SRSS or the CQC method (Wilson et al. 1981) of modal combination. The two methods are identical where applied to planar structures, or where zero damping is specified for the computation of the cross-modal coefficients in the CQC method. The modal damping specified in each mode for the CQC method should be equal to the damping level that was used in the development of the design response spectrum. For the spectrum in Section 11.4.6, the damping ratio is 0.05.

The SRSS or CQC method is applied to loading in one direction at a time. Where Section 12.5 requires explicit consideration of orthogonal loading effects, the results from one direction of loading may be added to 30% of the results from loading in an orthogonal direction. Wilson (2000) suggests that a more accurate approach is to use the SRSS method to combine 100% of the results from each of two orthogonal directions where the individual directional results have been combined by SRSS or CQC, as appropriate.

The CQC4 method, as modified by ASCE 4 (1998), is specified and is an alternative to the required use of the CQC method where there are closely spaced modes with significant cross-correlation of translational and torsional response. The CQC4 method varies slightly from the CQC method through the use of a parameter that forces a correlation in modal responses where they are partially or completely in phase with the input motion. This difference primarily affects structures with short fundamental periods, T, that have significant components of response that are in phase with the ground motion. In these cases, using the CQC method can be nonconservative. A general overview of the various modal response combination methods can be found in U.S Nuclear Regulatory Commission (2012).

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C12.9.1.4 Scaling Design Values of Combined Response. The modal base shear, V_t, may be less than the ELF base shear, V, because: (a) the calculated fundamental period, T, may be longer than that used in computing V, (b) the response is not characterized by a single mode, or (c) the ELF base shear assumes 100% mass participation in the first mode, which is always an overestimate.

C12.9.1.4.1 Scaling of Forces. The scaling required by Section 12.9.1.4.1 provides, in effect, a minimum base shear for design. This minimum base shear is provided because the computed fundamental period may be the result of an overly flexible (incorrect) analytical model. Recent studies of building collapse performance, such as those of FEMA P-695 (the ATC-63 project, 2009b), NIST GCR 10-917-8 (the ATC-76 project) and NIST GCR 12-917-20 (the ATC-84 project) show that designs based on the ELF procedure generally result in better collapse performance than those based on modal response spectrum analysis (MRSA) with the 15% reduction in base shear included. In addition, many of the designs using scaled MRSA did not achieve the targeted 10% probability of collapse given MCE ground shaking. Whereas scaling to 100% of the ELF base shear and to 100% of the drifts associated with Eq. (12.8-6) does not necessarily achieve the intended collapse performance, it does result in performance that is closer to the stated goals of this standard.

C12.9.1.4.2 Scaling of Drifts. Displacements from the modal response spectrum are only scaled to the ELF base shear where V_t is less than C_sW and C_s is determined based on Eq. (12.8-6). For all other situations, the displacements need not be scaled because the use of an overly flexible model will result in conservative estimates of displacement that need not be further scaled. The reason for requiring scaling when Eq. (12.8-6) controls the minimum base shear is to be consistent with the requirements for designs based on the ELF procedure.

C12.9.1.5 Horizontal Shear Distribution. Torsion effects in accordance with Section 12.8.4 must be included in the modal response spectrum analysis (MRSA) as specified in Section 12.9 by requiring use of the procedures in Section 12.8 for the determination of the seismic base shear, V. There are two basic approaches for consideration of accidental torsion.

The first approach follows the static procedure discussed in Section C12.8.4.2, where the total seismic lateral forces obtained from MRSA—using the computed locations of the centers of mass and rigidity—are statically applied at an artificial point offset from the center of mass to compute the accidental torsional moments. Most computer programs can automate this procedure for three-dimensional analysis. Alternatively, the torsional moments can be statically applied as separate load cases and added to the results obtained from MRSA.

Because this approach is a static approximation, amplification of the accidental torsion in accordance with Section 12.8.4.3 is required. MRSA results in a single, positive response, thus inhibiting direct assessment of torsional response. One method to circumvent this problem is to determine the maximum and average displacements for each mode participating in the direction being considered and then apply modal combination rules (primarily the CQC method) to obtain the total displacements used to check torsional irregularity and compute the amplification factor, A_x. The analyst should be attentive about how accidental torsion is included for individual modal responses.

The second approach, which applies primarily to three-dimensional analysis, is to modify the dynamic characteristics of the structure so that dynamic amplification of the accidental torsion is directly considered. This modification can be done, for example, by either reassigning the lumped mass for each floor and roof (rigid diaphragm) to alternate points offset from the initially calculated center of mass and modifying the mass moment of inertia, or physically relocating the initially calculated center of mass on each floor and roof by modifying the horizontal mass distribution (typically presumed to be uniformly distributed). This approach increases the computational demand significantly because all possible configurations would have to be analyzed, primarily two additional analyses for each principal axis of the structure. The advantage of this approach is that the dynamic effects of direct loading and accidental torsion are assessed automatically. Practical disadvantages are the increased bookkeeping required to track multiple analyses and the cumbersome calculations of the mass properties.

Where this "dynamic" approach is used, amplification of the accidental torsion in accordance with Section 12.8.4.3 is not required because repositioning the center of mass increases the coupling between the torsional and lateral modal responses, directly capturing the amplification of the accidental torsion.

Most computer programs that include accidental torsion in a MRSA do so statically (first approach discussed above) and do not physically shift the center of mass. The designer should be aware of the methodology used for consideration of accidental torsion in the selected computer program.

C12.9.1.6 P-Delta Effects. The requirements of Section 12.8.7, including the stability coefficient limit, θ_{max}, apply to modal response spectrum analysis.

C12.9.1.7 Soil–Structure Interaction Reduction. The standard permits including soil–structure interaction (SSI) effects in a modal response spectrum analysis in accordance with Chapter 19. The increased use of modal analysis for design stems from computer analysis programs automatically performing such an analysis. However, common commercial programs do not give analysts the ability to customize modal response parameters. This problem hinders the ability to include SSI effects in an automated modal analysis.

C12.9.1.8 Structural Modeling. Using modern software, it often is more difficult to decompose a structure into planar models than it is to develop a full three-dimensional model. As a result, three-dimensional models are now commonplace. Increased computational efficiency also allows efficient modeling of diaphragm flexibility. As a result, when modal response spectrum analysis is used, a three-dimensional model is required for all structures, including those with diaphragms that can be designated as flexible.

C12.9.2 Linear Response History Analysis

C12.9.2.1 General Requirements. The linear response history (LRH) analysis method provided in this section is intended as an alternate to the modal response spectrum (MRS) analysis method. The principal motivation for providing the LRH analysis method is that signs (positive–negative bending moments, tension–compression brace forces) are preserved, whereas they are lost in forming the SRSS and CQC combination in MRS analysis.

It is important to note that, like the ELF procedure and the MRS analysis method, the LRH analysis method is used as a basis for structural design, and not to predict how the structure will respond to a given ground motion. Thus, in the method provided in this section, spectrum-matched ground motions are used in lieu of amplitude-scaled motions. The analysis may be performed using modal superposition, or by analysis of the fully coupled equations of motion (often referred to as direct integration response history analysis).

As discussed in Section 12.9.2.3, the LRH analysis method requires the use of three sets of ground motions, with two orthogonal components in each set. These motions are then modified such that the response spectra of the modified motions closely match the shape of the target response spectrum. Thus, the maximum computed response in each mode is virtually identical to the value obtained from the target response spectrum. The only difference between the MRS analysis method and the LRH analysis method (as developed in this section using the spectrum-matched ground motions) is that in the MRS analysis method the system response is computed by statistical combination (SRSS or CQC) of the modal responses and in the LRH analysis method, the system response is obtained by direct addition of modal responses or by simultaneous solution of the full set of equations of motion.

C12.9.2.2 General Modeling Requirements. Three-dimensional (3D) modeling is required for conformance with the inherent and accidental torsion requirements of Section 12.9.2.2.2.

C12.9.2.2.1 P-Delta Effects. A static analysis is required to determine the stability coefficients using Eq. (12.8-17). Typically, the mathematical model used to compute the quantity Δ in Eq. (12.8-16) does not directly include P-delta effects. However, Section 12.8.7 provides a methodology for checking compliance with the θ_{max} limit where P-delta effects are directly included in the model. For dynamic analysis, an ex post facto modification of results from an analysis that does not include P-delta effects to one that does (approximately) include such effects is not rational.

Given that virtually all software that performs linear response history analysis has the capability to directly include P-delta effects, it is required that P-delta effects be included in all analyses, even when the maximum stability ratio at any level is less than 0.1. The inclusion of such effects causes a lengthening of the period of vibration of the structure, and this period should be used for establishing the range of periods for spectrum matching (Section 12.9.2.3.1) and for selecting the number of modes to include in the response (Section 12.9.2.2.4).

While the P-delta effect is essentially a nonlinear phenomenon (stiffness depends on displacements and displacements depend on stiffness), such effects are often "linearized" by forming a constant geometric stiffness matrix that is created from member forces generated from an initial gravity load analysis (Wilson and Habibullah 1987; Wilson 2004). This approach works for both the modal superposition method and the direct analysis method. It is noted, however, that there are some approximations in this method, principally the way the global torsional component of P-delta effects is handled. The method is of sufficient accuracy in analysis for which materials remain elastic. Where direct integration is used, a more accurate response can be computed by iteratively updating the geometric stiffness at each time step or by iteratively satisfying equilibrium about the deformed configuration. In either case, the analysis is in fact "nonlinear," but it is considered as a linear analysis in Section 12.9.2 because material properties remain linear.

For 3D models, it is important to use a realistic spatial distribution of gravity loads because such a distribution is necessary to capture torsional P-delta effects.

C12.9.2.2.2 Accidental Torsion. The required 5% offset of the center of mass need not be applied in both orthogonal directions at the same time. Direct modeling of accidental torsion by offsetting the center of mass is required to retain the signs (positive–negative bending moments, tension–compression forces in braces). In addition to the four mathematical models with mass offsets, a fifth model without accidental torsion (including only inherent torsion) must also be prepared. The model without accidental torsion is needed as the basis for scaling results as required in Section 12.9.2.5. Though not a requirement of the LRH analysis method, the analyst may also compare the modal characteristics (periods, mode shapes) to the systems with and without accidental mass eccentricity to gauge the sensitivity of the structure to accidental torsional response.

C12.9.2.2.3 Foundation Modeling. Foundation flexibility may be included in the analysis. Where such modeling is used, the requirements of Section 12.13.3 should be satisfied. Additional guidance on modeling foundation effects may be found in *Nonlinear Structural Analysis for Seismic Design: A Guide for Practicing Engineers* (NIST 2010).

C12.9.2.2.4 Number of Modes to Include in Modal Response History Analysis. Where modal response history analysis is used, it is common to analyze only a subset of the modes. In the past, the number of modes to analyze has been determined such that a minimum of 90% of the effective mass in each direction is captured in the response. An alternate procedure that produces participation of 100% of the effective mass is to represent all modes with periods less than 0.05 s in a single rigid body mode having a period of 0.05 s. In direct analysis, the question of the number of modes to include does not arise because the system response is computed without modal decomposition.

An example of a situation where it would be difficult to obtain 90% of the mass in a reasonable number of modes is reported in Chapter 4 of FEMA P-751 (2013), which presents the dynamic analysis of a 12-story building over a 1-story basement. When the basement walls and grade-level diaphragm were excluded from the model, 12 modes were sufficient to capture 90% of the effective mass. When the basement was modeled as a stiff first story, it took more than 120 modes to capture 90% of the total mass (including the basement and the ground-level diaphragm). It is noted in the Chapter 4 discussion that when the full structure was modeled and only 12 modes were used, the member forces and system deformations obtained were virtually identical to those obtained when 12 modes were used for the fixed-base system (modeled without the podium).

If modal response history analysis is used and it is desired to use a mathematical model that includes a stiff podium, it might be beneficial to use Ritz vectors in lieu of eigenvectors (Wilson 2004). Another approach is the use of the "static correction method," in which the responses of the higher modes are determined by a static analysis instead of a dynamic analysis (Chopra 2007). The requirement in Section 12.9.2.2.4 of including all modes with periods of less than 0.05 s as a rigid body mode is in fact an implementation of the static correction method.

C12.9.2.2.5 Damping. Where modal superposition analysis is used, 5% damping should be specified for each mode because it is equal to the damping used in the development of the response spectrum specified in Section 11.4.6 and in Section 21.1.3. Where direct analysis is used, it is possible but not common to form a damping matrix that provides uniform damping across all modes (Wilson and Penzien 1972). It is more common to use a mass and stiffness proportional damping matrix (i.e., Rayleigh damping), but when this is done, the damping ratio may be specified at only two periods. Damping ratios at other periods depend on the mass and stiffness proportionality constants. At periods associated with higher modes, the damping ratios may become excessive, effectively damping out important modes of response. To control this effect, Section 12.9.2.2.5 requires the damping in all included modes (with periods as low as T_{lower}) be less than or equal to 5% critical.

C12.9.2.3 Ground Motion Selection and Modification. Response spectrum matching (also called spectral matching) is the nonuniform scaling of an actual or artificial ground motion such that its pseudoacceleration response spectrum closely matches a target spectrum. In most cases, the target spectrum is the same spectrum used for scaling actual recorded ground motions (i.e., the ASCE 7 design spectrum). Spectral matching can be contrasted with amplitude scaling, in which a uniform scale factor is applied to the ground motion. The principal advantage of spectral matching is that fewer ground motions, compared to amplitude scaling, can be used to arrive at an acceptable estimate of the mean response as recommended in NIST GCR 11-918-15 (NIST 2011). Fig. C12.9-1(a) shows the response spectra of two ground motions that have been spectral matched, and Fig. C12.9-1(b) shows the response spectra of the original ground motions. In both cases, the ground motions are normalized to match the target response spectrum at a period of 1.10 s. Clearly, the two amplitude-scaled records will result in significantly different responses, whereas analysis using the spectrum-matched records will be similar. As described later, however, there is enough variation in the response using spectrum-matched records to require the use of more than one record in the response history analysis.

A variety of methods is available for spectrum matching, and the reader is referred to Hancock et al. (2006) for details. Additional information on use of spectrum-matched ground motions in response history analysis is provided by Grant and Diaferia (2012).

C12.9.2.3.1 Procedure for Spectrum Matching. Experience with spectrum matching has indicated that it is easier to get a good match when the matching period extends beyond the period range of

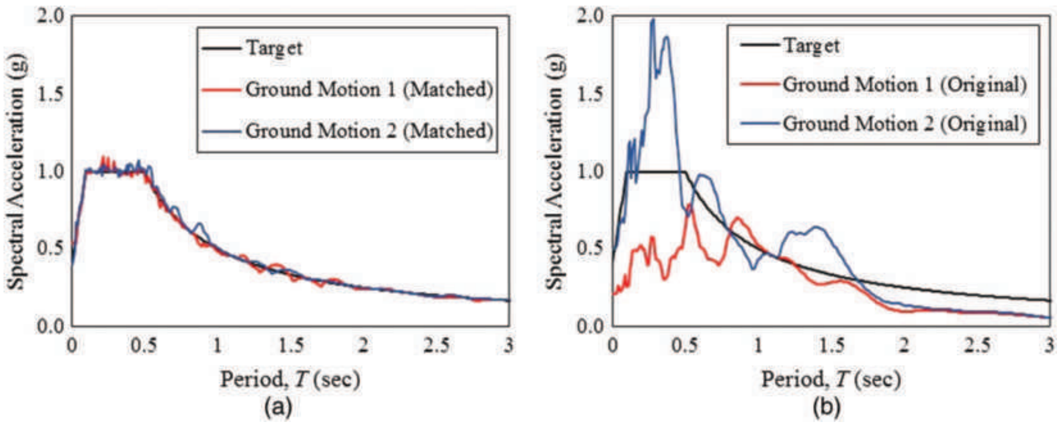

FIGURE C12.9-1. Spectral Matching vs. Amplitude-Scaled Response Spectra

interest. It is for this reason that spectrum matching is required over the range $0.8T_{\text{lower}}$ to $1.2T_{\text{upper}}$. For the purposes of this section, a good match is defined when the ordinates of the average (arithmetic mean) of the computed acceleration spectrum from the matched records in each direction does not fall above or below the target spectrum by more than 10% over the period range of interest.

C12.9.2.4 Application of Ground Acceleration Histories. One of the advantages of linear response history analysis is that analyses for gravity loads and for ground shaking may be computed separately and then combined in accordance with Section 12.4.2. Where linear response history analysis is performed in accordance with Section 12.9.2, it is required that each direction of response for each ground motion be computed independently. This requirement is based on the need to apply different scaling factors in the two orthogonal directions. Analyses with and without accidental torsion are required to be run for each ground motion. Thus, the total number of response histories that need to be computed is 18. (For each ground motion, one analysis is needed in each direction without mass eccentricity, and two analyses are needed in each direction to account for accidental torsion. These six cases times three ground motions give 18 required analyses.)

C12.9.2.5 Modification of Response for Design. The dynamic responses computed using spectrum-matched motions are elastic responses and must be modified for inelastic behavior.

For force-based quantities, the design base shear computed from the dynamic analysis must not be less than the base shear computed using the equivalent lateral force procedure. The factors η_X and η_Y, computed in Section 12.9.2.5.2, serve that purpose. Next, the force responses must be multiplied by I_e and divided by R. This modification, together with the application of the ELF scale factors, is accomplished in Section 12.9.2.5.3.

For displacement base quantities, it is not required to normalize to ELF, and computed response history quantities need be multiplied only by the appropriate C_d/R in the direction of interest. This step is accomplished in Section 12.9.2.5.4.

Whereas accidental torsion is not required for determining the maximum elastic base shear, which is used only for determining the required base shear scaling, it is required for all analyses that are used to determine design displacements and member forces.

C12.9.2.6 Enveloping of Force Response Quantities. Forces used in design are the envelope of forces computed from all analyses. Thus, for a brace, the maximum tension and the maximum compression forces are obtained. For a beam-column, envelope values of axial force and envelope values of bending moment are obtained, but these actions do not likely occur at the same time, and using these values in checking member capacity is not rational. The preferred approach is to record the histories of axial forces and bending moments, and to plot their traces together with the interaction diagram of the member. If all points of the force trace fall inside the interaction diagram, for all ground motions analyzed, the design is sufficient. An alternate is to record member demand to capacity ratio histories (also called usage ratio histories), and to base the design check on the envelope of these values.

C12.10 DIAPHRAGMS, CHORDS, AND COLLECTORS

This section permits choice of diaphragm design in accordance with either Sections 12.10.1 and 12.10.2 provisions or the new provisions of Section 12.10.3. Section 12.10.3 is mandatory for precast concrete diaphragms in buildings assigned to SDC C, D, E, or F and is optional for precast concrete diaphragms in SDC B buildings, cast-in-place concrete diaphragms, and wood diaphragms. The required mandatory use of Section 12.10.3 for precast diaphragm systems in SDC C through F buildings is based on recent research that indicates that improved earthquake performance can thus be attained. Many conventional diaphragm systems designed in accordance with Sections 12.10.1 and 12.10.2 have performed adequately. Continued use of Sections 12.10.1 and 12.10.2 is considered reasonable for diaphragm systems other than those for which Section 12.10.3 is mandated.

C12.10.1 Diaphragm Design. Diaphragms are generally treated as horizontal deep beams or trusses that distribute lateral forces to the vertical elements of the seismic force-resisting system. As deep beams, diaphragms must be designed to resist the resultant shear and bending stresses. Diaphragms are commonly compared to girders, with the roof or floor deck analogous to the girder web in resisting shear, and the boundary elements (chords) analogous to the flanges of the girder in resisting flexural tension and compression. As in girder design, the chord members (flanges) must be sufficiently connected to the body of the diaphragm (web) to prevent separation and to force the diaphragm to work as a single unit.

Diaphragms may be considered flexible, semirigid, or rigid. The flexibility or rigidity of the diaphragm determines how lateral forces are distributed to the vertical elements of the seismic force-resisting system (see Section C12.3.1). Once the distribution of lateral forces is determined, shear and moment diagrams are used to compute the diaphragm shear and chord forces. Where diaphragms are not flexible, inherent and accidental torsion must be considered in accordance with Section 12.8.4.

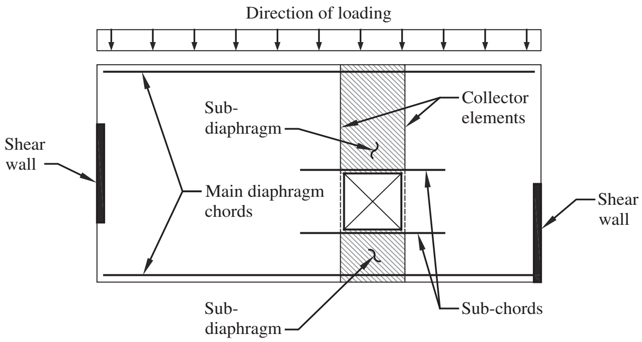

FIGURE C12.10-1 Diaphragm with an Opening

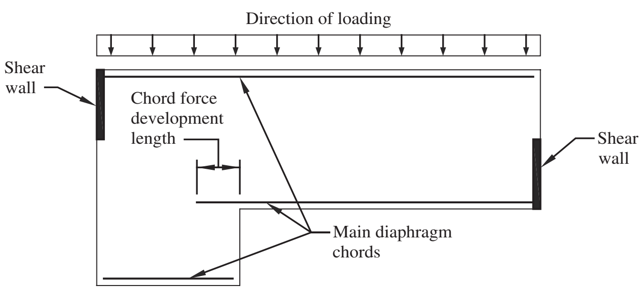

FIGURE C12.10-2 Diaphragm with a Reentrant Corner

Diaphragm openings may require additional localized reinforcement (subchords and collectors) to resist the subdiaphragm chord forces above and below the opening and to collect shear forces where the diaphragm depth is reduced (Fig. C12.10-1). Collectors on each side of the opening drag shear into the subdiaphragms above and below the opening. The subchord and collector reinforcement must extend far enough into the adjacent diaphragm to develop the axial force through shear transfer. The required development length is determined by dividing the axial force in the subchord by the shear capacity (in force/unit length) of the main diaphragm.

Chord reinforcement at reentrant corners must extend far enough into the main diaphragm to develop the chord force through shear transfer (Fig. C12.10-2). Continuity of the chord members also must be considered where the depth of the diaphragm is not constant.

In wood and metal deck diaphragm design, framing members are often used as continuity elements, serving as subchords and collector elements at discontinuities. These continuity members also are often used to transfer wall out-of-plane forces to the main diaphragm, where the diaphragm itself does not have the capacity to resist the anchorage force directly. For additional discussion, see Sections C12.11.2.2.3 and C12.11.2.2.4.

C12.10.1.1 Diaphragm Design Forces. Diaphragms must be designed to resist inertial forces, as specified in Eq. (12.10-1), and to transfer design seismic forces caused by horizontal offsets or changes in stiffness of the vertical resisting elements. Inertial forces are those seismic forces that originate at the specified diaphragm level, whereas the transfer forces originate above the specified diaphragm level. The redundancy factor, ρ, used for design of the seismic force-resisting elements also applies to diaphragm transfer forces, thus completing the load path.

C12.10.2.1 Collector Elements Requiring Load Combinations Including Overstrength for Seismic Design Categories C through F. The overstrength requirement of this section is intended to keep inelastic behavior in the ductile elements of the seismic force-resisting system (consistent with the response modification coefficient, R) rather than in collector elements.

C12.10.3 Alternative Design Provisions for Diaphragms, Including Chords and Collectors. The provisions of Section 12.10.3 are being mandated for precast concrete diaphragms in buildings assigned to SDC C, D, E, or F and are being offered as an alternative to those of Sections 12.10.1 and 12.10.2 for other precast concrete diaphragms, cast-in-place concrete diaphragms, and wood-sheathed diaphragms supported by wood framing. Diaphragms designed by Sections 12.10.1 and 12.10.2 have generally performed adequately in past earthquakes. The level of diaphragm design force from Sections 12.10.1 and 12.10.2 may not ensure, however, that diaphragms have sufficient strength and ductility to mobilize the inelastic behavior of vertical elements of the seismic force-resisting system. Analytical and experimental results show that actual diaphragm forces over much of the height of a structure during the design-level earthquake may be significantly greater than those from Sections 12.10.1 and 12.10.2, particularly when diaphragm response is near-elastic. There are material-specific factors that are related to overstrength and deformation capacity that may account for the adequate diaphragm performance in past earthquakes. The provisions of Section 12.10.3 consider both the significantly greater forces observed in near-elastic diaphragms and the anticipated overstrength and deformation capacity of diaphragms, resulting in an improved distribution of diaphragm strength over the height of buildings and among buildings with different types of seismic force-resisting systems.

Based on experimental and analytical data and observations of building performance in past earthquakes, changes are warranted to the procedures of Sections 12.10.1 and 12.10.2 for some types of diaphragms and for some locations within structures. Examples include the large diaphragms in some parking garages.

Section 12.10.3, Item 1, footnote *b* to Table 12.2-1 permits reduction in the value of Ω_0 for structures with flexible diaphragms. The lowered Ω_0 results in lower diaphragm forces, which is not consistent with experimental and analytical observations. Justification for footnote *b* is not apparent; therefore, to avoid the inconsistency, the reduction is eliminated when using the Section 12.10.3 design provisions.

Section 12.10.3, item 2: The ASCE 7-10 Section 12.3.3.4 provision requiring a 25% increase in design forces for certain diaphragm elements in buildings with several listed irregularities is eliminated when using the Section 12.10.3 design provisions because the diaphragm design force level in this section is based on realistic assessment of anticipated diaphragm behavior. Under the Sections 12.10.1 and 12.10.2 design provisions, the 25% increase is invariably superseded by the requirement to amplify seismic design forces for certain diaphragm elements by Ω_0;, the only exception is wood diaphragms, which are exempt from the Ω_0 multiplier.

Section 12.10.3, items 3 and 4: Section 12.10.3.2 provides realistic seismic design forces for diaphragms. Section 12.10.3.4 requires that diaphragm collectors be designed for 1.5 times the force level used for diaphragm in-plane shear and flexure. Based on these forces, the use of a ρ factor greater than one for collector design is not necessary and would overly penalize designs. The unit value of the redundancy factor is retained for diaphragms designed by the force level given in Sections 12.10.1 and 12.10.2. This value is reflected in the deletion of item 7 and the addition of diaphragms to item 5. For transfer diaphragms, see Section 12.10.3.3.

C12.10.3.1 Design. This provision is a rewrite of ASCE 7-10, Sections 12.10.1 and 12.10.2. The phrase "diaphragms including chords, collectors, and their connections to the vertical elements" is used consistently throughout the added or modified provisions, to emphasize that its provisions apply to all portions of a diaphragm. It is also emphasized that the diaphragm is to be designed for motions in two orthogonal directions.

C12.10.3.2 Seismic Design Forces for Diaphragms, Including Chords and Collectors. Eq. (12.10-4) makes the diaphragm seismic design force equal to the weight tributary to the diaphragm, w_{px}, times a diaphragm design acceleration coefficient, C_{px}, divided by a diaphragm design force reduction factor, R_s, which is material-dependent and whose background is given in Section C12.10.3.5. The background to the diaphragm design acceleration coefficient, C_{px}, is given below.

The diaphragm design acceleration coefficient at any height of the building can be determined from linear interpolation, as indicated in Fig. 12.10-2.

The diaphragm design acceleration coefficient at the building base, C_{p0}, equals the peak ground acceleration consistent with the design response spectrum in ASCE 7-10, Section 11.4.5, times the Importance Factor I_e. Note that the term $0.4S_{DS}$ can be calculated from Eq. (11.4-5) by making $T = 0$.

The diaphragm design acceleration coefficient at 80% of the structural height, C_{pi}, given by Eqs. (12.10-8) and (12.10-9), reflects the observation that at about this height, floor accelerations are largely, but not solely, contributed by the first mode of response. In an attempt to provide a simple design equation, coefficient C_{pi} was formulated as a function of the design base shear coefficient, C_s, of ASCE 7-10, which may be determined from equivalent static analysis or modal response spectrum analysis of the structure. Note that C_s includes a reduction by the response modification factor, R, of the seismic force-resisting system. It is magnified back up by the overstrength factor, Ω_0, of the seismic force-resisting system because overstrength will generate higher first-mode forces in the diaphragm. In many lateral systems, at 80% of the building height, the contribution of the second mode is negligible during linear response, and during nonlinear response it is typically small, though nonnegligible. In recognition of this observation, the diaphragm seismic design coefficient at this height has been made a function of the first mode of response only, and the contribution of this mode has been factored by $0.9\Gamma_{m1}$ as a weighed value between contributions at the first-mode effective height (approximately 2/3 of the building height) and the building height.

Systems that make use of high R-factors, such as buckling, restrained braced frames (BRBFs) and moment-resisting frames (MRFs), show that in the lower floors the higher modes add to the accelerations, whereas the contribution of the first mode is minimal. For this reason, the coefficient C_{pi} needs to have a lower bound. A limit of C_{p0} has been chosen; it makes the lower floor acceleration coefficients independent of R. Wall systems are unlikely to be affected by this lower limit on C_{pi}.

At the structural height, h_n, the diaphragm design acceleration coefficient, C_{pn}, given by Eq. (12.10-7), reflects the influence of the first mode, amplified by system overstrength, and of the higher modes without amplification on the floor acceleration at this height. The individual terms are combined using the square root of the sum of the squares. The overstrength amplification of the first mode recognizes that the occurrence of an inelastic mechanism in the first mode is an anticipated event under the design earthquake, whereas inelastic mechanisms caused by higher mode behavior are not anticipated. The higher mode seismic response coefficient, C_{s2}, is computed as the smallest of the values given by Eqs. (12.10-10), (12.10-11), and (12.10-12a) or (12.10-12b).

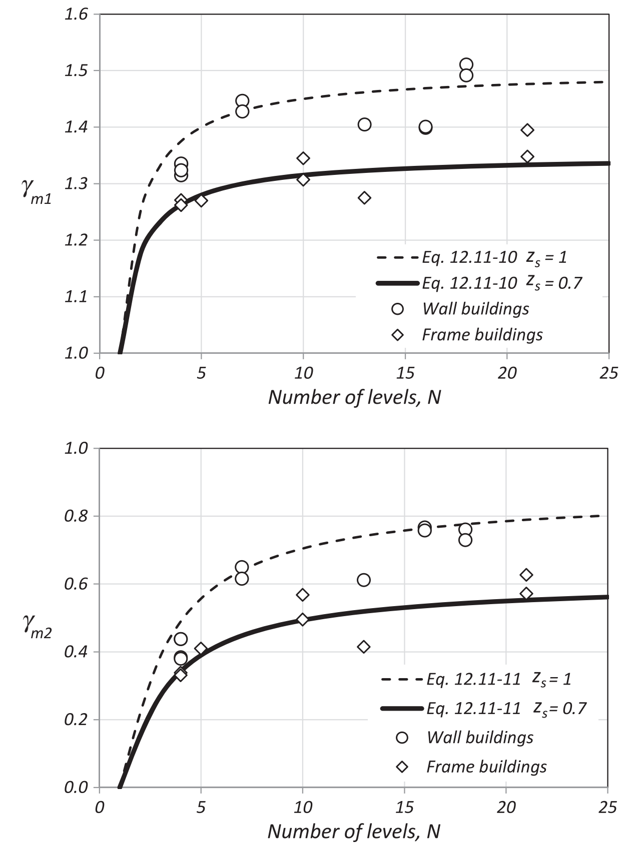

FIGURE C12.10-3 Comparison of Factors Γ_{m1} and Γ_{m2} Obtained from Analytical Models and Actual Structures with Those Predicted by Eqs. (12.10-13) and (12.10-14)

These four equations consider that the periods of the higher modes contributing to the floor acceleration can lie on the ascending, constant, or first descending branch of the design response spectrum shown in ASCE 7-10, Fig. 11.4-1. Users are warned against extracting higher modes from their modal analysis of buildings and using them in lieu of the procedure presented in Section 12.10.3.2.1 because the higher mode contribution to floor accelerations can come from a number of modes, particularly when there is lateral-torsional coupling of the modes.

Note that Eq. (12.10-7) makes use of the modal contribution factor defined here as the mode shape ordinate at the building height times the modal participation factor and is uniquely defined for each mode of response (Chopra 1995). A building database was compiled to obtain approximate equations for the first mode and higher mode contribution factors. The first and second translational modes, as understood in the context of two-dimensional modal analysis, were extracted from the mode shapes obtained from three-dimensional modal analysis by considering modal ordinates at the center of mass. These buildings had diverse lateral systems, and the number of stories ranged from 3 to 23. Eqs. (12.10-13) and (12.10-14) were empirically calibrated from simple two-dimensional models of realistic frame-type and wall-type buildings and then compared with data extracted from the database (Fig. C12.10-3). In Eq. (12.10-7), C_{pn} is required to be no less than C_{pi}, based on judgment, in order to eliminate instances where the design acceleration at roof level might be lower than that at $0.8h_n$. This cap will particularly affect low-z_s systems such as BRBFs.

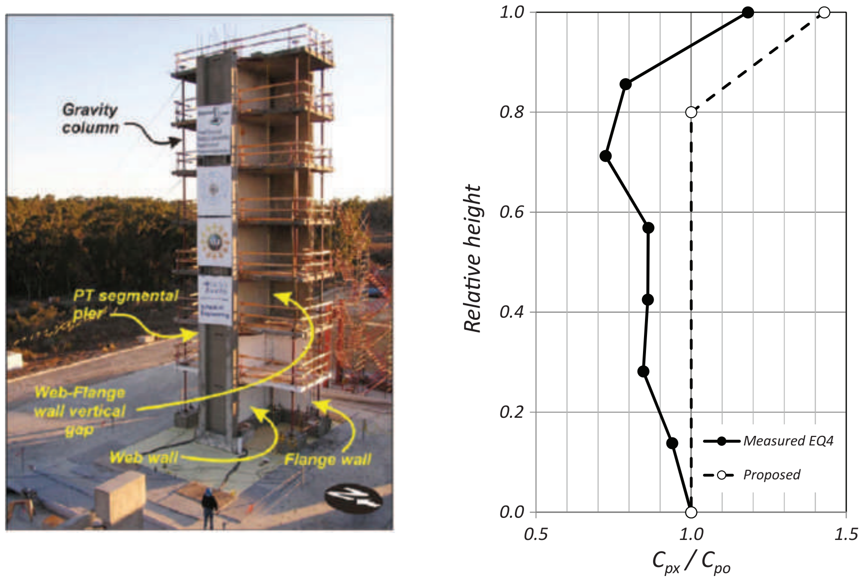

FIGURE C12.10-4 Comparison of Measured Floor Accelerations and Accelerations Predicted by Eq. (12.10-4) for a Seven-Story Bearing Wall Building

Source: Panagiotou et al. 2011.

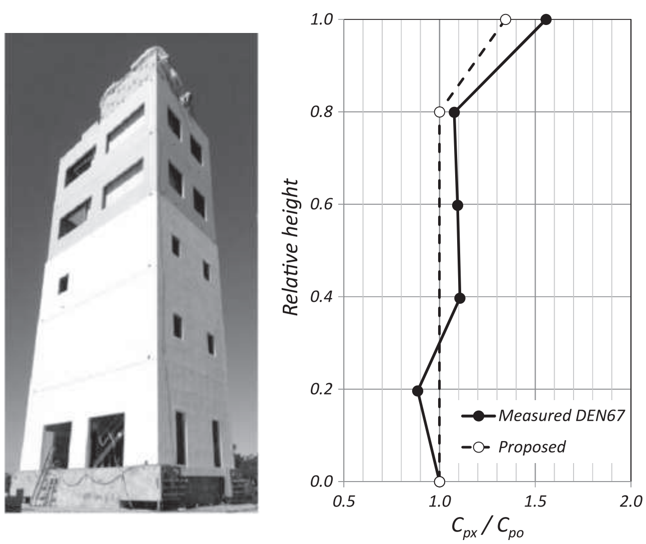

FIGURE C12.10-5 Comparison of Measured Floor Accelerations and Accelerations Predicted by Eq. (12.10-4) for a Five-Story Special Moment-Resisting Frame Building

Source: (left) Courtesy of Michelle Chen; (right) Adapted from Chen et al. (2015).

To validate Eq. (12.10-4), coefficients C_{px} were calculated for various buildings tested on a shake table. Figs. C12.10-4 and C12.10-5 plot the floor acceleration envelopes and the floor accelerations predicted from Eq. (12.10-4) with $R_s = 1$ for two buildings built at full scale and tested on a shake table (Panagiotou et al. 2011, Chen et al. 2015), with C_{p0} defined as the diaphragm design acceleration coefficient at the structure base and C_{px} defined as the diaphragm design acceleration coefficient at level x. Measured floor accelerations are reasonably predicted by Eq. (12.10-4). Research work by Choi et al. (2008) concluded that buckling-restrained braced frames are very effective in limiting floor accelerations in buildings arising from higher mode effects. This finding is reflected in this proposal, where the mode shape factor z_s has been made the smallest for buckling-restrained braced frame systems. Fig. C12.10-6 compares average floor accelerations obtained from the nonlinear time history analyses of four buildings (two steel buckling-restrained braced frame systems and two steel special moment frame systems) when subjected to an ensemble of spectrum-compatible earthquakes with floor accelerations computed from Eqs. (12.10-4) and (12.10-5). The proposed design equations predict the accelerations in the uppermost part of the building and in the lowest levels reasonably well.

The significant difference between a low-z_s system such as the BRBF and a high-z_s system such as a bearing wall system is that inelastic deformations are distributed throughout the height of the structure in a low-z_s system, whereas they are concentrated at the base of the structure in a high-z_s system. If rational analysis can be performed to demonstrate that inelastic deformations are in fact distributed along the height of the structure, as is often the case with eccentrically braced frame or coupled shear wall systems, then the use of a low z_s value, as has been assigned to the BRBF for such a system, would be justified.

During the calibration of the design procedure leading to Eq. (12.10-4), it was found that at intermediate levels in lateral systems designed using large response modification coefficients, diaphragm design forces given by this equation could be rather low. There was consensus within the BSSC PUC Issue Team that developed Section 12.10.3 that diaphragm design forces should not be taken as less than the minimum force currently prescribed by ASCE 7-10, and hence they developed Eq. (12.10-5).

The procedure presented in Section 12.10.3 is based on consideration of buildings and structures whose mass distribution is reasonably uniform along the building height. Buildings or structures with tapered mass distribution along their height or with setbacks in their upper levels may experience diaphragm forces in the upper levels that are greater than those derived from Eq. (12.10-4). In such buildings and structures, it is preferable to define an effective building height, h_{ne}, and a corresponding level, n_e, the level to which the structural effective height is measured. The effective number of levels in a building, n_e, is defined as level x where the ratio $\sum_{i=1}^{x} w_i / \sum_{i=1}^{n} w_i$ first exceeds 0.95. Level 1 is defined as the first level above the base. The effective structural

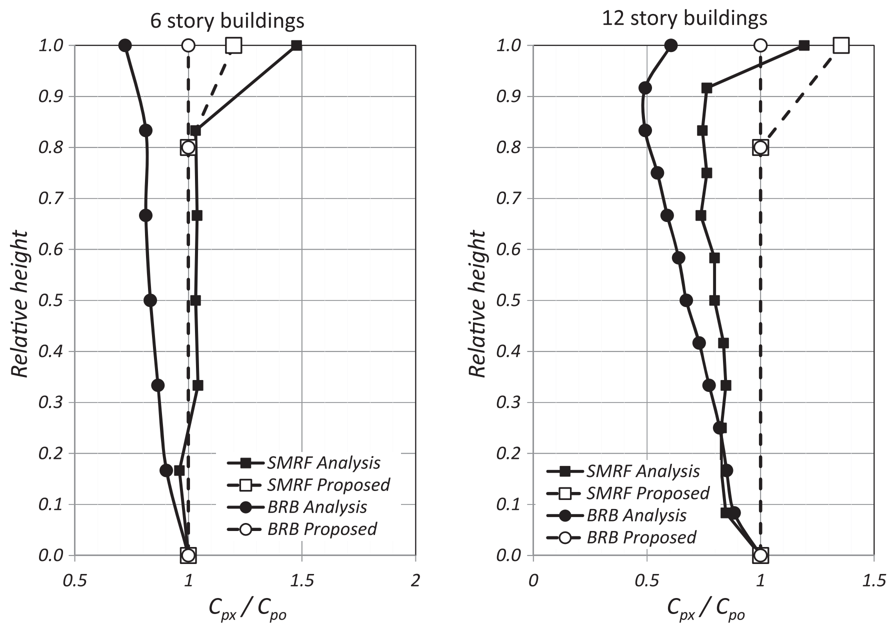

FIGURE C12.10-6 Comparison of Measured Floor Accelerations with Proposed Eqs. (12.10-4) and (12.10-5) for Steel Buckling-Restrained Braced Frame and Special Moment-Resisting Frame Buildings

Source: Adapted from Choi et al. 2008

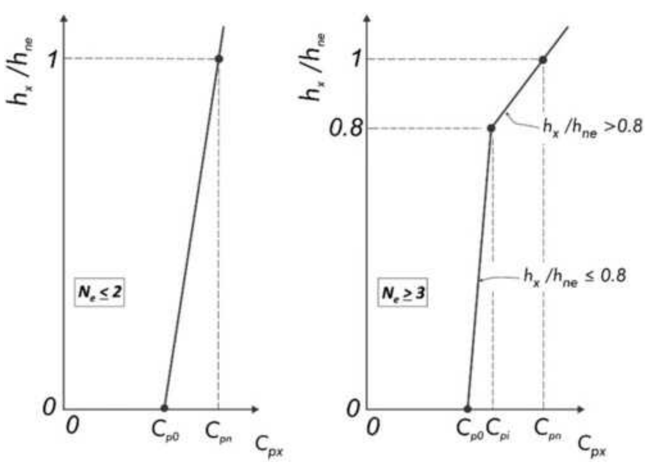

FIGURE C12.10-7 Diaphragm Design Acceleration Coefficient C_{px} for Buildings with Nonuniform Mass Distribution

height, h_{ne}, is the height of the building measured from the base to level n_e. In buildings with tapered mass distribution or setbacks, the diaphragm design acceleration coefficient, C_{pn}, is calculated by interpolation and extrapolation, as shown in Fig. C12.10-7, with n replaced by n_e in Eqs. (12.10-10) through (12.10-14).

C12.10.3.3 Transfer Forces in Diaphragms. All diaphragms are subject to inertial forces caused by the weight tributary to the diaphragm. Where the relative lateral stiffnesses of vertical seismic force-resisting elements vary from story to story, or the vertical seismic force-resisting elements have out-of-plane offsets, lateral forces in the vertical elements need to be transferred through the diaphragms as part of the load path between vertical elements above and below the diaphragm. These transfer forces are in addition to the inertial forces and can at times be quite large.

For structures that have a horizontal structural irregularity of Type 4 in Table 12.3-1, the magnitude of the transfer forces is largely dependent upon the overstrength in the offset vertical elements of the seismic force-resisting system. Therefore, the transfer force caused by the offset is required to be amplified by the overstrength factor, Ω_0, of the seismic force-resisting system. The amplified transfer force is to be added to the inertial force for the design of this portion of the diaphragm.

Transfer forces can develop in many other diaphragms, even within regular buildings; the design of diaphragms with such transfer forces can be for the sum of the transfer forces, unamplified, and the inertial forces.

C12.10.3.4 Collectors—Seismic Design Categories C through F. For structures in Seismic Design Categories C through F, ASCE 7-10, Section 12.10.2.1 specifies the use of forces including the overstrength factor, Ω_0, for design of diaphragm collectors and their connections to vertical elements of the seismic force-resisting system. The intent of this requirement is to increase collector forces in order to help ensure that collectors will not be the weak links in the seismic force-resisting system.

In this section the collector force is instead differentiated by using a multiplier of 1.5. This is a smaller multiplier than has been used in the past, but it is justified because the diaphragm forces are more accurately determined by Eq. (12.10-4). For collector elements of diaphragms that carry transfer forces caused by out-of-plane offsets of the vertical elements of the seismic force-resisting system, only the inertia force is amplified by 1.5; the transfer forces, already amplified by Ω_0, are not further amplified by 1.5.

Some seismic force-resisting systems, such as braced frames and moment frames, have a fairly well defined lateral strength corresponding to a well-defined yield mechanism. When collectors deliver seismic forces to such systems, it is not sensible to have to design the collectors for forces higher than those corresponding to the lateral strength of the supporting elements in the story below. This is why the cap on collector design forces is included. The lateral strength of a braced frame or moment frame may be calculated using the same methods as are used for determining whether a weak-story irregularity is present (Table 12.3-2). It should be noted that only the moment frames or braced frames below the collector are to be considered in calculating the upper-bound collector design force. The shear strength of the gravity columns and the lateral strength of the frames above are not included.

The design forces in diaphragms that deliver forces to collectors can also be limited by the maximum forces that can be generated in those collectors by the moment frames or braced frames below.

C12.10.3.5 Diaphragm Design Force Reduction Factor. Despite the fact that analytical and shake table studies indicate higher diaphragm accelerations than currently used in diaphragm design, many commonly used diaphragm systems, including diaphragms designed under a number of U.S. building codes and editions, have a history of excellent earthquake performance. With limited exceptions, diaphragms have not been reported to have performed below the life-safety intent of building code seismic design provisions in past earthquakes. Based on this history, it is felt that, for many diaphragm systems, no broad revision is required to the balance between demand and capacity used for design of diaphragms under current ASCE 7 provisions. In view of this observation, it was recognized that the analytical studies and diaphragm testing from which the higher accelerations and design forces were being estimated used diaphragms that were elastic or near-elastic in their response. Commonly used diaphragm systems are recognized to have a wide range of overstrength and inelastic displacement capacity (ductility). It was recognized that the effect of the varying diaphragm systems on seismic demand required evaluation and incorporation into the proposed diaphragm design forces. Eq. (12.10-4) incorporates the diaphragm overstrength and inelastic displacement capacity through the use of the diaphragm force reduction factor, R_s. This factor is most directly based on the global ductility capacity of the diaphragm system; however, the derivation of the global ductility capacity inherently also captures the effect of diaphragm overstrength.

For diaphragm systems with inelastic deformation capacity sufficient to permit inelastic response under the design earthquake, the diaphragm design force reduction factor, R_s, is typically greater than 1.0, so that the design force demand, F_{px}, is reduced relative to the force demand for a diaphragm that remains linear elastic under the design earthquake. For diaphragm systems that do not have sufficient inelastic deformation capacity, R_s should be less than 1.0, or even 0.7, so that linear-elastic force-deformation response can be expected under the risk-targeted maximum considered earthquake (MCE_R).

Diaphragms with R_s values greater than 1.0 shall have the following characteristics: (1) a well-defined, specified yield mechanism, (2) global ductility capacity for the specified yield mechanism, which exceeds anticipated ductility demand for the risk-targeted maximum considered earthquake, and (3) sufficient local ductility capacity to provide for the intended global ductility capacity, considering that the specified yield mechanism may require concentrated local inelastic deformations to occur. The following discussion addresses these characteristics and the development of R_s-factors in detail.

A diaphragm system with an R_s value greater than 1.0 should have a specified, well-defined yield mechanism, for which both the global strength and the global deformation capacity can be estimated. For some diaphragm systems, a shear-yield mechanism may be appropriate, whereas for other diaphragm systems, a flexural-yield mechanism may be appropriate.

Fig. C12.10-8(a) shows schematically the force-deformation ($F_{dia} - \Delta_{dia}$) response of a diaphragm with significant inelastic deformation capacity. The figure illustrates the response of a diaphragm system, such as a wood diaphragm or a steel deck diaphragm, which is not expected to exhibit a distinct yield point, so that an effective yield point ($F_{Y\text{-eff}}$ and $\Delta_{Y\text{-eff}}$) needs to be defined. For wood diaphragms and steel deck diaphragms, the figure illustrates one way to define the effective yield point. The stiffness of a test specimen is defined by the secant stiffness through a point corresponding to 40% of the peak strength (F_{peak}). The effective yield point ($F_{Y\text{-eff}}$ and $\Delta_{Y\text{-eff}}$) for a diaphragm is defined by the secant stiffness through $0.4F_{peak}$ and the nominal diaphragm strength reduced by a strength reduction factor (ϕF_n), as shown in the figure. The $F_{dia} - \Delta_{dia}$ response is then idealized with a bilinear model, using the effective yield point ($F_{Y\text{-eff}}$ and $\Delta_{Y\text{-eff}}$) and F_{peak} and the corresponding deformation Δ_{peak} as shown in the figure.

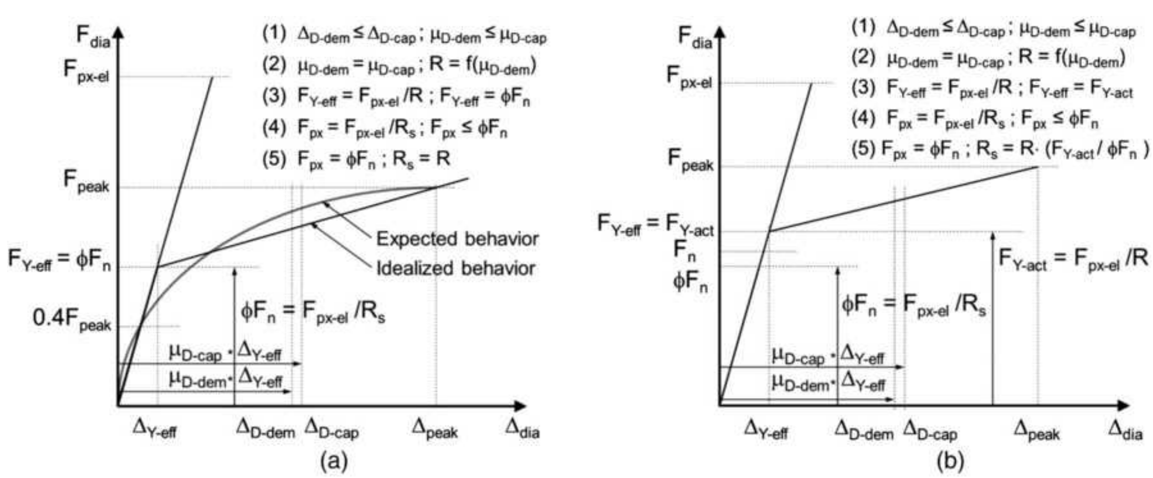

FIGURE C12.10-8. Diaphragm Inelastic Response Models for (a) a Diaphragm System That Is Not Expected to Exhibit a Distinct Yield Point and (b) a Diaphragm System That Does Exhibit a Distinct Yield Point

Fig. C12.10-8(b) shows schematically the force-deformation ($F_{dia} - \Delta_{dia}$) response of a diaphragm with significant inelastic deformation capacity, which is expected to have nearly linear $F_{dia} - \Delta_{dia}$ response up to a distinct yield point, such as a cast-in-place reinforced concrete diaphragm. For this type of diaphragm system, the effective yield point can be taken as the actual yield point ($F_{Y\text{-actual}}$ and $\Delta_{Y\text{-actual}}$) of the diaphragm (accounting for diaphragm material overstrength and not including a strength reduction factor (ϕ)).

The global (or system) deformation capacity of a diaphragm system (Δ_{cap}) should be estimated from analyses of test data. The force-deformation ($F_{dia} - \Delta_{dia}$) response shown schematically in Figs. C12.10-8(a) and C12.10-8(b) is the global force-deformation behavior.

In some cases, tests provide directly the global deformation capacity, but more often, tests provide only the local response, including the strength and deformation capacity, of diaphragm components and connections. When tests provide only the local deformation capacity, analyses of typical diaphragms should be made to estimate the global deformation capacity of these diaphragms. These analyses should consider: (1) the specified yield mechanism, (2) the local force-deformation response data from tests, (3) the typical distributions of design strength and internal force demands across the diaphragm, and (4) any other factors that may require concentrated local inelastic deformation to occur when the intended yield mechanism forms.

After the global force-deformation ($F_{dia} - \Delta_{dia}$) response of a diaphragm has been estimated, the global deformation capacity (Δ_{cap}) can be determined. In Fig. C12.10-8(a), for example, Δ_{cap} can be taken as Δ_{peak}, which is the deformation corresponding to the strength (F_{peak}). For some diaphragm systems, it may be acceptable to take the deformation corresponding to 80% of F_{peak} (i.e., postpeak) as Δ_{cap}.

Only a selected portion of the deformation capacity of a diaphragm (Δ_{cap}) should be used under the design earthquake in recognition of two major concerns: (1) the diaphragm must perform adequately under the MCE_R, which has a design spectrum 50% more intense than the design earthquake design spectrum, and (2) significant inelastic deformation under the design earthquake may result in undesirable damage to the diaphragm. As a rough estimate, the diaphragm deformation capacity under the design earthquake ($\Delta_{D\text{-cap}}$) should be limited to approximately one-half to two-thirds of the deformation capacity Δ_{cap}.

To develop the diaphragm force reduction factor, R_s, the diaphragm global deformation capacity should be expressed as a global ductility capacity (μ_{cap}), which equals the deformation capacity (Δ_{cap}) divided by the effective yield deformation ($\Delta_{Y\text{-eff}}$). The corresponding diaphragm design ductility capacity ($\mu_{D\text{-cap}}$) equals $\Delta_{D\text{-cap}}/\Delta_{Y\text{-eff}}$.

From the diaphragm global deformation capacity and corresponding ductility capacity, an appropriate R_s factor can be estimated. Use of the estimated R_s factor in design should result in diaphragm ductility demands that do not exceed the ductility capacity that was used to estimate R_s. The force reduction factor is ideally derived from system-specific studies. Where such studies are unavailable, however, some guidance on the conversion from global ductility to force reduction is available from past studies.

Expressions that provide the force reduction factor, R, for the seismic force-resisting system of a building corresponding to an expected ductility demand (μ_{dem}) have been proposed by numerous research teams. Numerous factors, including vibration period, inherent damping, deformation hardening (stiffness after the effective yield point), and hysteretic energy dissipation under cyclic loading, have been considered in developing these expressions. Two such expressions, which are based on elastoplastic force-deformation response under cyclic loading (Newmark and Hall 1982), are as follows: (1) $R = (2\mu_{dem} - 1)^{0.5}$, applicable to short-period systems, and (2) $R = \mu_{dem}$, applicable to systems with longer periods. The first function, known as the equal energy rule, gives a smaller value of R for a given value of μ_{dem}; the second function, known as the equal displacement rule, is also widely used.

Figs. C12.10-8(a) and C12.10-8(b) summarize an approach to estimating R_s as follows:

1. For the selected value of R_s, the diaphragm deformation demand under the design earthquake ($\Delta_{D\text{-dem}}$) should not exceed the diaphragm design deformation capacity ($\Delta_{D\text{-cap}}$). This design constraint, expressed in terms of diaphragm ductility, requires that the diaphragm ductility demand under the design earthquake ($\mu_{D\text{-dem}}$) should not exceed the diaphragm design ductility capacity ($\mu_{D\text{-cap}}$).
2. The largest value of R that can be justified for a given diaphragm design deformation capacity is obtained by setting the ductility demand ($\mu_{D\text{-dem}}$) equal to the design ductility capacity ($\mu_{D\text{-cap}}$) and determining R from a function that provides R for a given μ_{dem}. For example, if $\mu_{D\text{-cap}} = 2.5$, then $\mu_{D\text{-dem}}$ is set equal to 2.5 and the corresponding $R = 2$ from the equal energy rule or $R = 2.5$ from the equal displacement rule.
3. R from step (2) is the ratio of the force demand for a linear elastic diaphragm ($F_{px\text{-el}}$) to the effective yield strength of the diaphragm ($F_{Y\text{-eff}}$). For a diaphragm system that is not expected to exhibit a distinct yield point (Fig. C12.10-8a), $F_{Y\text{-eff}}$ equals the factored nominal diaphragm strength (ϕF_n). For a diaphragm system that is expected to exhibit a distinct yield point (Fig. C12.10-8b), $F_{Y\text{-eff}}$ equals the actual yield strength ($F_{Y\text{-actual}}$), accounting for diaphragm material overstrength and not including the strength reduction factor (ϕ).
4. R_s is, however, defined as the ratio of the force demand for a linear elastic diaphragm ($F_{px\text{-el}}$) to the design force demand (F_{px}). The diaphragm must be designed such that the design force demand (F_{px}) is less than or equal to the factored nominal diaphragm strength (ϕF_n).
5. For a diaphragm system without a distinct yield point (Fig. C12.10-8(a)) that has the minimum strength ($F_{px} = \phi F_n$), R_s equals R from step (2). For a diaphragm system with a distinct yield point (Fig. C12.10-8(b)), which has the minimum strength ($F_{px} = \phi F_n$), R_s equals R from step (2) multiplied by the ratio $F_{Y\text{-eff}}/\phi F_n$.

Diaphragm force reduction factors, R_s, have been developed for some commonly used diaphragm systems. The derivation of factors for each of these systems is explained in detail in the following commentary sections. For each, the specific design standard considered in the development of the R_s factor is specified. The resulting R_s factors are specifically tied to the design and detailing requirements of the noted standard because these play a significant role in setting the ductility and overstrength of the diaphragm system. For this reason, the applicability of the R_s factor to diaphragms designed using other standards must be specifically considered and justified.

Cast-in-Place Concrete Diaphragms. The R_s values in Table 12.10-1 address cast-in-place concrete diaphragms designed in accordance with ACI 318.

Intended Mechanism. Flexural yielding is the intended yield mechanism for a reinforced concrete diaphragm. Where this can be achieved, designation as a flexure-controlled diaphragm and use of the corresponding R_s factor in Table 12.10-1 is appropriate. There are many circumstances, however, where the development of a well-defined yielding mechanism is not possible because of diaphragm geometry (aspect ratio or complex diaphragm configuration), in which case, designation as a shear-controlled diaphragm and use of the lower R_s factor is required.

Derivation of Diaphragm Force Reduction Factor. Test results for reinforced concrete diaphragms are not available in the literature. Test results for shear walls under cyclic lateral loading were considered. The critical regions of shear wall test specimens usually have high levels of shear force, moment, and flexural deformation demands; high levels of shear force are known to degrade the flexural ductility capacity. The flexural ductility capacity of shear wall test specimens under cyclic lateral loading was used to estimate the flexural ductility capacity of reinforced concrete diaphragms, using the previously described method based on Newmark and Hall (1982).

Based on shear wall test results, the estimated global flexural ductility capacity of a reinforced concrete diaphragm is 3, based on the actual yield displacement ($\Delta_{Y\text{-actual}}$) of the test specimens. The design ductility capacity is taken as 2/3 of the ductility capacity; the design ductility capacity ($\mu_{D\text{-cap}}$) is 2.

Setting the ductility demand (μ_{dem}) equal to the design ductility capacity ($\mu_{D\text{-cap}}$) and using the equal energy rule, the force reduction factor R is $R = (2\mu_{\text{dem}} - 1)^{0.5} = 1.73$.

R_s equals R multiplied by the ratio $F_{Y\text{-eff}}/\phi F_n$. $F_{Y\text{-eff}}$ is taken equal to $F_{Y\text{-actual}}$, which is assumed to be $1.1F_n$ and ϕ equals 0.9. Therefore, $R_s = 2.11$, which is rounded to 2.

Because of the geometric characteristics of a building or other factors, such as minimum reinforcement requirements, it is not possible to design some reinforced concrete diaphragms to yield in flexure. Such diaphragms are termed "shear controlled" to indicate that they are expected to yield in shear. Shear-controlled reinforced concrete diaphragms should be designed to remain essentially elastic under the design earthquake, with their available global ductility held in reserve for safety under the MCE_R.

Based on the following considerations, R_s is specified as 1.5 for shear-controlled reinforced concrete diaphragms: Reinforced-concrete diaphragms have performed well in past earthquakes. ACI-318 specifies ϕ of 0.75 or 0.6 for diaphragm shear strength and limits the concrete contribution to the shear strength to only $2(f_c')^{0.5}$. In addition, reinforced concrete floor slabs often have gravity load reinforcement that is not considered in determining the diaphragm shear strength. Therefore, shear-controlled reinforced concrete diaphragms are expected to have significant overstrength. The ratio $F_{Y\text{-eff}}/\phi F_n$ for a reinforced concrete diaphragm, where $F_{Y\text{-eff}}$ is taken equal to $F_{Y\text{-actual}}$, is expected to exceed 1.5, which is the rationale for $R_s = 1.5$, even though μ_{dem} is assumed to be 1 for the design earthquake.

Table C12.10-1. Diaphragm Design Performance Targets

	Flexure		Shear
Options	DE	MCE_R	DE and MCE_R
EDO	Elastic	Elastic	Elastic
BDO	Elastic	Inelastic	Elastic
RDO	Inelastic	Inelastic	Elastic

Note: DE, design earthquake, MCE_R, risk-targeted maximum considered earthquake, EDO, elastic design objective; BDO, basic design objective; and RDO, reduced design objective.

Precast Concrete Diaphragms. The R_s values in Table 12.10-1 address precast concrete diaphragms designed in accordance with ACI 318.

Derivation of Diaphragm Force Reduction Factors. The diaphragm force reduction factors, R_s, in Table 12.10-1 for precast concrete diaphragms were established based on the results of analytical earthquake simulation studies conducted within a multiple-university project: Diaphragm Seismic Design Methodology (DSDM) for Precast Concrete Diaphragms (Fleischman et al. 2013). In this research effort, diaphragm design force levels have been aligned with the diaphragm deformation capacities specifically for precast concrete diaphragms. Three different design options were proposed according to different design performance targets, as indicated in Table C12.10-1. The relationships between diaphragm design force levels and diaphragm local/global ductility demands have been established in the DSDM research project. These relationships have been used to derive the R_s for precast concrete diaphragms in Table 12.10-1.

Diaphragm R_{dia}-μ_{global}-μ_{local} Relationships. Extensive analytical studies have been performed (Fleischman et al. 2013) to develop the relationship of R_{dia}-μ_{global}-μ_{local}. R_{dia} is the diaphragm force reduction factor (similar to the R_s in Table 12.10-1) measured from the required elastic diaphragm design force at MCE_R level. μ_{global} is the diaphragm global ductility demand, and μ_{local} is the diaphragm local connector ductility demand measured at MCE level. Fig. C12.10-9 shows the μ_{global}-μ_{local} and R_{dia}-μ_{global} analytical results for different diaphragm aspect ratios (AR) and proposed linear equations derived from the data.

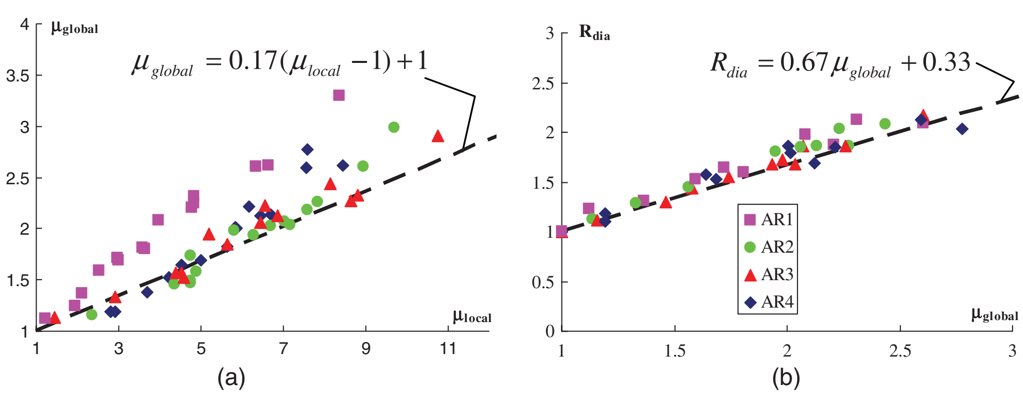

FIGURE C12.10-9. Relationships: (a) μ_{global}-μ_{local} and (b) R_{dia}-μ_{global}

Table C12.10-2 Diaphragm Force Reduction Factors, R_s

Options	Diaphragm Connector Category	δ_{local} (in.)	μ_{local}	μ_{global}	$\mu_{global,red}$	R_s
EDO	LDE	0.06	1.0	1.0	0.58	0.7
BDO	MDE	0.2	3.5	1.4	1.0	1.0
RDO	HDE	0.4	7.0	2.0	1.6	1.4

Note: EDO, elastic design objective; LDE, low deformability elements; BDO, basic design objective; MDE, moderate deformability elements; RDO reduced design objective, and HDE, high deformability elements.

Diaphragm Force Reduction Factor (R_s). Using the equations in Fig. C12.10-9, the R_s can be calculated for different diaphragm design options provided that the diaphragm local reinforcement ductility capacity is known. In the DSDM research, precast diaphragm connectors have been extensively tested (Fleischman et al. 2013) and have been qualified into three categories: high deformability elements (HDEs), moderate deformability elements (MDEs), and low deformability elements (LDEs), which are required as a minimum for designs using the reduced design objective (RDO), the basic design objective (BDO), and the elastic design objective (EDO), respectively. The local deformation and ductility capacities for diaphragm connector categories are shown in Table C12.10-2. Considering that the proposed diaphragm design force level [Eq. (12.10-1)] targets elastic diaphragm response at the design earthquake, which is equivalent to design using BDO where $\mu_{local} = 3.5$ at MCE_R (see Table C12.10-2), the available diaphragm global ductility capacity has to be reduced from Fig. C12.10-9(a), acknowledging more severe demands in the MCE_R,

$$\mu_{global,red} = 0.17(\mu_{local} - 3.5) + 1 \qquad \text{(C12.10-1)}$$

Accordingly, the R_s-factor can be modified from Fig. C12.10-9(b) (see Table C12.10-2):

$$R_s = 0.67\mu_{global,red} + 0.33 \qquad \text{(C12.10-2)}$$

Diaphragm Shear Overstrength Factor. Precast diaphragms typically exhibit ductile flexural response but brittle shear response. In order to avoid brittle shear failure, elastic shear response targets are required for both flexure-controlled and shear-controlled systems at design earthquake and MCE_R levels. Thus, a shear overstrength factor, Ω_v, is required for diaphragm shear design. For EDO design, since the diaphragm is expected to remain elastic under the MCE_R, no shear overstrength is needed. Fig. C12.10-10 shows the analytical results for required shear overstrength factors for BDO and RDO (shown as marks). A simplified conservative equation is proposed as (see black lines in Fig. C12.10-10):

$$\Omega_v = 1.4R_s \qquad \text{(C12.10-3)}$$

Wood-Sheathed Diaphragms. The R_s values given in Table 12.10-1 are for wood-sheathed diaphragms designed in accordance with *Special Design Provisions for Wind and Seismic* (AWC 2008).

Intended Mechanism. Wood-sheathed diaphragms are shear-controlled, with design strength determined in accordance with AWC (2008) and the shear behavior based on the sheathing-to-framing connections. Wood diaphragm chord members are unlikely to form flexural mechanisms (ductile or otherwise) because of the overstrength inherent in axially loaded members designed in accordance with applicable standards.

Derivation of Diaphragm Design Force Reduction Factor. An R_s factor of 3 is assigned in Table 12.10-1, based on diaphragm test data (APA 1966, 2000, DFPA 1954, 1963) and analytical studies. The available testing includes diaphragm spans (loaded as simple-span beams) ranging from 24 to 48 ft (7.3 to 14.6 m), aspect ratios ranging between 1 and 3.3, and diaphragm construction covering a range of construction types including blocked and unblocked construction, and regular and high-load diaphragms. The loading was applied with a series of point loads at varying spacing; however, the loading was reasonably close to uniform. Whereas available diaphragm testing was monotonic, based on shear wall loading protocol studies (Gatto and Uang 2002), it is believed that the monotonic load-deflection behavior is reasonably representative of the cyclic load-deflection envelope, suggesting that it is appropriate to use monotonic load-deflection behavior in the estimation of overstrength, ductility, and displacement capacity.

Analytical studies using nonlinear response history analysis evaluated the relationship between global ductility and diaphragm force reduction factor for a model wood building. The analysis identified the resulting diaphragm force reduction factor as ranging from just below 3 to significantly in excess of 5. A force reduction factor of 3 was selected so that diaphragm design force levels would generally not be less than determined in accordance with provisions of Section 12.10.

The calibration approach for selection of R_s of 3 was considered appropriate to limit conditions where diaphragm force levels would drop below those determined in accordance with

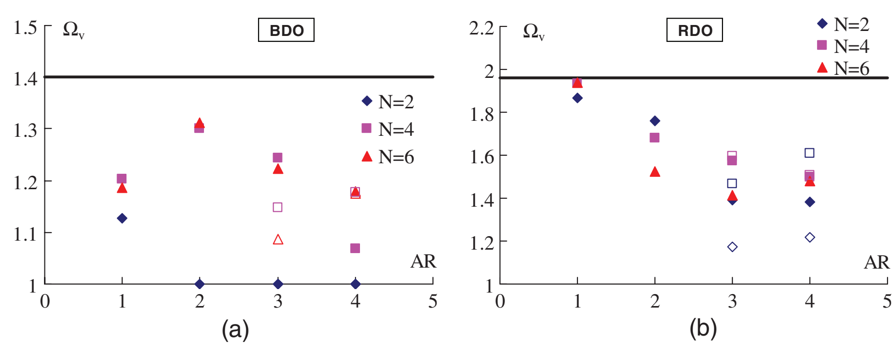

FIGURE C12.10-10 Diaphragm Shear Overstrength Factor, Ω_v vs. Aspect Ratio, AR, for Different Numbers of Stories, *N*: (a) BDO; (b) RDO
Source: Fleischman et al. 2013.

Section 12.10. This was due in part to historical experience of good diaphragm performance across a range of wood diaphragm types, even though test data showed varying levels of ductility and deformation capacity. Tests of nailed wood diaphragms showed significant but varying levels of overstrength. It is recognized that even further variation of overstrength will result from

- Presence of floor coverings or toppings and their attachment or bond to diaphragm sheathing,
- Presence of wall to floor framing nailing through diaphragm sheathing, and
- Presence of adhesives in combination with required sheathing nailing (commonly used for purposes of mitigating floor vibration, increasing floor stiffness for gravity loading, and reducing the potential for squeaking).

These sources of overstrength are not considered to be detrimental to overall diaphragm performance.

C12.11 STRUCTURAL WALLS AND THEIR ANCHORAGE

As discussed in Section C1.4, structural integrity is important not only in earthquake-resistant design but also in resisting high winds, floods, explosion, progressive failure, and even such ordinary hazards as foundation settlement. The detailed requirements of this section address wall-to-diaphragm integrity.

C12.11.1 Design for Out-of-Plane Forces. Because they are often subjected to local deformations caused by material shrinkage, temperature changes, and foundation movements, wall connections require some degree of ductility to accommodate slight movements while providing the required strength.

Although nonstructural walls are not subject to this requirement, they must be designed in accordance with Chapter 13.

C12.11.2 Anchorage of Structural Walls and Transfer of Design Forces into Diaphragms or Other Supporting Structural Elements. There are numerous instances in U.S. earthquakes of tall, single-story, and heavy walls becoming detached from supporting roofs, resulting in collapse of walls and supported bays of roof framing (Hamburger and McCormick 2004). The response involves dynamic amplification of ground motion by response of the vertical system and further dynamic amplification from flexible diaphragms. The design forces for Seismic Design Category D and higher have been developed over the years in response to studies of specific failures. It is generally accepted that the rigid diaphragm value is reasonable for structures subjected to high ground motions. For a simple idealization of the dynamic response, these values imply that the combined effects of inelastic action in the main framing system supporting the wall, the wall (acting out of plane), and the anchor itself correspond to a reduction factor of 4.5 from elastic response to an MCE_R motion, and therefore the value of the response modification coefficient, R, associated with nonlinear action in the wall or the anchor itself is 3.0. Such reduction is generally not achievable in the anchorage itself; thus, it must come from yielding elsewhere in the structure, for example, the vertical elements of the seismic force-resisting system, the diaphragm, or walls acting out of plane. The minimum forces are based on the concept that less yielding occurs with smaller ground motions and less yielding is achievable for systems with smaller values of R, which are permitted in structures assigned to Seismic Design Categories B and C. The minimum value of R in structures assigned to Seismic Design Category D, except cantilever column systems and light-frame walls sheathed with materials other than wood structural panels, is 3.25, whereas the minimum values of R for Categories B and C are 1.5 and 2.0, respectively.

Where the roof framing is not perpendicular to anchored walls, provision needs to be made to transfer both the tension and sliding components of the anchorage force into the roof diaphragm. Where a wall cantilevers above its highest attachment to, or near, a higher level of the structure, the reduction factor based on the height within the structure, $(1 + 2z/h)/3$, may result in a lower anchorage force than appropriate. In such an instance, using a value of 1.0 for the reduction factor may be more appropriate.

C12.11.2.1 Wall Anchorage Forces. Diaphragm flexibility can amplify out-of-plane accelerations so that the wall anchorage forces in this condition are twice those defined in Section 12.11.1.

C12.11.2.2 Additional Requirements for Anchorage of Concrete or Masonry Structural Walls to Diaphragms in Structures Assigned to Seismic Design Categories C through F

C12.11.2.2.1 Transfer of Anchorage Forces into Diaphragm. This requirement, which aims to prevent the diaphragm from tearing apart during strong shaking by requiring transfer of anchorage forces across the complete depth of the diaphragm, was prompted by failures of connections between tilt-up concrete walls and wood panelized roof systems in the 1971 San Fernando earthquake.

Depending on diaphragm shape and member spacing, numerous suitable combinations of subdiaphragms, continuous tie elements, and smaller sub-subdiaphragms connecting to larger subdiaphragms and continuous ties are possible. The configurations of each subdiaphragm (or sub-subdiaphragm) provided must comply with the simple 2.5-to-1 length-to-width ratio, and the continuous tie must have adequate member and connection strength to carry the accumulated wall anchorage forces. The 2.5-to-1 aspect ratio is applicable to subdiaphragms of all materials, but only when they serve as part of the continuous tie system.

C12.11.2.2.2 Steel Elements of Structural Wall Anchorage System. A multiplier of 1.4 has been specified for strength design of steel elements to obtain a fracture strength of almost 2 times the specified design force (where ϕ_t is 0.75 for tensile rupture).

C12.11.2.2.3 Wood Diaphragms. Material standards for wood structural panel diaphragms permit the sheathing to resist shear

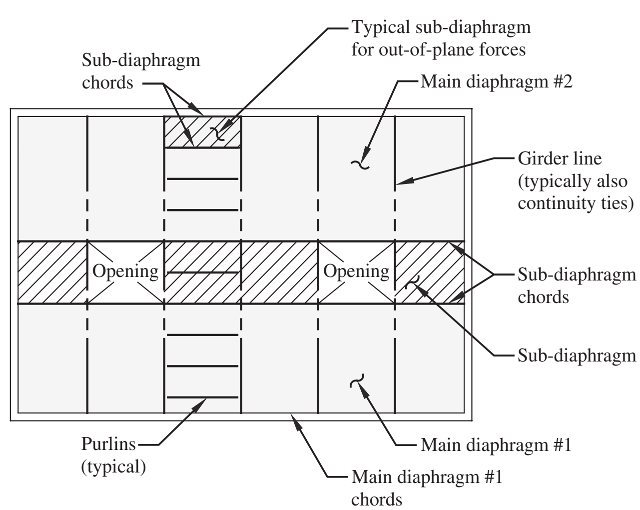

FIGURE C12.11-1 Typical Subdiaphragm Framing

forces only; use of diaphragm sheathing to resist direct tension or compression forces is not permitted. Therefore, seismic out-of-plane anchorage forces from structural walls must be transferred into framing members (such as beams, purlins, or subpurlins) using suitable straps or anchors. For wood diaphragms, it is common to use local framing and sheathing elements as subdiaphragms to transfer the anchorage forces into more concentrated lines of drag or continuity framing that carry the forces across the diaphragm and hold the building together. Fig. C12.11-1 shows a schematic plan of typical roof framing using subdiaphragms.

Fasteners that attach wood ledgers to structural walls are intended to resist shear forces from diaphragm sheathing attached to the ledger that act longitudinally along the length of the ledger but not shear forces that act transversely to the ledger, which tend to induce splitting in the ledger caused by cross-grain bending. Separate straps or anchors generally are provided to transfer out-of-plane wall forces into perpendicular framing members.

Requirements of Section 12.11.2.2.3 are consistent with requirements of AWC SDPWS, SDPWS-15 (2014) Section 4.1.5.1 but also apply to wood use in diaphragms that may fall outside the scope of AWC SDPWS. Examples include use of wood structural panels attached to steel bar joists or metal deck attached to wood nailers.

C12.11.2.2.4 Metal Deck Diaphragms. In addition to transferring shear forces, metal deck diaphragms often can resist direct axial forces in at least one direction. However, corrugated metal decks cannot transfer axial forces in the direction perpendicular to the corrugations and are prone to buckling if the unbraced length of the deck as a compression element is large. To manage diaphragm forces perpendicular to the deck corrugations, it is common for metal decks to be supported at 8- to 10-ft (2.4- to 3.0-m) intervals by joists that are connected to walls in a manner suitable to resist the full wall anchorage design force and to carry that force across the diaphragm. In the direction parallel to the deck corrugations, subdiaphragm systems are considered near the walls; if the compression forces in the deck become large relative to the joist spacing, small compression reinforcing elements are provided to transfer the forces into the subdiaphragms.

C12.11.2.2.5 Embedded Straps. Steel straps may be used in systems where heavy structural walls are connected to wood or steel diaphragms as the wall-to-diaphragm connection system. In systems where steel straps are embedded in concrete or masonry walls, the straps are required to be bent around reinforcing bars in the walls, which improve their ductile performance in resisting earthquake load effects (e.g., the straps pull the bars out of the wall before the straps fail by pulling out without pulling the reinforcing bars out). Consideration should be given to the probability that light steel straps have been used in past earthquakes and have developed cracks or fractures at the wall-to-diaphragm framing interface because of gaps in the framing adjacent to the walls.

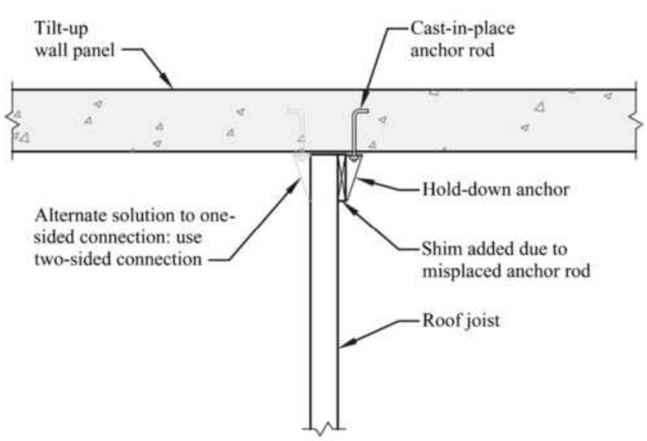

FIGURE C12.11-2 Plan View of Wall Anchor with Misplaced Anchor Rod

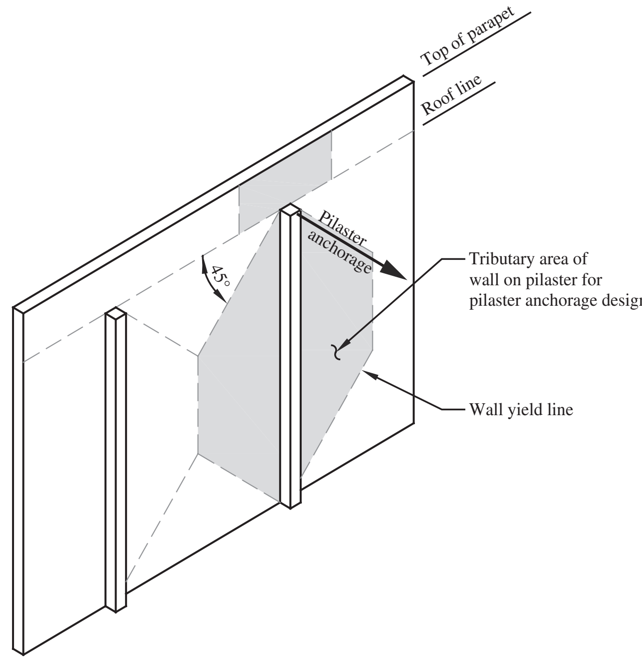

FIGURE C12.11-3 Tributary Area Used to Determine Anchorage Force at Pilaster

C12.11.2.2.6 Eccentrically Loaded Anchorage System. Wall anchors often are loaded eccentrically, either because the anchorage mechanism allows eccentricity or because of anchor bolt or strap misalignment. This eccentricity reduces the anchorage connection capacity and hence must be considered explicitly in design of the anchorage. Fig. C12.11-2 shows a one-sided roof-to-wall anchor that is subjected to severe eccentricity because of a misplaced anchor rod. If the detail were designed as a concentric two-sided connection, this condition would be easier to correct.

C12.11.2.2.7 Walls with Pilasters. The anchorage force at pilasters must be calculated considering two-way bending in wall panels. It is customary to anchor the walls to the diaphragms assuming one-way bending and simple supports at the top and bottom of the wall. However, where pilasters are present in the walls, their stiffening effect must be taken into account. The panels between pilasters are typically supported along all panel edges. Where this support occurs, the reaction at the top of the pilaster is the result of two-way action of the panel and is applied directly to the anchorage supporting the top of the pilaster. The anchor load at the pilaster generally is larger than the typical uniformly distributed anchor load between pilasters. Fig. C12.11-3 shows the tributary area typically used to determine the anchorage force for a pilaster.

Anchor points adjacent to the pilaster must be designed for the full tributary loading, conservatively ignoring the effect of the adjacent pilaster.

C12.12 DRIFT AND DEFORMATION

As used in the standard, deflection is the absolute lateral displacement of any point in a structure relative to its base, and design story drift, Δ, is the difference in deflection along the height of a story (i.e., the deflection of a floor relative to that of the floor below). The drift, Δ, is calculated according to the procedures of Section 12.8.6. (Sections 12.9.2 and 16.1 give procedures for calculating displacements for modal response spectrum and linear response history analysis procedures, respectively; the definition of Δ in Section 11.3 should be used).

Calculated story drifts generally include torsional contributions to deflection (i.e., additional deflection at locations of the center of rigidity at other than the center of mass caused by diaphragm rotation in the horizontal plane). The provisions allow these contributions to be neglected where they are not significant, such as in cases where the calculated drifts are much less than the allowable story drifts, Δ_a, no torsional irregularities exist, and more precise calculations are not required for structural separations (see Sections C12.12.3 and C12.12.4).

The deflections and design story drifts are calculated using the design earthquake ground motion, which is two-thirds of the risk-targeted maximum considered earthquake (MCE_R) ground motion. The resulting drifts are therefore likely to be underestimated.

The design base shear, V, used to calculate Δ is reduced by the response modification coefficient, R. Multiplying displacements by the deflection amplification factor, C_d, is intended to correct for this reduction and approximate inelastic drifts corresponding to the design response spectrum unreduced by R. However, it is recognized that use of values of C_d less than R underestimates deflections (Uang and Maarouf 1994). Also Sections C12.8.6.2 and C12.9.1.4 deal with the appropriate base shear for computing displacements.

For these reasons, the displacements calculated may not correspond well to MCE_R ground motions. However, they are appropriate for use in evaluating the structure's compliance with the story drift limits put forth in Table 12.12-1 of the standard.

There are many reasons to limit drift; the most significant are to address the structural performance of member inelastic strain and system stability and to limit damage to nonstructural components, which can be life-threatening. Drifts provide a direct but imprecise measure of member strain and structural stability. Under small lateral deformations, secondary stresses caused by the P-delta effect are normally within tolerable limits (see Section C12.8.7). The drift limits provide indirect control of structural performance.

Buildings subjected to earthquakes need drift control to limit damage to partitions, shaft and stair enclosures, glass, and other fragile nonstructural components. The drift limits have been established without regard to economic considerations such as a comparison of present worth of future repairs with additional structural costs to limit drift. These are matters for building owners and designers to address.

The allowable story drifts, Δ_a, of Table 12.12-1 reflect the consensus opinion of the ASCE 7 Committee taking into account the life-safety and damage control objectives described in the aforementioned commentary. Because the displacements induced in a structure include inelastic effects, structural damage as the result of a design-level earthquake is likely. This notion may be seen from the values of Δ_a stated in Table 12.12-1. For other structures assigned to Risk Category I or II, the value of Δ_a is $0.02h_{sx}$, which is about 10 times the drift ordinarily allowed under wind loads. If deformations well in excess of Δ_a were to occur repeatedly, structural elements of the seismic force-resisting system could lose so much stiffness or strength that they would compromise the safety and stability of the structure.

To provide better performance for structures assigned to Risk Category III or IV, their allowable story drifts, Δ_a, generally are more stringent than for those assigned to Risk Category I or II. However, those limits are still greater than the damage thresholds for most nonstructural components. Therefore, though the performance of structures assigned to Risk Category III or IV should be improved, there may be considerable damage from a design-level earthquake.

The allowable story drifts, Δ_a, for structures a maximum of four stories above the base are relaxed somewhat, provided that the interior walls, partitions, ceilings, and exterior wall systems have been designed to accommodate story drifts. The type of structure envisioned by footnote *d* in Table 12.12-1 would be similar to a prefabricated steel structure with metal skin.

The values of Δ_a set forth in Table 12.12-1 apply to each story. For some structures, satisfying strength requirements may produce a system with adequate drift control. However, the design of moment-resisting frames and of tall, narrow shear walls or braced frames often is governed by drift considerations. Where design spectral response accelerations are large, seismic drift considerations are expected to control the design of midrise buildings.

C12.12.3 Structural Separation. This section addresses the potential for impact from adjacent structures during an earthquake. Such conditions may arise because of construction on or near a property line or because of the introduction of separations within a structure (typically called "seismic joints") for the purpose of permitting their independent response to earthquake ground motion. Such joints may effectively eliminate irregularities and large force transfers between portions of the building with different dynamic properties.

The standard requires the distance to be "sufficient to avoid damaging contact under total deflection." It is recommended that the distance be computed using the square root of the sum of the squares of the lateral deflections. Such a combination method treats the deformations as linearly independent variables. The deflections used are the expected displacements (e.g., the anticipated maximum inelastic deflections including the effects of torsion and diaphragm deformation). Just as these displacements increase with height, so does the required separation. If the effects of impact can be shown not to be detrimental, the required separation distances can be reduced.

For rigid shear wall structures with rigid diaphragms whose lateral deflections cannot be reasonably estimated, the NEHRP provisions (FEMA 2009a) suggest that older code requirements for structural separations of at least 1 in. (25 mm) plus 1/2 in. (13 mm) for each 10 ft (3 m) of height above 20 ft (6 m) be followed.

C12.12.4 Members Spanning between Structures. Where a portion of the structure is seismically separated from its support, the design of the support requires attention to ensure that support is maintained as the two portions move independently during earthquake ground motions. To prevent loss of gravity support for members that bridge between the two portions, the relative displacement must not be underestimated. Displacements computed for verifying compliance with drift limits [Eq. (12.8-15)] and structural separations [Eq. (12.12-1)] may be insufficient for this purpose.

The provision gives four requirements to ensure that displacement is not underestimated:

1. The deflections calculated using Eq. (12.8-15) are multiplied by $1.5R/C_d$ to correct for likely underestimation of displacement by the equation. The factor of 1.5 corrects for the 2/3 factor that is used in the calculation of seismic base shear, V, by reducing the base shear from the value based

on the MCE_R ground motion (Section 11.4.4). Multiplying by R/C_d corrects for the fact that values of C_d less than R underestimate deflections (Uang and Maarouf 1994).

2. The deflections are calculated for torsional effects, including amplification factors. Diaphragm rotation can add significantly to the center-of-mass displacements calculated using Eq. (12.8-15).
3. Displacements caused by diaphragm deformations are required to be calculated, as in some types of construction where the deformation during earthquake ground motions of the diaphragm can be considerable.
4. The absolute sum of displacements of the two portions is used instead of a modal combination, such as with Eq. (12.12-2), which would represent a probable value.

It is recognized that displacements so calculated are likely to be conservative. However, the consequences of loss of gravity support are likely to be severe, and some conservatism is deemed appropriate.

C12.12.5 Deformation Compatibility for Seismic Design Categories D through F. In regions of high seismicity, many designers apply ductile detailing requirements to elements that are intended to resist seismic forces but neglect such practices for nonstructural components, or for structural components that are designed to resist only gravity forces but must undergo the same lateral deformations as the designated seismic force-resisting system. Even where elements of the structure are not intended to resist seismic forces and are not detailed for such resistance, they can participate in the response and may suffer severe damage as a result. This provision requires the designer to provide a level of ductile detailing or proportioning to all elements of the structure appropriate to the calculated deformation demands at the design story drift, Δ. This provision may be accomplished by applying details in gravity members similar to those used in members of the seismic force-resisting system or by providing sufficient strength in those members, or by providing sufficient stiffness in the overall structure to preclude ductility demands in those members.

In the 1994 Northridge earthquake, such participation was a cause of several failures. A preliminary reconnaissance report of that earthquake (EERI 1994) states the following:

> *Of much significance is the observation that six of the seven partial collapses (in modern precast concrete parking structures) seem to have been precipitated by damage to the gravity load system. Possibly, the combination of large lateral deformation and vertical load caused crushing in poorly confined columns that were not detailed to be part of the lateral load resisting system. Punching shear failures were observed in some structures at slab-to-column connections, such as at the Four Seasons building in Sherman Oaks. The primary lateral load resisting system was a perimeter ductile frame that performed quite well. However, the interior slab–column system was incapable of undergoing the same lateral deflections and experienced punching failures.*

This section addresses such concerns. Rather than relying on designers to assume appropriate levels of stiffness, this section explicitly requires that the stiffening effects of adjoining rigid structural and nonstructural elements be considered and that a rational value of member and restraint stiffness be used for the design of structural components that are not part of the seismic force-resisting system.

This section also includes a requirement to address shears that can be induced in structural components that are not part of the seismic force-resisting system because sudden shear failures have been catastrophic in past earthquakes.

The exception is intended to encourage the use of intermediate or special detailing in beams and columns that are not part of the seismic force-resisting system. In return for better detailing, such beams and columns are permitted to be designed to resist moments and shears from unamplified deflections. This design approach reflects observations and experimental evidence that well-detailed structural components can accommodate large drifts by responding inelastically without losing significant vertical load-carrying capacity.

C12.13 FOUNDATION DESIGN

C12.13.1 Design Basis. In traditional geotechnical engineering practice, foundation design is based on allowable stresses, with allowable foundation load capacities, Q_{as}, for dead plus live loads based on limiting static settlements, which provides a large factor of safety against exceeding ultimate capacities. In this practice, allowable soil stresses for dead plus live loads often are increased arbitrarily by one-third for load combinations that include wind or seismic forces. That approach is overly conservative and not entirely consistent with the design basis prescribed in Section 12.1.5, since it is not based on explicit consideration of the expected strength and dynamic properties of the site soils. Strength design of foundations in accordance with Section 12.13.5 facilitates more direct satisfaction of the design basis.

Section 12.13.1.1 provides horizontal load effect, E_h, values that are used in Section 12.4.2 to determine foundation load combinations that include seismic effects. Vertical seismic load effects are still determined in accordance with Section 12.4.2.2.

Foundation horizontal seismic load effect values specified in Section 12.13.1.1 are intended to be used with horizontal seismic forces, Q_E, defined in Section 12.4.2.1.

C12.13.3 Foundation Load-Deformation Characteristics. For linear static and dynamic analysis methods, where foundation flexibility is included in the analysis, the load-deformation behavior of the supporting soil should be represented by an equivalent elastic stiffness using soil properties that are compatible with the soil strain levels associated with the design earthquake motion. The strain-compatible shear modulus, G, and the associated strain-compatible shear wave velocity, v_s, needed for the evaluation of equivalent elastic stiffness are specified in Chapter 19 of the standard or can be based on a site-specific study. Although inclusion of soil flexibility tends to lengthen the fundamental period of the structure, it should not change the maximum period limitations applied when calculating the required base shear of a structure.

A mathematical model incorporating a combined superstructure and foundation system is necessary to assess the effect of foundation and soil deformations on the superstructure elements. Typically, frequency-independent linear springs are included in the mathematical model to represent the load-deformation characteristics of the soil, and the foundation components are either explicitly modeled (e.g., mat foundation supporting a configuration of structural walls) or are assumed to be rigid (e.g., spread footing supporting a column). In specific cases, a spring may be used to model both the soil and the foundation component (e.g., grade beams or individual piles).

For dynamic analysis, the standard requires a parametric evaluation with upper and lower bound soil parameters to account for the uncertainty in as-modeled soil stiffness and in situ soil variability and to evaluate the sensitivity of these variations on the superstructure. Sources of uncertainty include variability in the rate of loading, including the cyclic nature of building response, level of strain associated with loading at the design earthquake (or stronger), idealization of potentially nonlinear soil properties as elastic, and variability in the estimated soil properties. To a lesser extent, this variation accounts for variability in the performance of the foundation components, primarily when a rigid foundation is assumed or distribution of cracking of concrete elements is not explicitly modeled.

Commonly used analysis procedures tend to segregate the "structural" components of the foundation (e.g., footing, grade beam, pile, and pile cap) from the supporting (e.g., soil) components. The "structural" components are typically analyzed using standard strength design load combinations and methodologies, whereas the adjacent soil components are analyzed using allowable stress design (ASD) practices, in which earthquake forces (that have been reduced by R) are considered using ASD load combinations, to make comparisons of design forces versus allowable capacities. These "allowable" soil capacities are typically based on expected strength divided by a factor of safety, for a given level of potential deformations.

When design of the superstructure and foundation components is performed using strength-level load combinations, this traditional practice of using allowable stress design to verify soil compliance can become problematic for assessing the behavior of foundation components. The 2009 NEHRP provisions (FEMA 2009a) contain two resource papers (RP 4 and RP 8) that provide guidance on the application of ultimate strength design procedures in the geotechnical design of foundations and the development of foundation load-deformation characterizations for both linear and nonlinear analysis methods. Additional guidance on these topics is contained in ASCE 41 (2014b).

C12.13.4 Reduction of Foundation Overturning. Since the vertical distribution of horizontal seismic forces prescribed for use with the equivalent lateral force procedure is intended to envelope story shears, the resulting base overturning forces can be exaggerated in some cases. (See Section C12.13.3.) Such overturning will be over-estimated where multiple vibration modes are excited, so a 25 percent reduction in overturning effects is permitted for verification of soil stability. This reduction is not permitted for inverted pendulum or cantilevered column type structures, which typically have a single mode of response.

Since the modal response spectrum analysis procedure more accurately reflects the actual distribution of base shear and overturning moment, the permitted reduction is reduced to 10 percent.

C12.13.5 Strength Design for Foundation Geotechnical Capacity. This section provides guidance for determination of nominal strengths, resistance factors, and acceptance criteria when the strength design load combinations of Section 12.4.2 are used, instead of allowable stress load combinations, to check stresses at the soil–foundation interface.

C12.13.5.1.1 Soil Strength Parameters. If soils are saturated or anticipated to become so, undrained soil properties might be used for transient seismic loading, even though drained strengths may have been used for static or more sustained loading. For competent soils that are not expected to degrade in strength during seismic loading (e.g., due to partial or total liquefaction of cohesionless soils or strength reduction of sensitive clays), use of static soil strengths is recommended for determining the nominal foundation geotechnical capacity, Q_{ns}, of foundations. Use of static strengths is somewhat conservative for such soils because rate-of-loading effects tend to increase soil strengths for transient loading. Such rate effects are neglected because they may not result in significant strength increase for some soil types and are difficult to estimate confidently without special dynamic testing programs. The assessment of the potential for soil liquefaction or other mechanisms for reducing soil strengths is critical, because these effects may reduce soil strengths greatly below static strengths for susceptible soils.

The best estimated nominal strength of footings, Q_{ns}, should be determined using accepted foundation engineering practice. In the absence of moment loading, the ultimate vertical load capacity of a rectangular footing of width B and length L may be written as $Q_{ns} = q_c(BL)$, where q_c = ultimate soil bearing pressure.

For rigid footings subject to moment and vertical load, contact stresses become concentrated at footing edges, particularly as footing uplift occurs. Although the nonlinear behavior of soils causes the actual soil pressure beneath a footing to become nonlinear, resulting in an ultimate foundation strength that is slightly greater than the strength that is determined by assuming a simplified trapezoidal or triangular soil pressure distribution with a maximum soil pressure equal to the ultimate soil pressure, q_c, the difference between the nominal ultimate foundation strength and the effective ultimate strength calculated using these simplified assumptions is not significant.

Lateral resistance may be determined from test data, or by a combination of lateral bearing, lateral friction, and cohesion values. The lateral bearing values may represent values determined from the passive strength values of soil or rock, or they may represent a reduced "allowable" value determined to meet a defined deformation limit. Lateral friction values may represent side-friction values caused by uplift or movement of a foundation against soils, such as for pile uplift or a side friction caused by lateral foundation movement, or they may represent the lateral friction resistance that may be present beneath a foundation caused by the gravity weight of loads that is bearing upon the supporting material.

The lateral foundation geotechnical capacity of a footing may be assumed to be equal to the sum of the best estimated soil passive resistance against the vertical face of the footing plus the best estimated soil friction force on the footing base. The determination of passive resistance should consider the potential contribution of friction on the vertical face.

For piles, the best estimated vertical strength (for both axial compression and axial tensile loading) should be determined using accepted foundation engineering practice. The moment capacity of a pile group should be determined assuming a rigid pile cap, leading to an initial triangular distribution of axial pile loading from applied overturning moments. However, the full expected axial capacity of piles may be mobilized when computing moment capacity, in a manner analogous to that described for a footing. The strength provided in pile caps and intermediate connections should be capable of transmitting the best estimated pile forces to the supported structure. When evaluating axial tensile strength, consideration should be given to the capability of pile cap and splice connections to resist the factored tensile loads.

The lateral foundation geotechnical capacity of a pile group may be assumed to be equal to the best estimated passive resistance acting against the face of the pile cap plus the additional resistance provided by piles.

When the nominal foundation geotechnical capacity, Q_{ns}, is determined by in situ testing of prototype foundations, the test program, including the appropriate number and location of test specimens, should be provided to the authority having jurisdiction by a registered design professional, based on the scope and variability of geotechnical conditions present at the site.

C12.13.5.2 Resistance Factors. Resistance factors, ϕ, are provided to reduce nominal foundation geotechnical capacities,

Q_{ns}, to design foundation geotechnical capacities, $\phi\, Q_{ns}$, to verify foundation acceptance criteria. The values of ϕ recommended here have been based on the values presented in the AASHTO *LRFD Bridge Design Specifications* (2010). The AASHTO values have been further simplified by using the lesser values when multiple values are presented. These resistance factors account not only for unavoidable variations in design, fabrication, and erection, but also for the variability that often is found in site conditions and test methods (AASHTO 2010).

C12.13.5.3 Acceptance Criteria. The design foundation geotechnical capacity, $\phi\, Q_{ns}$, is used to assess acceptability for the linear analysis procedures. The mobilization of ultimate capacity in nonlinear analysis procedures does not necessarily lead to unacceptable performance because structural deformations caused by foundation displacements may be tolerable. For the nonlinear analysis procedures, Section 12.13.3 also requires evaluation of structural behavior using parametric variation of foundation strength to identify potential changes in structural ductility demands.

C12.13.6 Allowable Stress Design for Foundation Geotechnical Capacity. In traditional geotechnical engineering practice, foundation design is based on allowable stresses, with allowable foundation load capacities, Q_{as}, for dead plus live loads based on limiting static settlements, which provides a large factor of safety against exceeding ultimate capacities. In this practice, allowable soil stresses for dead plus live loads often are increased arbitrarily by one-third for load combinations that include wind or seismic forces. That approach may be both more conservative and less consistent than the strength design basis prescribed in Section 12.1.5, since it is not based on explicit consideration of the expected strength and dynamic properties of the site soils.

C12.13.7 Requirements for Structures Assigned to Seismic Design Category C

C12.13.7.1 Pole-Type Structures. The high contact pressures that develop between an embedded pole and soil as a result of lateral loads make pole-type structures sensitive to earthquake motions. Pole-bending strength and stiffness, the soil lateral bearing capacity, and the permissible deformation at grade level are key considerations in the design. For further discussion of pole–soil interaction, see Section C12.13.8.7.

C12.13.7.2 Foundation Ties. One important aspect of adequate seismic performance is that the foundation system acts as an integral unit, not permitting one column or wall to move appreciably to another. To attain this performance, the standard requires that pile caps be tied together. This requirement is especially important where the use of deep foundations is driven by the existence of soft surface soils.

Multistory buildings often have major columns that run the full height of the building adjacent to smaller columns that support only one level; the calculated tie force should be based on the heavier column load.

The standard permits alternate methods of tying foundations together when appropriate. Relying on lateral soil pressure on pile caps to provide the required restraint is not a recommended method because ground motions are highly dynamic and may occasionally vary between structure support points during a design-level seismic event.

C12.13.7.3 Pile Anchorage Requirements. The pile anchorage requirements are intended to prevent brittle failures of the connection to the pile cap under moderate ground motions. Moderate ground motions can result in pile tension forces or bending moments that could compromise shallow anchorage embedment. Loss of pile anchorage could result in increased structural displacements from rocking, overturning instability, and loss of shearing resistance at the ground surface. A concrete bond to a bare steel pile section usually is unreliable, but connection by means of deformed bars properly developed from the pile cap into concrete confined by a circular pile section is permitted.

C12.13.8 Requirements for Structures Assigned to Seismic Design Categories D through F

C12.13.8.1 Pole-Type Structures. See Section C12.13.7.1.

C12.13.8.2 Foundation Ties. See Section C12.13.7.2. For Seismic Design Categories D through F, the requirement is extended to spread footings on soft soils (Site Class E or F).

C12.13.8.3 General Pile Design Requirement. Design of piles is based on the same response modification coefficient, R, used in design of the superstructure; because inelastic behavior results, piles should be designed with ductility similar to that of the superstructure. When strong ground motions occur, inertial pile–soil interaction may produce plastic hinging in piles near the bottom of the pile cap, and kinematic soil–pile interaction results in bending moments and shearing forces throughout the length of the pile, being higher at interfaces between stiff and soft soil strata. These effects are particularly severe in soft soils and liquefiable soils, so Section 14.2.3.2.1 requires special detailing in areas of concern.

The shears and curvatures in piles caused by inertial and kinematic interaction may exceed the bending capacity of conventionally designed piles, resulting in severe damage. Analysis techniques to evaluate pile bending are discussed by Margason and Holloway (1977) and Mylonakis (2001), and these effects on concrete piles are further discussed by Sheppard (1983). For homogeneous, elastic media and assuming that the pile follows the soil, the free-field curvature (soil strains without a pile present) can be estimated by dividing the peak ground acceleration by the square of the shear wave velocity of the soil. Considerable judgment is necessary in using this simple relationship for a layered, inelastic profile with pile–soil interaction effects. Norris (1994) discusses methods to assess pile–soil interaction.

Where determining the extent of special detailing, the designer must consider variation in soil conditions and driven pile lengths, so that adequate ductility is provided at potentially high curvature interfaces. Confinement of concrete piles to provide ductility and maintain functionality of the confined core pile during and after the earthquake may be obtained by use of heavy spiral reinforcement or exterior steel liners.

C12.13.8.4 Batter Piles. Partially embedded batter piles have a history of poor performance in strong ground shaking, as shown by Gerwick and Fotinos (1992). Failure of battered piles has been attributed to design that neglects loading on the piles from ground deformation or assumes that lateral loads are resisted by axial response of piles without regard to moments induced in the pile at the pile cap (Lam and Bertero 1990). Because batter piles are considered to have limited ductility, they must be designed using the load combinations including overstrength. Moment-resisting connections between pile and pile cap must resolve the eccentricities inherent in batter pile configurations. This concept is illustrated clearly by EQE Engineering (1991).

C12.13.8.5 Pile Anchorage Requirements. Piles should be anchored to the pile cap to permit energy-dissipating mechanisms, such as pile slip at the pile–soil interface, while maintaining a competent connection. This section of the standard sets forth a capacity design approach to achieve that objective. Anchorages occurring at pile cap corners and edges should be reinforced to preclude local failure of plain concrete sections caused by pile shears, axial loads, and moments.

C12.13.8.6 Splices of Pile Segments. A capacity design approach, similar to that for pile anchorage, is applied to pile splices.

C12.13.8.7 Pile–Soil Interaction. Short piles and long slender piles embedded in the earth behave differently when subjected to lateral forces and displacements. The response of a long slender pile depends on its interaction with the soil considering the nonlinear response of the soil. Numerous design aid curves and computer programs are available for this type of analysis, which is necessary to obtain realistic pile moments, forces, and deflections and is common in practice (Ensoft 2004b). More sophisticated models, which also consider inelastic behavior of the pile itself, can be analyzed using general-purpose nonlinear analysis computer programs or closely approximated using the pile–soil limit state methodology and procedure given by Song et al. (2005).

Each short pile (with length-to-diameter ratios no more than 6) can be treated as a rigid body, simplifying the analysis. A method assuming a rigid body and linear soil response for lateral bearing is given in the current building codes. A more accurate and comprehensive approach using this method is given in a study by Czerniak (1957).

C12.13.8.8 Pile Group Effects. The effects of groups of piles, where closely spaced, must be taken into account for vertical and horizontal response. As groups of closely spaced piles move laterally, failure zones for individual piles overlap and horizontal strength and stiffness response of the pile–soil system is reduced. Reduction factors or "*p*-multipliers" are used to account for these groups of closely spaced piles. For a pile center-to-center spacing of three pile diameters, reduction factors of 0.6 for the leading pile row and 0.4 for the trailing pile rows are recommended by Rollins et al. (1999). Computer programs are available to analyze group effects assuming nonlinear soil and elastic piles (Ensoft 2004a).

C12.13.9 Requirements for Foundations on Liquefiable Sites. This new section provides requirements for foundations of structures that are located on sites that have been determined to have the potential to liquefy when subjected to Geomean Maximum Considered Earthquake ground motions. This section complements the requirements of Section 11.8, which provides requirements for geotechnical investigations in areas with significant seismic ground motion hazard with specific requirements for additional geotechnical information and recommendations for sites that have the potential to liquefy when subjected to the Geomean Maximum Considered Earthquake ground motion.

Before the 2010 edition of ASCE 7 (which was based on the *2009 NEHRP Recommended Seismic Provisions for New Buildings and Other Structures*, FEMA 2009a), the governing building code requirements for foundations where potentially liquefiable soil conditions were present was Chapter 18 of the International Building Code (ICC 2009). Chapter 18 of the IBC (ICC 2009) specified the use of the design earthquake (DE) ground motions for all structural and geotechnical evaluations for buildings. Chapter 18 of IBC (ICC 2012) references ASCE 7-10 (2010) and deletes reference to the DE. Chapter 11 of ASCE 7-10 (2010) has new requirements that specify that Maximum Considered Earthquake (MCE) rather than the DE ground motions should be used for geotechnical (liquefaction-related) evaluations that are specified in IBC (ICC 2009).

The reason that the change to MCE ground motions for liquefaction evaluations was made in ASCE 7-10 (2010) was to make the ground motions used in the evaluations consistent with the ground motions used as the basis for the design of structures. Starting with the 2000 edition of the IBC (ICC 2000), the ground motion maps provided in the code for seismic design were MCE mapped values and not DE values. Although design values for structures in the IBC are based on DE ground motions, which are two-thirds of the MCE, studies (FEMA 2009b) have indicated that structures designed for DE motions had a low probability of collapse at MCE level motions. However, these studies presumed nonliquefiable soil conditions. It should also be noted that most essential structures, such as hospitals, are required to be explicitly designed for MCE motions. Whereas ASCE 7-10 has specific requirements for MCE-level liquefaction evaluations, it has no specific requirements for foundation design when these conditions exist. This lack of clear direction was the primary reason for the development of this new section.

The requirements of this section, along with the seismic requirements of this standard, are intended to result in structure foundation systems that satisfy the performance goals stated in Section 1.1 of the 2009 *NEHRP Recommended Seismic Provisions for New Buildings and Other Structures* for structure sites that have been determined to be liquefiable per Section 11.8. They require mitigation of significant liquefaction-induced risks, either through ground improvement or structural measures, aimed at preventing liquefaction-induced building collapse and permitting the structure and its nonstructural system to satisfy the Section 1.1 performance goals. With the exception of Risk Category IV Essential Facilities, the provisions do not seek to control non-life-threatening damage to buildings that may occur as a result of liquefaction-induced settlement. For Risk Category IV Essential Facilities, the provisions seek to limit damage attributable to liquefaction to levels that would permit post-earthquake use. For example, settlement is controlled to levels that would be expected to allow for continued operation of doors.

There is nothing in these provisions that is intended to preclude the Authority Having Jurisdiction from enacting more stringent planning regulations for building on sites susceptible to potential geologic hazards, in recognition of losses that may occur in the event of an earthquake that triggers liquefaction.

In the first paragraph of Section 12.13.9, it is stated that the foundation must also be designed to resist the effects of design earthquake seismic load effects assuming that liquefaction does not occur. This additional requirement is imposed since maximum seismic loads on a foundation during an earthquake can occur before liquefaction. This additional requirement provides assurance that the foundation will be adequate regardless of when liquefaction occurs during the seismic event.

Observed Liquefaction-Related Structural Damage in Past Earthquakes

Damage to structures from liquefaction-related settlement, punching failure of footings, and lateral spreading has been common in past earthquakes. Whereas total postliquefaction settlement values have varied from several inches to several feet (depending on the relative density and thickness of saturated sand deposits), differential settlements depend on the uniformity of site conditions and the depth of liquefied strata. For example, in the 1995 Kobe, Japan, earthquake, total settlements of 1.5 to 2.5 ft (0.46 to 0.76 m) were observed but with relatively small differential settlements.

In the 1989 Loma Prieta, California, earthquake, settlements of as much as 2 ft (0.61 m) and lateral spreading that ranged between 0.25 and 5 ft (0.08 and 1.5 m) were observed on the

Moss Landing spit. The Monterey Bay Aquarium Research Institute's (MBARI's) technology building was supported on shallow foundation with ties and located some 30 ft (9.14 m) away from the edge of the Moss Landing South Harbor. Whereas 0.25 ft (0.76 m) of lateral spreading was measured at the MBARI building, it suffered only minor cracks. On the other hand, the Moss Landing Marine Lab (MLML) building was located on a different part of the spit where between 4 and 5 ft (1.22 and 1.52 m) of lateral spreading was measured. The MLML building, which was supported on shallow foundations without ties, collapsed as the building footings were pulled apart. The MBARI research pier, located at the harbor, across the street from the Technology Building, suffered no damage except for minor spalling at the underside of the concrete deck, where the 16-in. (406.4-mm) diameter cylindrical driven piles for the pier interfaced with the overlying concrete deck.

The 1999 Kocaeli, Turkey, earthquake provided numerous examples of the relationship between liquefaction-induced soil deformations and building and foundation damage in the city of Adapazari. Examples include a five-story reinforced concrete frame building on a mat foundation that settled about 0.5 ft (0.15 m) at one corner and 5 ft (1.5 m) at the opposite corner with related tilting associated with rigid body motion. Essentially no foundation or structural damage was observed. In contrast, several buildings on mat foundations underwent bearing capacity failures and overturned. The foundation soil strength loss, evidenced by bulging around the building perimeter, initiated the failures, as opposed to differential settlement caused by post-liquefaction volume change in the former case history. In addition, lateral movements of building foundations were also observed. Movements were essentially rigid body for buildings on stiff mat foundations, and they led to no significant building damage. For example, a five-story building experienced about 1.5 ft (0.46 m) of settlement and 3 ft (0.91 m) of lateral displacement.

In the 2011 and 2012 Christchurch, New Zealand, earthquakes, significant differential settlement occurred for several buildings on spread footings. Values of differential settlement of 1 to 1.5 ft (0.31 to 0.46 m) were measured for three- to five-story buildings, resulting in building tilt of 2 to 3 deg. Structural damage was less for cases where relatively strong reinforced concrete ties between footings were used to minimize differential settlement. Footing punching failures also occurred leading to significant damage. For taller buildings on relatively rigid raft foundations, differential ground settlement resulted in building tilt, but less structural damage. In contrast, structures on pile foundations performed relatively well.

C12.13.9.1 Foundation Design. Foundations are not allowed to lose the strength capacity to support vertical reactions after liquefaction. This requirement is intended to prevent bearing capacity failure of shallow foundations and axial load failure of deep foundations. Settlement in the event of such failures cannot be accurately estimated and has potentially catastrophic consequences. Such failures can be prevented by using ground improvement or adequately designed deep foundations.

Liquefaction-induced differential settlement can result from variations in the thickness, relative density, or fines content of potentially liquefiable layers that occur across the footprint of the structure. When planning a field exploration program for a potentially liquefiable site, where it is anticipated that shallow foundations may be used, the geotechnical engineer must have information on the proposed layout of the building(s) on the site. This information is essential to properly locating and spacing exploratory holes to obtain an appropriate estimate of anticipated differential settlement. One acceptable method for dealing with unacceptable liquefaction-induced settlements is by performing ground improvement. There are many acceptable methods for ground improvement.

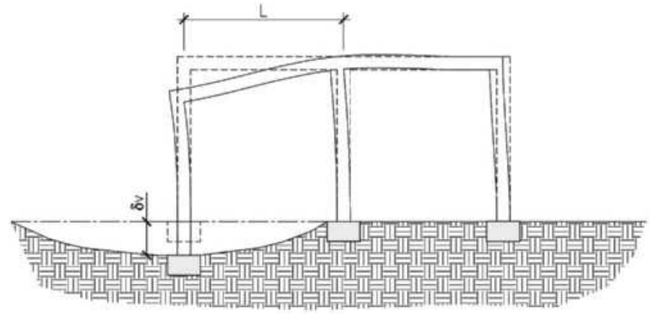

FIGURE C12.13-1 Example Showing Differential Settlement Terms δ_v and L

C12.13.9.2 Shallow Foundations. Shallow foundations are permitted where individual footings are tied together so that they have the same horizontal displacements, and differential settlements are limited or where the expected differential settlements can be accommodated by the structure and the foundation system. The lateral spreading limits provided in Table 12.13-2 are based on engineering judgment and are the judged upper limits of lateral spreading displacements that can be tolerated while still achieving the desired performance for each Risk Category, presuming that the foundation is well tied together. Differential settlement is defined as δ_v/L, where δ_v and L are illustrated for an example structure in Fig. C12.13-1. The differential settlement limits specified in Table 12.13-3 are intended to provide collapse resistance for Risk Category II and III structures.

The limit for one-story Risk Category II structures with concrete or masonry structural walls is consistent with the drift limit in ASCE 41 (2014b) for concrete shear walls to maintain collapse prevention. The limit for taller structures is more restrictive because of the effects that the tilt would have on the floors of upper levels. This more restrictive limit is consistent with the "moderate to severe damage" for multistory masonry structures, as indicated in Boscardin and Cording (1989).

The limits for structures without concrete or masonry structural walls are less restrictive and are consistent with the drift limits in ASCE 41 (2014b) for high-ductility concrete frames to maintain collapse prevention. Frames of lower ductility are not permitted in Seismic Design Categories C and above, which are the only categories where liquefaction hazards need to be assessed.

The limits for Risk Category III structures are two-thirds of those specified for Risk Category II.

The limits for Risk Category IV are intended to maintain differential settlements less than the distortion that will cause doors to jam in the design earthquake. The numerical value is based on the median value of drift (0.0023) at the onset of the damage state for jammed doors developed for the ATC-58 project (ATC 2012), multiplied by 1.5 to account for the dispersion and scaled to account for the higher level of shaking in the MCE relative to the DE.

Shallow foundations are required to be interconnected by ties, regardless of the effects of liquefaction. The additional detailing requirements in this section are intended to provide moderate ductility in the behavior of the ties because the adjacent foundations may settle differentially. The tie force required to accommodate lateral ground displacement is intended to be a

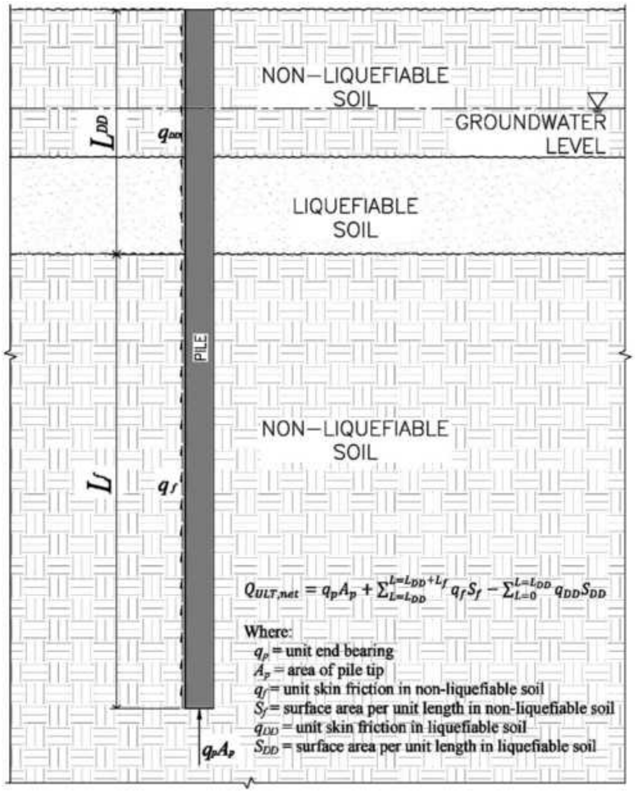

FIGURE C12.13-2 Determination of Ultimate Pile Capacity in Liquefiable Soils

conservative assessment to overcome the maximum frictional resistance that could occur between footings along each column or wall line. The tie force assumes that the lateral spreading displacement occurs abruptly midway along the column or wall line. The coefficient of friction between the footings and underlying soils may be taken conservatively as 0.50. This requirement is intended to maintain continuity throughout the substructure in the event of lateral ground displacement affecting a portion of the structure. The required tie force should be added to the force determined from the lateral loads for the design earthquake in accordance with Sections 12.8, 12.9, 12.14, or Chapter 16.

C12.13.9.3 Deep Foundations. Pile foundations are intended to remain elastic under axial loadings, including those from gravity, seismic, and downdrag loads. Since geotechnical design is most frequently performed using allowable stress design (ASD) methods, and liquefaction-induced downdrag is assessed at an ultimate level, the requirements state that the downdrag is considered as a reduction in the ultimate capacity. Since structural design is most frequently performed using load and resistance factor design (LRFD) methods, and the downdrag is considered as a load for the pile structure to resist, the requirements clarify that the downdrag is considered as a seismic axial load, to which a factor of 1.0 would be applied for design.

The ultimate geotechnical capacity of the pile should be determined using only the contribution from the soil below the liquefiable layer. The net ultimate capacity is the ultimate capacity reduced by the downdrag load (Fig. C12.13-2).

Lateral resistance of the foundation system includes resistance of the piles as well as passive pressure acting on walls, pile caps, and grade beams. Analysis of the lateral resistance provided by these disparate elements is usually accomplished separately. In order for these analyses to be applicable, the displacements used must be compatible. Lateral pile analyses commonly use nonlinear soil properties. Geotechnical recommendations for passive pressure should include the displacement at which the pressure is applicable, or they should provide a nonlinear mobilization curve. Liquefaction occurring in near-surface layers may substantially reduce the ability to transfer lateral inertial forces from foundations to the subgrade, potentially resulting in damaging lateral deformations to piles. Ground improvement of surface soils may be considered for pile-supported structures to provide additional passive resistance to be mobilized on the sides of embedded pile caps and grade beams, as well as to increase the lateral resistance of piles. Otherwise, the check for transfer of lateral inertial forces is the same as for structures on nonliquefiable sites.

IBC (ICC 2012), Section 1810.2.1, requires that deep foundation elements in fluid (liquefied) soil be considered unsupported for lateral resistance until a point 5 ft (1.5 m) into stiff soil or 10 ft (3.1 m) into soft soil unless otherwise approved by the authority having jurisdiction on the basis of a geotechnical investigation by a registered design professional. Where liquefaction is predicted to occur, the geotechnical engineer should provide the dimensions (depth and length) of the unsupported length of the pile or should indicate if the liquefied soil will provide adequate resistance such that the length is considered laterally supported in this soil. The geotechnical engineer should develop these dimensions by performing an analysis of the nonlinear resistance of the soil to lateral displacement of the pile (i.e., *p-y* analysis).

Concrete pile detailing includes transverse reinforcing requirements for columns in ACI 318-14 (2014). This is intended to provide ductility within the pile similar to that required for columns.

Where permanent ground displacement is indicated, piles are not required to remain elastic when subjected to this displacement. The provisions are intended to provide ductility and maintain vertical capacity, including flexure-critical behavior of concrete piles.

The required tie force specified in Section 12.13.9.3.5 should be added to the force determined from the lateral loads for the design earthquake in accordance with Sections 12.8, 12.9, 12.14, or Chapter 16.

C12.14 SIMPLIFIED ALTERNATIVE STRUCTURAL DESIGN CRITERIA FOR SIMPLE BEARING WALL OR BUILDING FRAME SYSTEMS

C12.14.1 General. In recent years, engineers and building officials have become concerned that the seismic design requirements in codes and standards, though intended to make structures perform more reliably, have become so complex and difficult to understand and implement that they may be counterproductive. Because the response of buildings to earthquake ground shaking is complex (especially for irregular structural systems), realistically accounting for these effects can lead to complex requirements. There is a concern that the typical designers of small, simple buildings, which may represent more than 90% of construction in the United States, have difficulty understanding and applying the general seismic requirements of the standard.

The simplified procedure presented in this section of the standard applies to low-rise, stiff buildings. The procedure, which was refined and tested over a five-year period, was developed to be used for a defined set of buildings deemed to be sufficiently regular in structural configuration to allow a reduction of prescriptive requirements. For some design

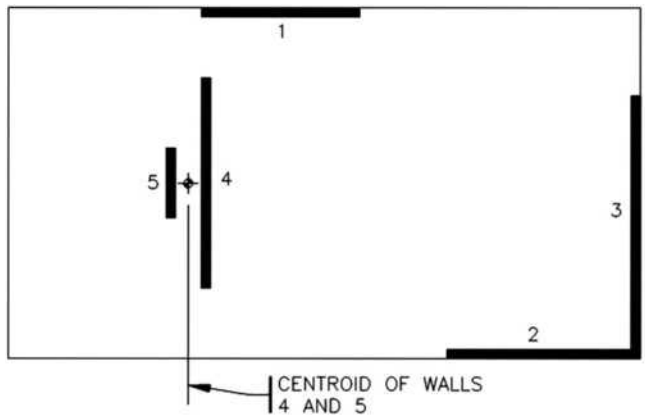

FIGURE C12.14-1 Treatment of Closely Spaced Walls

elements, such as foundations and anchorage of nonstructural components, other sections of the standard must be followed, as referenced within Section 12.14.

C12.14.1.1 Simplified Design Procedure. Reasons for the limitations of the simplified design procedure of Section 12.14 are as follows:

1. The procedure was developed to address adequate seismic performance for standard occupancies. Because it was not developed for higher levels of performance associated with structures assigned to Risk Categories III and IV, no Importance Factor (I_e) is used.
2. Site Class E and F soils require specialized procedures that are beyond the scope of the procedure.
3. The procedure was developed for stiff, low-rise buildings, where higher mode effects are negligible.
4. Only stiff systems where drift is not a controlling design criterion may use the procedure. Because of this limitation, drifts are not computed. The response modification coefficient, R, and the associated system limitations are consistent with those found in the general Chapter 12 requirements.
5. To achieve a balanced design and a reasonable level of redundancy, two lines of resistance are required in each of the two major axis directions. Because of this stipulation, no redundancy factor (ρ) is applied.
6. When combined with the requirements in items 7 and 8, this requirement reduces the potential for dominant torsional response.
7. Although concrete diaphragms may be designed for even larger overhangs, the torsional response of the system would be inconsistent with the behavior assumed in development of Section 12.14. Large overhangs for flexible diaphragm buildings can also produce a response that is inconsistent with the assumptions associated with the procedure.
8. Linear analysis shows a significant difference in response between flexible and rigid diaphragm behavior. However, nonlinear response history analysis of systems with the level of ductility present in the systems permitted in Table 12.14-1 for the higher Seismic Design Categories has shown that a system that satisfies these layout and proportioning requirements provides essentially the same probability of collapse as a system with the same layout but proportioned based on rigid diaphragm behavior (BSSC 2015). This procedure avoids the need to check for torsional irregularity, and calculation of accidental torsional moments is not required. Fig. C12.14-1 shows a plan with closely spaced walls in which the method permitted in subparagraph (c) should be implemented. In that circumstance, the flexible diaphragm analysis would first be performed as if there were one wall at the location of the centroid of walls 4 and 5, then the force computed for that group would be distributed to walls 4 and 5 based on an assessment of their relative stiffnesses.
9. An essentially orthogonal orientation of lines of resistance effectively uncouples response along the two major axis directions, so orthogonal effects may be neglected.
10. Where the simplified design procedure is chosen, it must be used for the entire design in both major axis directions.
11. Because in-plane and out-of-plane offsets generally create large demands on diaphragms, collectors, and discontinuous elements, which are not addressed by the procedure, these irregularities are prohibited.
12. Buildings that exhibit weak-story behavior violate the assumptions used to develop the procedure.

C12.14.3 Seismic Load Effects and Combinations. The equations for seismic load effects in the simplified design procedure are consistent with those for the general procedure, with one notable exception: The overstrength factor (corresponding to Ω_0 in the general procedure) is set at 2.5 for all systems, as indicated in Section 12.14.3.2.1. Given the limited systems that can use the simplified design procedure, specifying unique overstrength factors was deemed unnecessary.

C12.14.7 Design and Detailing Requirements. The design and detailing requirements outlined in this section are similar to those for the general procedure. The few differences include the following:

1. Forces used to connect smaller portions of a structure to the remainder of the structures are taken as 0.20 times the short-period design spectral response acceleration, S_{DS}, rather than the general procedure value of 0.133 (Section 12.14.7.1).
2. Anchorage forces for concrete or masonry structural walls for structures with diaphragms that are not flexible are computed using the requirements for nonstructural walls (Section 12.14.7.5).

C12.14.8 Simplified Lateral Force Analysis Procedure

C12.14.8.1 Seismic Base Shear. The seismic base shear in the simplified design procedure, as given by Eq. (12.14-11), is a function of the short-period design spectral response acceleration, S_{DS}. The value for F in the base shear equation addresses changes in dynamic response for buildings that are two or three stories above grade plane (see Section 11.2 for definitions of "grade plane" and "story above grade plane"). As in the general procedure (Section 12.8.1.3), S_{DS} may be computed for short, regular structures with S_S taken as no greater than 1.5.

C12.14.8.2 Vertical Distribution. The seismic forces for multistory buildings are distributed vertically in proportion to the weight of the respective floor. Given the slightly amplified base shear for multistory buildings, this assumption, along with the limit of three stories above grade plane for use of the procedure, produces results consistent with the more traditional triangular distribution without introducing that more sophisticated approach.

C12.14.8.5 Drift Limits and Building Separation. For the simplified design procedure, which is restricted to stiff shear wall and braced frame buildings, drift need not be calculated. Where drifts are required (such as for structural separations and cladding design) a conservative drift value of 1% is specified.

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